



State-of-the-Art Review of Numerical Simulation of Punching Shear in Slabs

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Abstract

The study of punching shear has gained prominence in structural engineering, particularly over the past decade, due to the abrupt nature of this failure mode and its potentially severe consequences. While certain aspects of punching shear have been extensively investigated, notable gaps persist in the literature. These include the analysis of punching under specific boundary conditions, slabs constructed with alternative materials to conventional reinforced concrete, and solutions such as shear reinforcements and structural strengthening techniques. Although experimental studies dominate the field, many numerical investigations using the finite element method have emerged worldwide, offering valuable insights into this complex phenomenon. This review aims to identify and highlight these research gaps by examining published studies on the numerical simulation of punching shear in slabs. Key aspects of numerical modeling are presented alongside the main conclusions achieved so far while outlining promising future research directions. Potential future research avenues include the numerical simulation of punching shear under dynamic loading conditions, such as seismic actions; the analysis of slabs incorporating novel construction materials, such as lightweight concrete or fiber-reinforced composites; the development and assessment of advanced shear reinforcement systems; and the exploration of punching in complex slab systems, including ribbed slabs, sandwich panels, and biaxial hollow-core slabs. This review seeks to contribute to advancing numerical modeling techniques and understanding punching shear behavior in contemporary structural applications, by addressing these topics.

1 Introduction

Punching shear is a sudden and critical failure mode that can cause significant damage to slabs and, in extreme cases, progressive collapse and complete failure of the structure. This phenomenon is caused by the concentration of stresses around columns in slabs, leading to the formation of cone-shaped cracks. Numerous studies have focused on this failure mode in response to the risks associated with punching

shear, particularly in flat reinforced concrete slabs. The last decade has seen a significant increase in the publication of high-quality scientific papers on this topic, as seen in Fig. 1. This increase is attributed to factors such as the increasing use of flat slabs in construction, advances in experimental testing capabilities, the development of sophisticated numerical tools, and the need to predict structural responses to such failures accurately. In this regard, numerical analyses have also gained prominence in punching studies, primarily due to their lower cost and the ability to evaluate various parameters quickly and efficiently.

Flat slabs offer several advantages, including cost and time savings, increased ceiling height, improved ventilation and lighting, and reduced foundation loads. However, like other slab systems—such as ribbed, hollow-core, sandwich, and biaxial hollow-core slabs—they are also susceptible to punching shear, as reflected in the literature. Research in this area includes various scenarios with different loading conditions, geometries, and construction materials. Studies range from asymmetric punching to alternative materials, shear reinforcement systems, progressive collapse, and

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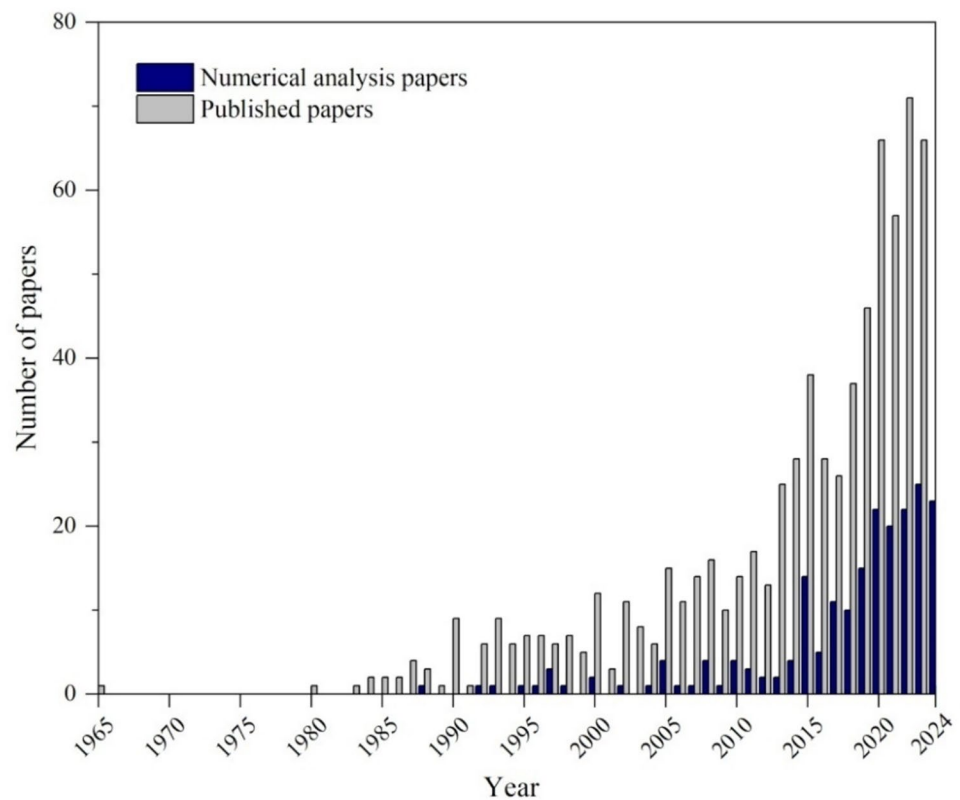
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Fig. 1 Published high-quality scientific papers on punching shear: a comparison between all types of analyses and numerical analyses only



more. These investigations utilize experimental, numerical, and analytical methods to determine the rupture load, evaluate the influence of parameters on the structural response, propose new design approaches, refine existing standards, and validate or compare results with code provisions. Some papers focus on specific aspects of punching shear, while others integrate multiple types of analysis.

Although all types of analyses have provided essential conclusions for this field, this literature review focuses exclusively on numerical simulations of punching shear. Numerical analyses bridge theoretical studies and real-world engineering applications, validating complex models with experimental data obtained in laboratories or field tests. Simulations in this field offer a significantly lower cost than experimental analyses and enable various parameter variations across punching shear scenarios. They allow for the simulation of critical loading conditions, complex behaviors of alternative materials, various column and slab geometries, different flexural and shear reinforcement arrangements, the modeling of strengthening techniques, and alternative structural systems. Numerical tools make it possible to assess optimal or critical scenarios before implementation in design, verify whether code provisions apply under specific conditions, and generate input data for artificial intelligence and Machine Learning simulations. This approach ensures both the safety and cost-effectiveness of structures by enabling the evaluation of multiple design alternatives.

In punching shear scenarios, these simulations often involve computational models validated against experimental data and offer the potential to provide detailed insights into structural responses. However, numerical modeling is a significant challenge and requires in-depth knowledge of programming, constitutive material theories, and the finite element method (FEM). Other important aspects include modeling boundary conditions, the simulation of the behavior of innovative materials, the construction of element meshes for complex geometries, and the selection of suitable methods for solving equations, for example. The techniques used in numerical analysis are very diverse and include, for example, hexahedral or tetrahedral 3D elements, shell, beam, and truss elements. Advanced constitutive models for concrete are often used, considering mechanisms such as cracking, material damage, plasticity, viscosity, and dynamic effects. Researchers use commercial software with calibrated parameters, custom routines, or custom-developed computational tools to simulate material behavior accurately.

Although numerical simulations are widely used to analyze punching shear, specific topics have been explored more intensively than others, and researchers have preferred some modeling techniques over others. On the other hand, some conclusions obtained through numerical studies have been crucial to the current understanding of punching shear. With this in mind, this review aims to comprehensively synthesize the current state-of-the-art about numerical simulations of

punching shear in slabs. To the best of the authors' knowledge, no other study has provided such a comprehensive and focused review of numerical analyses on punching shear, making this paper a valuable contribution to advancing knowledge in this critical area of structural engineering. This study stands out from previously published works by placing numerical analyses at its core, offering not only an in-depth but also a broad examination of the various research approaches to punching shear in slabs. In this regard, it provides unique insights and perspectives, highlighting the applicability of numerical tools used by researchers, the contributions of numerical studies to the understanding of punching shear, and the identification of both key topics extensively discussed in the literature and research gaps—less-represented yet crucial areas for the advancement of structural engineering. Additionally, the text is followed by a discussion outlining the profile of published research, consistently emphasizing numerical analysis as the central focus.

2 Literature Review

The review focuses on presenting, reviewing, and discussing studies that include numerical analyses of punching shear in slabs. This review has analyzed and categorized numerous papers from the literature based on their topics and the type of analysis performed. Only papers published in high-quality journals were selected to elucidate better the state of research on numerical analyses related to punching shear, highlighting those that provided critical analyses and conclusions for punching shear in slabs.

In this context, since the selected studies primarily employed models based on the FEM, particularly involving three-dimensional elements, it is appropriate to provide a brief theoretical overview of the method before presenting the studies that analyzed punching shear and their results. It is important to note that this overview is intended to introduce the general foundations of the FEM in the context of static loading, providing readers with a theoretical basis. However, the method's versatility allows its application to various scenarios, such as dynamic loading, contact problems, and viscoelastic behavior, which will not be covered in this theoretical background.

The FEM is a widely used numerical technique for solving complex problems in structural analysis, solid mechanics, heat transfer, fluid dynamics, and other fields of engineering and applied sciences. Structural engineering uses this method to analyze various systems, including bridges, foundations, soil-structure interaction problems, tunnels, slabs, columns, beams, frames, and trusses [1–10]. Essentially, the FEM discretizes a continuous domain into a finite number of elements interconnected at nodes. This

discretization allows the formulation and solution of equilibrium equations for each element, which can then be assembled to model the behavior of the entire structure. Using the Principle of Virtual Work, it is possible to derive these equilibrium equations from a variational framework, a common approach in solid mechanics.

As illustrated in Eq. 1, the displacement vector $\underline{u}$ of a material particle in the body can be expressed as the difference between the position vectors $\underline{x}$ and $\underline{X}$, which represents the spatial position of the particle in the deformed (Ω_t) and undeformed (Ω_0) configurations of the body, respectively.

$$\underline{u} = \underline{x} - \underline{X} \quad (1)$$

The static equilibrium equations can be expressed considering the material density in the undeformed configuration (ρ_0), the gravity acceleration vector ($\underline{f}$), the traction force vector ($\underline{T}^d$) acting on the surface S_{T_0} , the stress tensor ($\underline{\sigma}$), and the vector of direction cosines of the normal to the surface where the traction acts ($\underline{n}$), as presented in Eq. 2.

$$\begin{cases} \text{div}_x \underline{\sigma} + \rho_0 \underline{f} = \underline{0}(\Omega_0) \\ \underline{\sigma} \cdot \underline{n} = \underline{T}^d(S_{T_0} \subset \partial\Omega_0) \end{cases} \quad (2)$$

The weak formulation of the equilibrium equations can be derived from the Principle of Virtual Work by applying a virtual displacement field ($\underline{u}'$), as presented in Eq. 3. Applying Green's Theorem to the weak form shown in Eq. 3, one obtains Eq. 4, which is suitable for the implementation of the FEM. In this expression, $\underline{\varepsilon}'$ is the virtual strain tensor associated with $\underline{u}'$.

$$\int_{\Omega_0} [\text{div}_x \underline{\sigma} + \rho_0 \underline{f}] \cdot \underline{u}' d\Omega_0 = 0 \quad (3)$$

$$\int_{\Omega_0} \underline{\sigma} : \underline{\varepsilon}' d\Omega_0 = \int_{\Omega_0} \rho_0 \underline{f} \cdot \underline{u}' d\Omega_0 + \int_{S_{T_0}} \underline{u}' \cdot \underline{T}^d dS \quad (4)$$

Furthermore, in the spatial discretization of FEM, the continuous medium is approximated by a discrete system composed of finite elements. This allows the displacement field $\hat{\underline{u}}$ and the coordinates $\underline{X}$ to be approximated within each element using the nodal values and interpolation functions, as shown in Eqs. 5 and 6. In this context, $\underline{N}$ is the matrix containing the interpolation (shape) functions associated with each node of the element, $\tilde{\underline{X}}^e$ is the vector of nodal coordinates, and $\tilde{\underline{u}}^e$ is the vector of nodal displacements for the current element e .

$$\underline{X} = \underline{N} \tilde{\underline{X}}^e \quad (5)$$

$$\underline{\hat{u}} = N \underline{\tilde{u}}^e \quad (6)$$

By applying the compatibility equations and expressing the strain tensor $\underline{\varepsilon}$ in Voigt notation as $\underline{\varepsilon} = [\varepsilon_x, \varepsilon_y, \varepsilon_z, \gamma_{xy}, \gamma_{yz}, \gamma_{xz}]^T$, the strain–displacement relationship can be defined as shown in Eq. 7. Furthermore, the stress–strain relationship can be adopted for linear elastic material behavior, as presented in Eq. 8. In these expressions, $\mathbf{B}$ is the strain–displacement matrix that relates the strain components to the nodal displacements of the element, $\mathbf{D}$ is the constitutive matrix defined in terms of the material's Young's modulus and Poisson's ratio, and the stress vector is given by $\underline{\sigma} = [\sigma_x, \sigma_y, \sigma_z, \sigma_{xy}, \sigma_{yz}, \sigma_{xz}]^T$.

$$\underline{\varepsilon} = \mathbf{B} \underline{\tilde{u}}^e \quad (7)$$

$$\underline{\sigma} = \mathbf{D} \underline{\varepsilon} \quad (8)$$

When the material exhibits nonlinear behavior, an incremental form of the stress–strain relationship must be adopted, as expressed in Eq. 9, where $\mathbf{D}_{ep}$ is the elastoplastic constitutive matrix.

$$\Delta \underline{\sigma} = \mathbf{D}_{ep} \Delta \underline{\varepsilon} \quad (9)$$

By substituting the previously defined relationships into Eq. 4, one arrives at Eq. 10, where the left-hand side represents the internal force vector. This term can be further expressed as in Eq. 11.

$$\int_{\Omega_0} \mathbf{B}^T \underline{\sigma} d\Omega_0 = \int_{\Omega_0} \mathbf{N}^T \rho_0 \underline{f} d\Omega_0 + \int_{S_{T_0}} \mathbf{N}^T \underline{T}^d dS \quad (10)$$

$$\underline{f}_{-int}^e = \int_{\Omega_0} \mathbf{B}^T \underline{\sigma} d\Omega_0 = \left(\int_{\Omega_0} \mathbf{B}^T \mathbf{D} \mathbf{B} d\Omega_0 \right) \underline{\tilde{u}}^e \quad (11)$$

In this expression, the term in parentheses corresponds to the element stiffness matrix, as indicated in Eq. 12. In the non-linear case, the linear elastic constitutive matrix $\mathbf{D}$ can be replaced by the elastoplastic matrix $\mathbf{D}_{ep}$.

$$\mathbf{K}^e = \int_{\Omega_0} \mathbf{B}^T \mathbf{D} \mathbf{B} d\Omega_0 \quad (12)$$

The element external force vector $\underline{f}_{-t}^e$ at the right-hand side of Eq. 10 can be further subdivided into body forces ($\underline{f}_{-v}^e$) and surface forces ($\underline{f}_{-s}^e$), as presented in Eqs. 13 and 14, respectively.

$$\underline{f}_{-v}^e = \int_{\Omega_0} \mathbf{N}^T \rho_0 \underline{f} d\Omega_0 \quad (13)$$

$$\underline{f}_{-s}^e = \int_{S_{T_0}} \mathbf{N}^T \underline{T}^d dS \quad (14)$$

Thus, the static equilibrium at the element level, as given by Eq. 10, can finally be written in the form of Eq. 15, where $\underline{\tilde{u}}^e = [u_x^1 u_y^1 u_z^1 \dots u_x^n u_y^n u_z^n]^T$, with u_x, u_y, u_z being the displacement components in the Cartesian x, y , and z directions, respectively.

$$\mathbf{K}^e \underline{\tilde{u}}^e = \underline{f}_{-t}^e \quad (15)$$

The interpolation matrix $\mathbf{N}$ and strain–displacement matrix $\mathbf{B}$ are expressed in Eqs. 16 and 17, respectively, where $N_{ij} = \frac{\partial N_i}{\partial x_j}$ represents the derivative of the shape function N_i with respect to the Cartesian coordinate j .

$$\mathbf{N} = \begin{bmatrix} N_1 & 0 & 0 & N_n & 0 & 0 \\ 0 & N_1 & 0 & \dots & 0 & N_n & 0 \\ 0 & 0 & N_1 & & 0 & 0 & N_n \end{bmatrix} \quad (16)$$

$$\mathbf{B} = \begin{bmatrix} N_{1,x} & 0 & 0 & & N_{n,x} & 0 & 0 \\ 0 & N_{1,y} & 0 & & 0 & N_{n,y} & 0 \\ 0 & 0 & N_{1,z} & \dots & 0 & 0 & N_{n,z} \\ N_{1,y} & N_{1,x} & 0 & \dots & N_{n,y} & N_{n,x} & 0 \\ 0 & N_{1,z} & N_{1,y} & & 0 & N_{n,z} & N_{n,y} \\ N_{1,z} & 0 & N_{1,x} & & N_{n,z} & 0 & N_{n,x} \end{bmatrix} \quad (17)$$

In this context, for reinforced concrete modeling, it is common to compute the stiffness matrix $\mathbf{K}_c^e$ by accounting for the contributions of the concrete and the embedded reinforcing bars. Assuming perfect bond between the materials at their interface, the total stiffness matrix of the reinforced concrete element can be computed as shown in Eq. 18, where $\mathbf{K}_c^*$ is the stiffness matrix of the concrete component as evaluated in Eq. 12, and $\mathbf{K}_b$ is the stiffness matrix of a bar element of length l_s , as defined in Eq. 19. The parameter nb represents the number of embedded bars within the current element. Additionally, E_s, A_s and $\mathbf{B}_b$ denote the Young's modulus, cross-sectional area, and strain–displacement matrix of the reinforcing bar, respectively.

$$\mathbf{K}_c^e = \mathbf{K}_c^* + \sum_{nb} \mathbf{K}_b \quad (18)$$

$$\mathbf{K}_b = \int_{l_s} \mathbf{B}_b^T E_s A_s \mathbf{B}_b ds \quad (19)$$

With this in mind, it is worth mentioning that the FEM offers several advantages when applied to punching shear analysis, which justifies its predominant use in studies

addressing this type of structural failure. Among these advantages is the ability to create detailed models that account for geometric complexity, use materials with different properties, and apply various boundary conditions. Furthermore, numerical simulations developed using this tool allow for an in-depth analysis of localized stresses and strains and consider complex material behaviors through refined constitutive models.

However, it is common for such simulations to raise specific issues that must be carefully addressed in the numerical model. These include the sensitivity of the results to the adopted constitutive model—which may also introduce mesh dependency—the accurate representation of the interaction between concrete and reinforcement, the complexity involved in model construction (such as reinforcing bar placement or the modeling of diverse geometries), and the high computational cost associated with certain types of analyses, particularly those employing refined meshes with a high number of nodes.

Simplified modeling approaches are often adopted to overcome some of these disadvantages. While these simplifications can reduce computational costs, they must always be accompanied by rigorous validation to ensure reliable results, carefully assessing any loss of accuracy and confidence in the numerical models. It is also important to note that the accuracy achieved through these models can be high, as can the associated computational expense, depending on the chosen constitutive model, the type of finite elements used, and the consideration of effects such as bond and contact between materials.

Specifically concerning punching shear in slabs, numerical analyses are applied in various scenarios involving different construction systems, loading types, and materials. Accordingly, this study aims to present the current state of the art in research related to this type of failure, which is detailed in the following sections. To show the studies clearly, the topic of punching shear was divided into eight sections, covering numerical studies developed up until early 2025.

The first section addresses the simplest case of punching shear, described as symmetric punching in flat slabs made of conventional reinforced concrete with no openings or shear reinforcement. This section briefly discusses the studies in this area, highlighting the type of modeling used and the analyses performed. As this is a particular case, it is noteworthy that relatively few studies focus exclusively on this topic. However, most numerical models include a validation step for this more straightforward scenario.

The second section examines asymmetric punching shear, including unbalanced or eccentric loads, punching in corner or edge connections, and punching in connections with columns or slabs of different geometries. The third section presents studies on cyclic lateral loading and seismic loads,

while the fourth addresses openings near the column area. The fifth section discusses the use of alternative materials for both conventional concrete and steel reinforcement bars. The sixth section explores structural strengthening techniques, and the seventh focuses on shear reinforcements in slabs subjected to punching shear. In all these cases, the numerical studies are presented clearly and concisely, complemented by a table summarizing each relevant work, its objectives, key points, and significant conclusions. The primary focus of these sections is an in-depth discussion of the topics analyzed from the perspective of numerical simulations. Each section also includes a brief introduction and a concluding discussion in which specific studies from the literature are further elaborated to provide the reader with insights into existing numerical models.

The eighth section deals briefly with other relevant punching shear scenarios, outlining the topics and highlighting critical aspects of numerical analyses performed on them. These scenarios include progressive collapse, impact, prestressed slabs, voided slabs, hollow-core slabs, ribbed slabs, sandwich slabs, composite slabs, slabs with drop panels, and bridge decks; all analyzed through the lens of punching shear in these structures.

It is essential to point out that while many sections and subsections incorporate experimental studies relevant to understanding the topic and identifying what has been previously produced, numerical studies remain at the forefront of the discussion. When a single paper includes multiple types of analyses (experimental, analytical, and numerical), its presentation always focuses on the numerical component of the study. If an article provides relevant information for two or more sections of this review, it may also appear in multiple sections, focusing on the pertinent conclusions to each topic analyzed.

2.1 The Simplest Case of Punching Shear in Flat Slabs

The simplest case of punching shear in flat slabs involves the analysis of an interior connection subjected to symmetric concentric loading on an isolated slab. This load can be applied directly onto the slab (fixing the column ends) or the column (fixing the slab). Symmetric punching shear refers to cases where stresses around the column are evenly distributed, as observed in interior connections of square slabs and square or circular columns, without openings and subjected solely to vertical loading. In this particular scenario, cracks initiate radially from the top surface of the slab, marking the end of the elastic phase of the structure and changing the slope of the load–displacement curve. Subsequently, the radial cracks continue to propagate under intermediate loading while circumferential cracks begin to form. These

circumferential cracks at load levels approaching failure lead to a punching shear failure cone.

In this simplest case, symmetry is considered alongside the use of conventional reinforced concrete, i.e., steel bars and ordinary concrete are employed. No additional features such as shear reinforcement, prestressing, openings, or drop panels are included. Consequently, the structural response is predominantly governed by the characteristics of the concrete (e.g. strength and modulus of elasticity), the reinforcement ratio of the steel, the slab's effective depth (d), and the column dimensions. This case, which represents the fundamental research in the field and continues to be studied, has consistently yielded converging conclusions in the literature regarding how geometric, material, and boundary condition parameters affect the structural response.

In numerical analyses, due to the symmetry of both the loading and geometry, only one-quarter of the structure is typically modeled to reduce computational effort. A variety of element types can be used, such as 3D solid elements [11–20], shell elements [13, 21, 22], thick-shell layered finite elements [23], 2D axisymmetric elements [24, 25], grid beam macro-modeling procedures [26], among others. Additionally, material modeling can range from simple to complex constitutive models for concrete, with reinforcement models as embedded, discrete, or distributed. Most studies have shown satisfactory results in numerical modeling, with a good approximation of the slab's failure load. However, certain aspects, such as stiffness and the load corresponding to initial cracking, occasionally deviate from experimental data.

Numerical analyses enable the evaluation of stress and strain distributions, crack patterns, maximum displacements, and failure loads, which provide a deeper understanding of this type of failure. Consequently, many numerical studies focus on validating a numerical model by comparing its results with experimental data [11–18, 20, 21, 23, 24]. Once these models are validated, they often serve as the basis for parametric analyses [14, 17, 21, 24, 27], where the influence of individual parameters, such as concrete properties, flexural reinforcement ratio, steel yield stress, slab thickness, column dimensions, boundary conditions, and size effects, can be isolated and evaluated.

In this context, it is worth noting that this simple case of punching shear remains prevalent in most numerical studies, either for an in-depth investigation of specific modeling aspects or for validating experimental data. This simple structure allows researchers to evaluate the parameters of the constitutive model [12, 13, 16, 18–20], assess whether the chosen elements adequately describe the punching shear phenomenon [11, 13, 20], analyze codes, computer programs, and software [21], and ensure that further numerical data are developed based on an adequately validated model. Therefore, although this case represents a particular scenario

regarding boundary conditions, materials, and geometry, it provides valuable insights into the general behavior of punching shear.

2.2 Asymmetric Punching Shear

Although symmetric punching shear has been extensively studied, it represents a particular case of loading and geometry that rarely occurs in practical design. This is because the asymmetry in the distribution of shear stresses around the column becomes evident in various situations, which have been increasingly studied over the past decade. Among these, noteworthy cases include loading due to unbalanced moments [28–37], punching shear with eccentric loading [38–50], punching in slabs supported by edge columns [32, 49, 51–65] and corner columns [32, 51, 61, 65–68], columns with different shapes [35, 69–72], or slabs supported by walls [73–75], as well as slabs that are not axisymmetric around the column axis [35, 61, 65, 76]. Furthermore, studies present analyses of asymmetric punching in slabs caused by openings or lateral loads. These will be discussed in more detail in other sections of this study.

The number of corner and edge columns is more significant in buildings with a small number of bays, and punching shear in these connections is more critical due to eccentric loads and moment transfer. However, published research has focused significantly more on internal connections [51, 52]. Moreover, in design practice, slab loading is rarely symmetric due to variations in the stiffness or strength of supporting columns or walls, asymmetric loading, irregular column layouts, differing span lengths, and creep and differential shrinkage between two continuous slabs [28, 51]. Amiri and Esfahani [38] also note that unbalanced moments can result from unequal vertical loads in adjacent spans or lateral forces caused by wind or earthquakes. Also, it is well known that special-shaped columns are widely used in slab-column structures [69] to meet architectural and structural layout requirements, altering the slab's stress distribution and cracking patterns.

Loading conditions can cause asymmetric punching. Some experimental studies have applied unbalanced loads, using different proportions of loading in various directions of the slab [28, 29] (Fig. 2a), applying eccentric loads by extending the column base [38–40] (Fig. 2b), or using both horizontal and vertical loads at the column ends [30, 31, 59] (Fig. 2c) to induce asymmetric punching. Regarding structural support, studies either restrained vertical displacement along the slab edges and applied the load at the column or fixed the displacement at the column ends and applied the load directly to the slab.

Regarding asymmetric punching shear caused by the geometry of the connection, experimental tests have been carried out under various boundary conditions to investigate

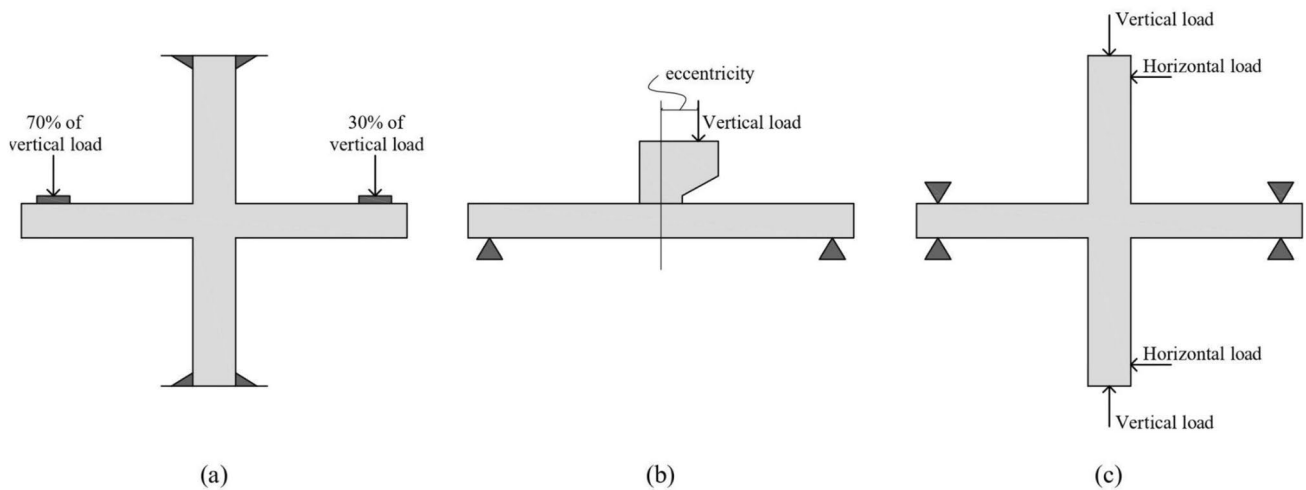


Fig. 2 Examples of different ways to apply asymmetric loading in internal connections: **(a)** unbalanced due to the application of loads with different proportions for each direction of the slab (for example, 30% of the load in one direction and 70% in the other), **(b)** eccentric

through the application of a load positioned at a certain distance from the center of the column, and **(c)** lateral loading, through the application of horizontal load, with the possibility of also applying vertical load

edge and corner connections. Hernández Fraile et al. [52] highlight that the most common test type for edge connections involves the slab portion surrounding an edge column, bounded by lines of contraflexure (Fig. 3a). In this case, the slab is supported at three edges so that the eccentricity can be applied to the column by vertical loads. However,

the authors also point out that some researchers performed tests or numerical simulations on slabs spanning the distance between two opposite edge columns (Fig. 3b) [62], tested slabs covering regions around the edge and adjacent internal columns (Fig. 3c), or represented slab continuity using steel profiles and specific boundary conditions (Fig. 3d) [52].

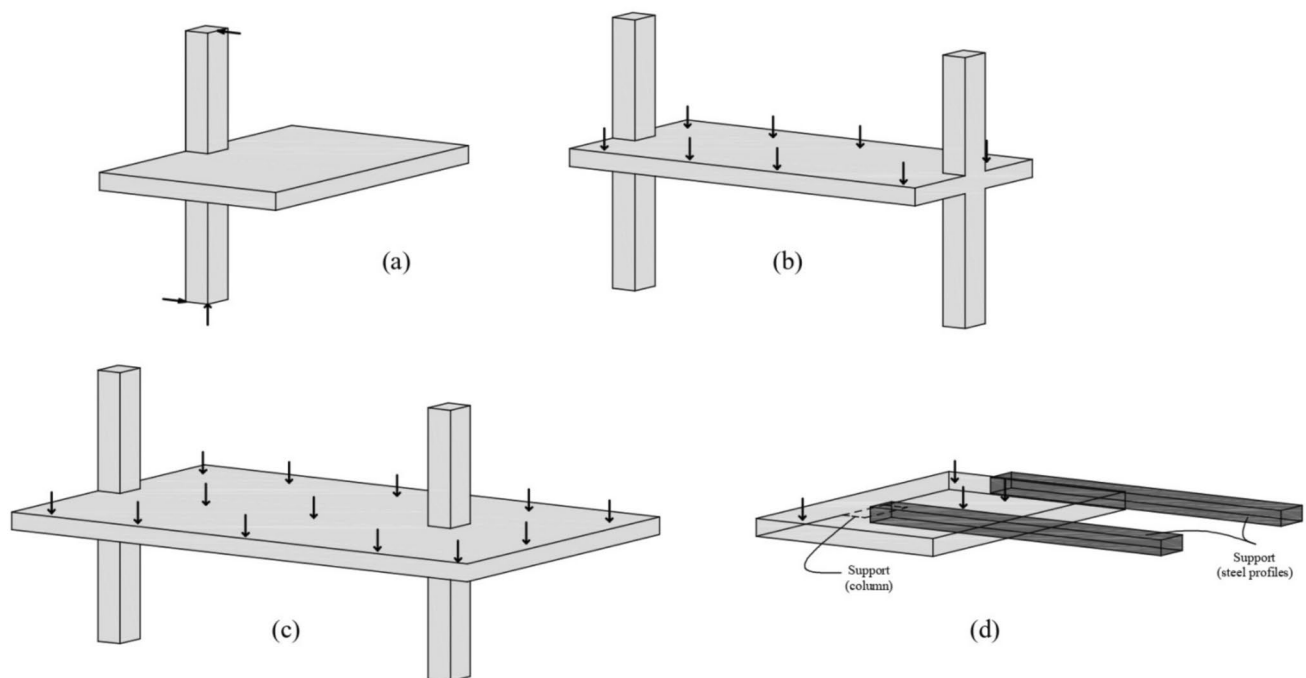


Fig. 3 Example of different ways of applying asymmetric loading in edge and corner connections: **(a)** slab portion surrounding an edge column, **(b)** slabs spanning the distance between two opposite edge

columns, **(c)** regions around the edge and adjacent internal columns, and **(d)** represented slab continuity using steel profiles

Regarding asymmetric punching due to column geometry, studies have evaluated rectangular [28, 35, 65, 72, 77], T-shaped, L-shaped, and cruciform columns [69, 70]. Milligan and Polak [70] and Xue et al. [71] also observed that the stress distribution around columns with different cross-sections is non-uniform, making punching in these structures more complex due to stress concentrations. Additionally, Milligan et al. [73] demonstrated that slabs supported by rectangular walls exhibit a significant directional shear impact on structural behavior.

Some studies have explored unbalanced or eccentric loading concurrently with factors like openings [28, 44, 46, 47], shear reinforcement [29, 30, 33, 34, 36, 42], alternative concretes [38, 39], alternative reinforcements [30], or slabs with spherical voids [36, 45] or ribbed slabs [40]. For edge or corner connections, research has also addressed alternative concretes [54], alternative reinforcements [57, 63, 64], shear reinforcement [52, 59, 63, 64, 66, 67], prestressing [53, 55], or capital heads [48, 61] and ribbed slabs [56]. Studies on columns with different cross-sections have explored using steel capitals [69] or openings [70].

Numerical simulations generally aimed to validate experimental data from the authors or the literature, evaluating whether the models adequately described asymmetric punching behavior. Most studies employed 3D solid modeling elements, although some used plate elements [31, 59, 61, 65, 78], plate and 3D joint elements [43, 74, 75], or grid models [32]. Hernández Fraile et al. [52] also investigated a testing system considering slab continuity in edge connections.

After validation, studies compared load predictions with design codes [31, 34, 51, 70, 72–74, 76] or conducted parametric analyses to understand the influence of various parameters [31, 41, 43, 46, 47, 50, 59, 62, 65, 71, 73, 75–77]. Wang and Teng [61] analyzed an irregular slab supported by seven columns, comparing it with experimental data, while Matseevich [65] investigated an entire floor system. Among asymmetric punching topics, numerical studies have significantly focused on modeling unbalanced moments, eccentric loading, and edge connections. These studies concluded that increased eccentricity or moment imbalance significantly reduces punching shear capacity and ductility, making asymmetric punching a critical issue for internal and edge connections [31, 36, 44, 48–50, 63, 64]. Few studies have addressed columns with different cross-sections, slabs with various geometries, or corner connections.

Table 1 provides a detailed summary of modeling aspects, including boundary conditions in numerical simulations. It also outlines the scope of each article, the type of asymmetry studied, key findings, and conclusions about asymmetric punching. Numerical analyses significantly contribute to understanding asymmetric punching, allowing for accurate modeling of geometry, loading application, supports, and material behavior. They enable estimations of cone

formation location, crack patterns, load capacity, and stiffness in such structures.

As evidenced in recent publications, numerous significant insights have been achieved through numerical analyses of asymmetric punching. The primary focus in this field has been on asymmetric punching caused by eccentric loading or its investigation in edge connections. Regarding eccentric or unbalanced loading, various studies such as Al-Hamd et al. [42], Mendes et al. [50], and Schimanowski et al. [36] highlight that such loading reduces punching shear capacity when compared to symmetric loading. The punching resistance decreases as the degree of load imbalance increases. Thus, special attention must be given to modeling the structure's geometry and the boundary conditions of support modeling and load application for the numerical analysis of asymmetric punching.

In this regard, Sanabria Díaz et al. [49] examined the behavior of edge connections under eccentric loading using a 3D solid model. The boundary conditions included structural symmetry, a common approach in similar studies. Supports were modeled with spring elements, incorporating a degree of stiffness into the system. As shown in Fig. 4, which compares numerical modeling with experimental behavior for different eccentricities, the eccentricity direction alters the cracking pattern in the slab, both on the top and bottom surfaces. According to the authors, for slab L1 (with inward eccentricity), the highest stress occurred in the top rebar perpendicular to the free edge, close to the column's inner face. In contrast, slab L3 (with outward eccentricity) presented the most significant stress in the top rebar near the free edge.

Recent numerical analyses often utilize advanced 3D models with complex constitutive laws for concrete, capturing the material's nonlinearity. A growing trend in asymmetric punching research is evident, exploring new conditions such as alternative materials, shear reinforcement, openings in slabs, and structural strengthening techniques. However, significant gaps remain, particularly concerning asymmetric punching due to column or slab geometry, slabs supported on walls, and corner connections, which demand further investigation.

2.3 Punching in Slabs Due to Cyclic Lateral Loads or Seismic Actions

Cyclic lateral loads are primarily associated with dynamic forces caused by seismic events, necessitating in-depth investigations to assess the structural behavior under such conditions. Abolhelm and Hrynyk [79] emphasize that inter-story drift demands resulting from lateral loads can adversely affect the shear resistance of slab-column connections, making punching failure a critical concern. While vertical load analyses have yielded consolidated conclusions regarding the influence of various parameters on punching

Table 1 Review of numerical analysis of slabs subjected to asymmetric punching shear

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Loo and Guan (1997) [78]	(Numerical analysis) FE Program/ Custom Program	To validate a layered finite element model for slabs supported on edge and corner columns with spandrel beams	The simulation was validated using experimental data obtained from the extant literature. Eight-node shell elements were employed, with each layer assigned specific material properties. The reinforcement was modeled as a distributed layer, and the load was applied as a distributed force on a structure with six columns. Material and geometric nonlinearities were considered in the analysis	The numerical results regarding punching shear resistance, load-displacement behavior, and crack patterns agreed with the experimental data. The simulation effectively captured the structural behavior of slabs with spandrel beams or torsion strips
Megally and Ghali (2000) [58]	(Numerical analysis) FE Program/ ANACAP	To calculate the moment transferred due to the eccentricity of the shear force	The validation step compared the numerical results with empirical data from extant literature concerning slabs supported on corner columns. Twenty-node hexahedral elements were employed to model the concrete, while bar elements were utilized to model the reinforcement as embedded elements. The investigation focused on edge connections under loads applied at the column, with varying eccentricities. Different parameters and theoretical formulas were analyzed	The model showed good agreement with the experimental results regarding failure loads. The ratio of the projections of the critical section contributed most to the fraction of the moment transferred by shear, among other parameters
Polak (2005) [59]	(Numerical analysis) FE Program	To investigate slabs supported on edge columns, with or without shear reinforcement, and with or without openings near the column	The study considered slabs with two different moment-to-shear ratios. The author's numerical and experimental data were compared. Shell elements with nine nodes were used in the numerical modeling. The slab edge containing the column was modeled as free, while other edges were simply supported. A lateral and vertical loading was applied to the column edges	The numerical model accurately represented the experimental failure load data but showed higher stiffness for slabs supported on edge columns with shear reinforcement. Increasing the moment-to-shear (M/S) ratio reduced the ultimate load and ductility of the slab

Table 1 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Park and Choi (2006) [31]	(Numerical analysis) FE Program/ Custom Program	To investigate the application of unbalanced moments in slabs supported on interior columns	The simulation was validated using experimental datasets developed under various test systems from the extant literature. In some cases, lateral loads were applied at the column ends, while in others, the unbalanced moments were imposed through a pair of oppositely directed displacements. Concrete was modeled with nine-node shell elements, and the reinforcement was modeled as a distributed layer	The numerical validation results closely matched the experimental data. A parametric analysis enabled the investigation of the distribution and intensity of internal forces near the slab-column connection
Marinković and Alendar (2008) [60]	(Numerical analysis) FE Program/ DIANA	To study punching shear in prestressed slabs supported on edge columns	The modeling was validated against experimental data. Twenty-node isoparametric solid elements were used. As in the experimental tests, the column was modeled with eight-node shell elements, incorporating a hollow steel collar. The load was applied at the steel collar, while the slab was fixed at the edge opposite the column and at specific points along the sides. Material models were tested with different concrete tensile softening relationships: brittle cracking and linear softening	The comparison with experimental data revealed that employing the brittle cracking concrete tensile softening provided better alignment with the results. High-stress concentrations were observed at the corners of the steel collar. Punching shear failure predominantly occurs due to concrete cracking rather than crushing
Wang and Teng (2008) [61]	(Numerical analysis) FE Program/ Custom Program	To analyze an irregular slab supported by seven columns and isolated slabs with and without column drop panels, subjected to concentric loading	Experimental data from the literature were used to validate the numerical simulation. Shell elements considering transverse shear deformations were used. For slabs with concentric loading, the load was applied at the column, and the slab was fixed at the edges with vertical restraints. All seven columns were modeled for the irregular slab model, and the load was applied as a uniformly distributed force	The numerical model agreed well with experimental results regarding failure load, displacements, and crack patterns. Local punching failure was observed, and the global behavior of the slab was reasonably well captured

Table 1 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Coronelli (2010) [32]	(Numerical analysis) Beam-column Frames/ OpenSees	To investigate punching shear in slabs supported on edge and corner columns subjected to unbalanced moments	A grid model, an intermediate approach between 3D finite elements and 2D frame models, was used to analyze experimental slabs from the literature. The slab was divided into a grid of beam finite elements, and frame elements were also used. Unbalanced loads were analyzed in slabs supported on corner, edge, or interior columns, with and without shear reinforcement	Overall, the model agreed well with the experimental data concerning failure loads and corresponding displacements. However, the model proved to be slightly conservative
Ebead and Saeed (2010) [33]	(Numerical analysis) FE Program/ ADINA	To investigate unbalanced moments in interior connections with steel plates and bolts	A three-dimensional hexahedral element was employed to model the concrete, while truss elements represented the reinforcement. Steel plates were modeled with shell elements, and the interface elements were used between the plates and the concrete. Concentric loading was applied vertically to the columns, and unbalanced moments were applied through horizontal forces at the column ends	The validation demonstrated good agreement in failure loads and corresponding displacements across all slabs analyzed. The failure modes observed in the simulation were the same as in the experimental tests. The addition of steel plates with bolts enhanced the ductility of the slabs
Farzam et al. (2010) [41]	(Numerical analysis) FE Program/ ATENA	To model slabs supported on interior columns subjected to eccentric loads	The model was validated against experimental data from the literature. Eight-node 3D hexahedral elements were used for the concrete, and truss elements represented the reinforcement. The column end was modeled with a larger cross-section to apply eccentric loads. A parametric analysis was conducted, varying the concrete's compressive strength, reinforcement ratio, and yield strength	The numerical validation agreed well with the experimental data regarding failure loads, displacements, and slab rotations. The yield strength of the reinforcement significantly influenced failure. Increasing the reinforcement ratio enhanced punching capacity and reduced slab ductility

Table 1 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Soares and Vollum (2016) [62]	(Numerical analysis) FE Program/ DIANA	To evaluate the effect of continuity in slabs supported on edge columns	The numerical simulation was validated with experimental data from the literature for slabs supported on edge columns, both isolated and continuous. Eight-node hexahedral elements were used for the concrete, and truss elements were used for the reinforcement. Vertical and horizontal loads were applied to the column, and a parametric analysis was performed, varying the eccentricity	In isolated slab models, failure loads and rotations were close to experimental results, with increased eccentricity reducing the failure load. In contrast, for continuous slabs, the numerical model exhibited higher levels of rotation and lower failure loads than the experimental data
Al-Hamd et al. (2019) [42]	(Numerical analysis) FE Program/ ABAQUS	To investigate interior connections with shearheads subjected to eccentric loads	Three-dimensional eight-node elements were used for the concrete, and truss elements represented the reinforcement. The column end was modeled with a larger cross-section for load application to account for eccentricity. The model was calibrated using various meshes and constitutive material properties. Also, the effect of fire on the structure was analyzed	Eccentric loading reduced punching shear capacity in slabs with and without shear reinforcement. The presence of shearhead reinforcement increased punching resistance. The models showed good agreement with experimental data across all analyses
Setiawan et al. (2019) [43]	(Numerical analysis) FE Program/ ADAPTIC and ATENA	To investigate the impact of different parameters on the structural behavior under eccentric loading	Nonlinear 3D joint and shell elements were employed in the ADAPTIC software. The results were then compared with models developed using 3D elements in the ATENA software. Eccentric loading was modeled by applying moments at the column ends or vertical forces at lever arms. Numerical results were then compared with experimental cases available in the literature	The employed modeling approach agreed well with experimental results for both concentric and eccentric loading. Furthermore, the joint model showed good agreement with the robust 3D model

Table 1 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Živković et al. (2019) [44]	(Numerical analysis) FE Program/ ABAQUS	To study the impact of eccentric loading on slabs supported by internal columns, with or without openings near the columns	The slab was supported at the edges, and the load was applied at the central column in the numerical model. The column end was modeled with an enlarged cross-section to account for eccentricity. In cases involving openings, these were positioned either on the same or the opposite side of the eccentricity. Eight-node 3D elements were used for the concrete, and truss elements were used for the reinforcement	Punching shear strength was affected even for small eccentricities in loading. Openings positioned in the direction of eccentricity resulted in more critical conditions
Oukaili and Hussein (2020) [45]	(Numerical analysis) FE Program/ ANSYS	To investigate slabs with spherical voids supported by internal columns and subjected to eccentric loading	The model validated the experimental data provided by the authors. The slabs varied in void presence, thickness, concrete compressive strength, and load eccentricity. Tetrahedral elements were used for the concrete, and bar elements were used for the reinforcement. An enlarged column cross-section at the load application point was used to model the loading eccentricity	The numerical validation showed results closely matching experimental data. Increased slab thickness enhanced stiffness and failure load. The presence of voids and/or load eccentricity significantly reduced the failure load
Salama et al. (2020) [63] and Salama et al. (2020) [64]	(Numerical analysis) FE Program/ ANSYS	To investigate slabs supported by edge columns reinforced with glass fiber-reinforced polymer (GFRP) bars	The numerical model was validated with experimental data, and parametric analysis was performed. Hexahedral elements were used for the concrete, and bar elements were used for the reinforcement. GFRP bars were employed for flexural and shear reinforcement (closed stirrups). Vertical and horizontal loads were applied to the column ends. In the parametric analysis, the properties of the stirrups, the M/S ratio, the reinforcement ratio, and the concrete strength were varied	The model showed good agreement with experimental data regarding ultimate load and displacement. The use of stirrups increased the failure load. Higher M/S ratios resulted in lower punching shear capacities, with high eccentricities causing abrupt failures. Increased concrete strength improved punching resistance

Table 1 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Yu and Wang (2020) [66] and Wang (2020) [67]	(Numerical analysis) FE Program/ ABAQUS	To study a slab-column connection using steel tubes in slabs supported by internal or corner columns	A steel column connected to a flat slab was experimentally tested and numerically analyzed. The slab-column connection included steel plates, shear studs, and bent-up bars. The modeling utilized hexahedral and tetrahedral elements for the concrete and truss elements for the reinforcement. For corner connections, the structure was fixed at two edges, and the load was applied to the steel column	The authors concluded that the proposed connection is suitable for linking concrete slabs to steel columns. The numerical modeling yielded results consistent with experimental data regarding failure load and crack patterns
Setiawan et al. (2020) [74] and Setiawan et al. (2019) [75]	(Numerical analysis) FE Program/ ATENA and ADAPTIC	To analyze punching in slabs supported by rectangular columns and wall corners	One study validated experimental data from slabs supported by elongated rectangular columns, while the other focused on slabs supported by wall corners. Simulations were validated using nonlinear finite element analysis, employing 3D solid elements and the joint-shell punching model (combining 2D nonlinear layered-shell elements with nonlinear joint elements). Analytical formulations for load capacity prediction and parametric analyses were proposed	Punching failure was found to initiate at support corners, with peak support reactions developing before full mobilization of shear resistance around a control perimeter at 0.5d from the column face. The joint-shell punching model proved much more computationally efficient than solid element modeling
Alrousan and Alnemrawi (2022) [46] and Alrousan and Alnemrawi (2022) [47]	(Numerical analysis) FE Program/ ABAQUS	To verify slabs with openings supported by internal columns and subjected to eccentric loads	The study was validated against experimental data from the literature, and a parametric analysis was performed. Different eccentricities, opening positions, concrete compressive strength, and reinforcement materials were investigated. Eight-node 3D elements were used for the concrete, and truss elements were used for the reinforcement	Increased eccentricity reduced elastic and plastic stiffness, energy absorption, ductility, and punching shear capacity. Higher concrete compressive strength resulted in greater failure loads

Table 1 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Jia and Chiang (2022) [34]	(Numerical analysis) FE Program/ ABAQUS	To investigate the impact of shear reinforcement on slabs supported by internal columns with unbalanced moments	Experimental data from the literature were used to validate the model. Slabs with different M/S ratios and shear reinforcement configurations were investigated. Hexahedral elements were used for the concrete, and truss elements were used for the reinforcement. Unbalanced loads or lateral forces at the column ends were used to apply eccentric conditions	The numerical model agreed well with the experimental data. The number of shear studs significantly influenced punching resistance. The authors proposed a new formulation for the coefficient of bending moment transferred by eccentric shear force
Xue et al. (2022) [71]	(Numerical analysis) FE Program/ ABAQUS	To study slabs supported by L-shaped columns with capitals or inverted T-shaped steel for shear reinforcement	The study involved experimental and numerical analyses. 3D solid elements were used for the concrete, truss elements for steel reinforcement, and plate elements for the T-shaped steel. The article investigated the impact of increased slab thickness, the use of capitals, and inverted T-shaped steel for shear reinforcement. After calibration and validation, the model was used for parametric analysis	The numerical validation showed excellent agreement with experimental data. For the same column cross-sectional area, L-shaped columns exhibited greater punching shear capacity than square columns due to a larger critical perimeter. Column shape changes did not affect the failure mode
Milligan et al. (2022) [73]	(Numerical analysis) FE Program/ ABAQUS	To investigate the punching and unidirectional shear in slabs supported by walls	The study validated the simulation with experimental data from the literature for slabs supported by rectangular columns. Subsequently, a parametric study was conducted on punching in wall-supported slabs, varying the support dimensions and type of loading. Connections subjected to a combined concentric vertical load and an unbalanced moment or concentric vertical load only were examined	The numerical validation produced good approximations. For slab-wall connections subjected to concentric vertical loading, one-way shear behavior significantly influences the overall response, with the dominant direction of one-way shear predicted to be parallel to the longer side of the wall

Table 1 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Zheng et al. (2023) [35]	(Numerical analysis) FE Program/ABAQUS	To study the behavior of square and rectangular slabs supported by square or rectangular internal columns	The study addressed eccentric loading caused by the column shape and asymmetrical load application. Truss elements were used to model the reinforcement, while hexahedral elements modeled the concrete. Numerical models were validated and calibrated using experimental data from the literature, followed by a parametric analysis	The authors provided recommendations on parameter values for material modeling. The slab rectangularity index significantly affected the ultimate load, failure type, and cracking patterns. The numerical models showed good agreement with experimental data
Wang et al. (2023) [48]	(Numerical analysis) FE Program/LS-DYNA	To investigate the slabs' punching and post-punching shear behavior under eccentric loading with column capitals	The authors carried out experimental and numerical studies on slabs with capitals of different thicknesses. The column ends were modeled with an enlarged cross-section to apply eccentric loads. Hexahedral 3D elements were used for the concrete, and beam elements for the reinforcement	Excellent agreement was observed between experimental and numerical results. Increased eccentricity significantly reduced punching resistance
Matseevich (2023) [65]	(Numerical analysis) FE Program/LIRACAD	To investigate a floor structure composed of multiple slabs and columns	Concrete was modeled using shell elements without reinforcement discretization, while columns were modeled as high-stiffness rod elements. Contact elements were also utilized. The load was applied as a distributed load over the entire floor. The results were compared with analytical calculations	The study revealed that concrete properties, column shape, and slab thickness significantly impact punching resistance
Milligan and Polak (2023) [70], Milligan and Polak (2021) [72] and Milligan et al. (2020) [77]	(Numerical analysis) FE Program/ABAQUS	To study slabs supported by rectangular, L-shaped, T-shaped, and cruciform column sections	One study validated experimental data from the literature and assessed the impact of column rectangularity. Another study used validated models to analyze how column shape, dimensions, and openings influence structural behavior. Hexahedral 3D elements modeled the concrete, and truss elements represented the reinforcement	The column aspect ratio significantly influences punching in slabs supported by rectangular columns. Stress distribution around L-shaped, T-shaped, and cruciform columns is non-uniform

Table 1 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Sanabria Díaz et al. (2024) [49]	(Numerical analysis) FE Program/ DIANA	To study slab-edge column connections subjected to outward and inward eccentricity	Numerical validation with experimental data from the literature was performed. 3D solid elements were used for modeling, and concentrated loads were applied at various points on the structure. Vertical springs were employed at the supports	Numerical validation closely matched experimental data. Significant differences were found in crack propagation, reinforcement stresses, and compression strut formation for slabs subjected to inward and outward eccentricity
Mendes et al. (2024) [50]	(Numerical analysis) FE Program/ ATENA	To analyze slabs supported by internal columns under concentric and eccentric loading	Numerical modeling was validated using literature data on slabs with unbalanced loads in different proportions. Parametric analysis and comparison with code predictions followed. Solid 3D elements modeled the concrete, while unidirectional elements represented the reinforcement. Load effects were considered through imposed displacement disproportion	Good agreement was found between experimental and numerical results for load capacity and crack maps under concentric and eccentric loading. Increased unbalanced moments reduced punching resistance
Schimanowski et al. (2024) [36]	(Numerical analysis) FE Program/ ANSYS	To investigate slabs with spherical voids supported by internal columns under concentric and eccentric loading	Tetrahedral elements were used for the concrete, and bar elements were used for the embedded reinforcement. Validation was performed using experimental data. Loading was analyzed by imposing displacement disproportion, forming asymmetric punching. The effects of symmetric and asymmetric loading on slab resistance were compared	Good agreement with the experimental data was observed for rupture load, displacements, and deformations. Increased load imbalance reduced punching resistance, and spherical voids significantly lowered load capacity, particularly under asymmetric punching
Zheng et al. (2025) [76]	(Numerical analysis) FE Program/ ABAQUS	To investigate punching shear in rectangular slabs supported by interior columns	The numerical model was validated against experimental data over a range of parameters. Mesh and parameters of the constitutive model were investigated. A parametric analysis verified punching shear in rectangular slabs with the same reinforcement ratio in both directions. The numerical analysis employed 3D elements for the concrete and truss elements for the reinforcement	The FEM results were consistent with experimental findings. The study revealed that the slab aspect ratio significantly influenced load-rotation curves, tensile reinforcement stress, cracking patterns, shear forces in both directions, and their symmetry

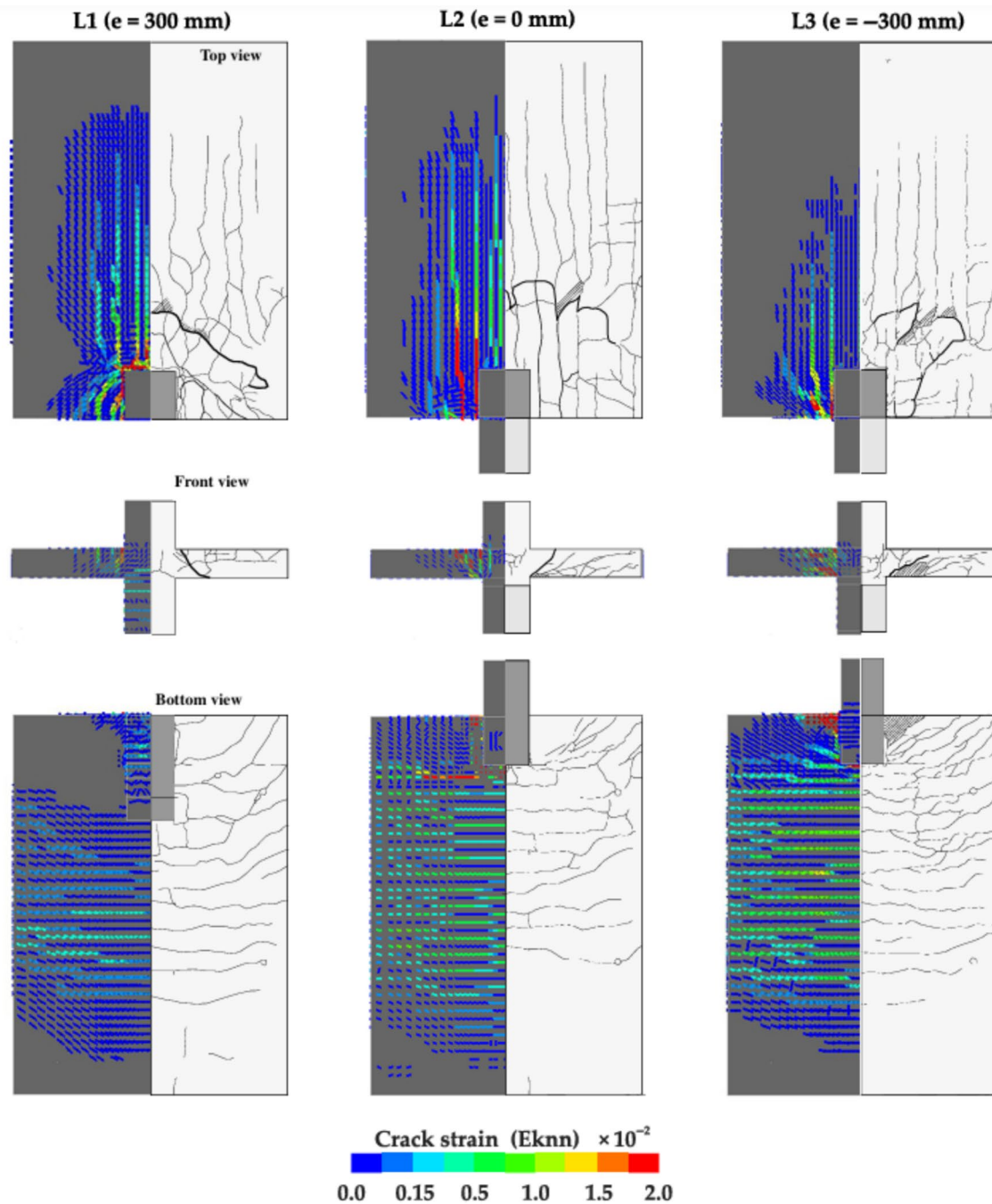


Fig. 4 The crack pattern of edge connections subjected to inward ($e = 300$ mm) and outward ($e = -300$ mm) eccentricities [49]

failure, the behavior under lateral loads still requires further clarification. Genikomsou [80] highlights real cases of punching failure due to earthquakes and summarizes the experimental data available in the literature, noting that this topic has been under study for approximately 60 years.

Almeida et al. [81] indicate that the critical condition for punching occurs when lateral and gravitational loads are combined. Coronelli et al. [82] discuss that various

experimental studies have employed different tests to simulate slabs under lateral loads. Among these, the most common involve isolated slab-column connections [81, 83–96], while a smaller number of studies have examined entire floors or multi-story structures at full scale [97, 98]. The limited number of full-scale tests is attributed to challenges in assembling test systems, constructing large-scale structures, and associated costs. Regarding loading, most studies

address seismic dynamic loads through cyclic lateral loading [81, 83–97, 99], while fewer studies have employed shake table tests [100]. Additionally, studies on isolated slabs have examined punching behavior in slabs supported by edge columns [32, 88–90], corner columns [32], or interior columns [81, 83, 85–87, 90–95, 101].

The behavior of structures under lateral loading has also been evaluated with shear reinforcement, such as post-installed bolts [81, 97], shearheads [102, 103], studs [84, 93], steel sheets [96], or stirrups [32, 85, 86]; in prestressed slabs [95]; in slabs with and without openings [104]; and in slabs made with advanced materials such as High-Performance Fiber Reinforced Concrete (HPFRC) [83], High-Strength Concrete (HSC) [87], Steel Fiber Reinforced Concrete (SFRC) [92], or Fiber Reinforced Polymer (FRP) bars [85, 88–90, 105]. Amiri and Esfahani [38] point out that solutions such as column capitals, drop panels, or ultra-high-performance concrete (UHPC) increase the shear capacity of slab-column connections but do not significantly improve ductility, a key factor for slabs subjected to cyclic loading. Overall, studies have identified critical factors influencing load capacity, drift ratio, and structural degradation, including flexural reinforcement ratio and shear reinforcement [84–86, 89, 90, 94, 106]. Another significant factor affecting structural behavior under seismic loads is the gravitational load applied to the slabs [88].

From a numerical perspective, modeling these structures often requires significant computational resources due to the cyclic loading and unloading behavior, especially when using refined methods like 3D solid finite elements. Consequently, many studies have adopted simplified numerical approaches, such as frame models [100, 106–109], beam-column frames [32, 110], or less sophisticated elements like thick-shell models [79]. While these approaches involve simplifications, they can still provide adequate modeling results.

Moreover, modeling cyclic loading poses challenges, as load cycles can be difficult to apply, and the structure's hysteretic behavior is tied to specific constitutive models that capture material degradation. Some studies have validated experimental cyclic data but used monotonic loading for numerical modeling [99, 111]. Generally, for numerical models evaluating isolated slab-column connections, cyclic loading is applied through displacement increments at one or both ends of the column.

It is worth noting that most cases investigate increasing cyclic loading, starting with small displacements and progressing to failure. Only a few studies investigated different loading conditions: Kang et al. [100] have evaluated structures using shake tables; Tian et al. [110] examined the effects of vertical ground motions, simulating actual conditions; Coronelli et al. [108] investigated structural behavior using natural signals compatible with serviceability and ultimate limit state response spectra; and Najafgholipour et al.

[109] studied the seismic behavior of structures with flat slab-column frames and structural walls as the seismic load-bearing elements through nonlinear static and time history analyses. Overall, the primary aim of published studies has been to validate experimental data, a complex study in itself. Few investigations have sought to analyze various parameters in numerical modeling, a common practice in other punching shear research areas.

As a result, most conclusions regarding this topic focus on whether a specific numerical modeling approach can effectively evaluate experimental data, often leading to limited study findings. Conversely, using 3D modeling has yielded consistent conclusions in recent research, particularly concerning parameters such as flexural reinforcement ratio, slab thickness, and column shape. Table 2 provides a detailed overview of studies on cyclic lateral loading and their relevant conclusions.

The phenomenon of punching due to lateral cyclic loading has emerged as a prominent topic in recent years, with an increasing number of studies addressing it. It is also worth noting that various modeling techniques have been employed, highlighting the complexity of analyzing such problems. In this context, the study by Coronelli et al. [108] is particularly noteworthy, as it explores different approaches to modeling the same problem. The study is based on a blind competition, wherein three teams submitted distinct numerical models to reproduce the behavior of a two-story structure tested experimentally. The experimental tests involved distinct load scenarios: one simulating seismic loading with response spectra for serviceability and ultimate limit states (SLS and ULS) and another involving lateral cyclic loading.

Two groups employed slab-frame modeling approaches. Group 1 used OpenSees and developed a three-dimensional nonlinear fiber-based finite element model (distributed plasticity), as illustrated in Fig. 5. Columns were modeled with confined concrete for the core and unconfined concrete for the cover. Multilayer shell elements were used to model the slabs. The software's material models for concrete and steel were employed, with hysteretic parameters implemented for the steel based on theoretical formulations. Regarding punching, the OpenSees model could not capture the out-of-plane nonlinearities associated with the punching shear response of flat slabs. Therefore, the first punching failure was detected using analytical formulations that accounted for the slab's rotation and the associated displacements at the top of the structural system.

Group 2 used the Ideea platform, combining nonlinear shell elements, elastic and nonlinear beam-column elements, and linear spring elements, as shown in Fig. 6. The reinforcement was distributed within the shell elements following the specified reinforcement layout. Nonlinear beam-column elements consisted of displacement-based elements with fiber discretization of the cross-section.

Table 2 Review of punching shear in slabs under cyclic lateral loading and seismic actions

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Kang et al. (2009) [100]	(Numerical analysis) Planar Frame Model/ OpenSees	To verify whether the proposed modeling approach adequately describes the experimental behavior of slabs	Isolated slab-column connections and structures subjected to shake table tests were investigated. Columns were modeled as fiber elements, and slab-column connections were modeled using spring elements. A limit state model was implemented in OpenSees to account for punching shear failure	The proposed model replicated experimental data for slabs subjected to shake table tests or lateral cyclic loading in isolated slab-column connections. The modeling approach is comprehensive for flat plate systems, particularly under dynamic loads
Coronelli (2010) [32]	(Numerical analysis) Beam-column Frames/ OpenSees	To employ a Beam-Column Frame model to investigate punching shear under various scenarios	The simulation employed a grid model, an intermediate approach between 3D finite elements and 2D frame models. The slab was subdivided into a grid of beam finite elements, and frame elements were also utilized. Unbalanced and cyclic loads were analyzed for slabs supported on corner, edge, or interior columns, as well as for concentric loading. The use of shear reinforcement was also studied	For monotonic loading, punching resistance was reasonably approximated. Under cyclic loading and varying levels of gravitational loading, flexural capacity deterioration was only partially captured, and ultimate drift capacity was underestimated. For concentric loading cases, the model achieved a good approximation
Eder et al. (2011) [102] and Eder et al. (2012) [103]	(Numerical analysis) FE Program/ DIANA	To analyze the use of shearheads in punching shear failure of reinforced concrete slab-to-steel column connections under lateral cyclic loading	The study employed experimental and numerical analysis. Lateral cyclic loading was modeled by applying prescribed displacements at the column's end. Respectively, concrete, shearheads, and reinforcement were modeled with hexahedral, shell, and truss elements. The modeling approach could not capture the degradation of concrete strength under cyclic loading	The hysteretic behavior of the structure agreed well with experimental data. Shearheads increased energy dissipation and enhanced the lateral resistance of the connection due to cyclic strain hardening of the shear arms

Table 2 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Genikomsou and Polak (2015) [99]	(Numerical analysis) FE Program/ABAQUS	To validate a numerical model for slabs subjected to punching shear	To validate a numerical ABAQUS model for slabs subjected to punching shear. The modeling aimed to validate experimental data from the literature. Cyclic loading was modeled monotonically, excluding the structure's hysteretic behavior. Eight-node hexahedral elements were used for concrete, and truss elements were used for reinforcement	The model approximated the ultimate load well but could not represent stiffness degradation from the hysteretic response. Modeling concrete was identified as the most challenging aspect of finite element simulations
Setiawan et al. (2019) [101]	(Numerical analysis) FE Program/ATENA	To study the degradation of slab-column connections under lateral cyclic loading	The modeling validated experimental data from the literature. Cyclic loading was applied through prescribed displacements on steel levers transmitting the load. Respectively, hexahedral, tetrahedral, and truss elements were used for concrete, levers, and reinforcement. Monotonic and cyclic loadings were analyzed, incorporating the Bauschinger effect in the steel's constitutive model	The modeling closely matched experimental results regarding cyclic degradation, slab sectoral rotations, crack patterns, and reinforcement deformations. Cyclic degradation was associated with plastic strain accumulation in the flexural reinforcement near the column
Tian et al. (2019) [110]	(Numerical analysis) Beam-column Frames/OpenSees	To analyze the effect of vertical ground motion on flat slab structures' behavior	The model was validated using experimental data from the literature for concentric and lateral cyclic loading. A 3D grid model discretized slabs into interconnected beams. Perimeter frame beams and columns were modeled with nonlinear elements, and beam-column connections were modeled as elastic. The model subsequently investigated the effect of ground motion on structural behavior	Vertical ground excitations near the source significantly increased rotational deformation demands in slabs, causing early punching failures and reducing the structure's lateral drift capacity. Vertical acceleration induced punching failures, which were not observed under horizontal ground motion alone

Table 2 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Isufi et al. (2020) [106]	(Numerical analysis) Frame Model/OpenSees	To assess the impact of shear reinforcement on slab-column connections under seismic loading	The frame model was validated against experimental data from the literature. Columns were modeled with fiber elements, while slab-column connections used spring and unidimensional finite elements. Three-story structures were studied under seismic loading using pushover analyses and fragility curves	In the absence of shear reinforcement, punching failures occurred at remarkably low spectral accelerations. Enhancing the ultimate capacity is achievable if the design of punching reinforcement enables the slab-column connection to achieve its full potential to transfer the unbalanced moments
El-Genidy and El-Salakawy (2021) [105]	(Numerical analysis) FE Program/ATENA	To investigate slabs supported on edge columns under lateral cyclic loading	The study validated the numerical analysis with experimental data and included a parametric analysis. Respectively, steel and FRP bars were used in the columns and the slabs. Concrete was modeled with tetrahedral elements, while truss elements represented the reinforcement. A constitutive model for steel considering cyclic behavior was adopted, with prescribed displacements applied to the column's end	Numerical validation yielded excellent agreement with experimental data. The parametric study indicated that increased slab thickness or flexural reinforcement enhanced load capacity and stiffness. Drift ratios were higher with increased flexural reinforcement ratios or reduced slab thickness
Najafgholipour and Sarhadi (2022) [111]	(Numerical analysis) FE Program/ABAQUS	To investigate various parameters in slabs supported by internal columns and subjected to lateral cyclic loading	The study validated the numerical simulation with experimental data and provided a parametric analysis. The cyclic load was simplified in the modeling process as monotonic, excluding the hysteretic structure's behavior, with prescribed displacements applied at the column ends. Eight-node 3D elements were used for the concrete, and truss elements were used for the reinforcement	The validation agreed well regarding the ultimate load; however, the corresponding drift was underestimated. In the parametric analysis, higher applied vertical loads reduced the drift capacity. There are significant variations in ultimate displacement with changes in slab thickness, flexural reinforcement ratio, and column size
El-Mandouh and El-Hassan (2022) [104]	(Numerical analysis) FE Program/ABAQUS	To investigate ordinary and lightweight concrete slabs supported by edge columns with openings and subjected to lateral cyclic loading	The numerical study was validated with experimental data provided by the authors. Eight-node 3D elements were used for the concrete, and surface elements were used for the slab reinforcement. In the modeling, the bottom end of the column was fixed, and reverse loading was applied at the top end	The validation results agreed well with experimental data for most cases. However, the modeled structure's hysteretic response was stiffer than the experimental results

Table 2 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Panahi and Genikomsou (2023) [107]	(Numerical analysis) Frame Model/ OpenSees	Machine Learning tools were utilized to develop a material model for cyclic loading analyses	A constitutive material model capable of describing the hysteretic behavior of structures was created with Machine Learning tools. A neural network was trained and integrated into the OpenSees source code to evaluate the seismic performance of connections. The model was validated by simulating slab-column connections in 2D frame models	The proposed material model significantly improved accuracy in capturing the hysteretic response of slabs compared to other theoretical models. A parametric analysis revealed that varying the flexural reinforcement ratio led to punching-shear-flexure failures in slabs with low reinforcement ratios
Coronelli et al. (2023) [108]	(Numerical analysis) ABAQUS, OpenSees, Ideea	Results from three numerical models submitted to a blind competition were presented. Lateral cyclic loads and seismic events were simulated in a full-scale structure	Each participating group employed a software package (ABAQUS, OpenSees, or Ideea). OpenSees and Ideea were used to model the structure as slab-frame systems, while ABAQUS utilized 3D hexahedral elements. Multiple experimental results were presented and compared with the numerical models	The 3D model reproduced the experimental data with higher accuracy but required a meticulous selection of material parameters and a detailed geometric description. In contrast, the slab-frame models were less accurate but more conservative and required lower computational costs
Abolhelm and Hrynyk (2024) [79]	(Numerical analysis) FE Program/ VecTor4	To simulate punching in slabs under various loading conditions with thick-shell elements	The study validated slabs subjected to concentric and lateral cyclic loading with experimental data from the extant literature. The modeling utilized nine-node layered thick-shell elements, with the reinforcement modeled by altering the properties of the corresponding layer	Thick-shell elements can effectively evaluate punching without model calibration or material parameter refinement, but elastic stiffness was overestimated. The modeling results closely matched experimental data for most slabs
Najafgholipour et al. (2024) [109]	(Numerical analysis) Frame Model/ OpenSees	To simulate flat slabs under seismic loading and evaluate punching shear occurrence at the connections	The study was validated with experimental data from the literature. The seismic behavior of reinforced concrete structures, where shear walls are the main lateral load-resisting elements, was investigated through nonlinear static and time-history analyses. A multiple-vertical-line-element model was utilized to model the global response of the walls	Almost all the connections that experienced punching shear in the pushover or time-history analysis were exterior. According to time-history analyses, incorporating slab-column frames into the modeling of structures significantly altered the predicted lateral displacements

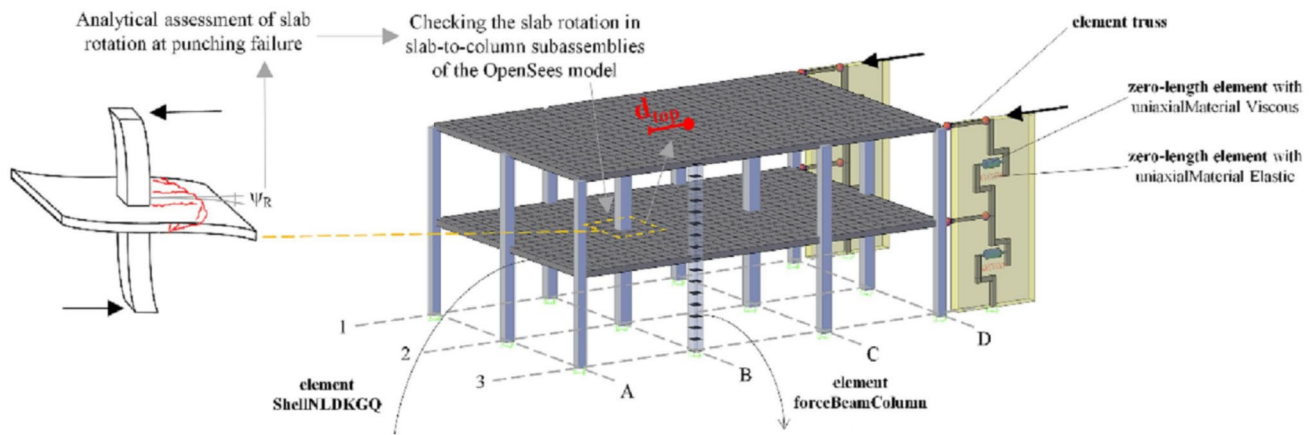
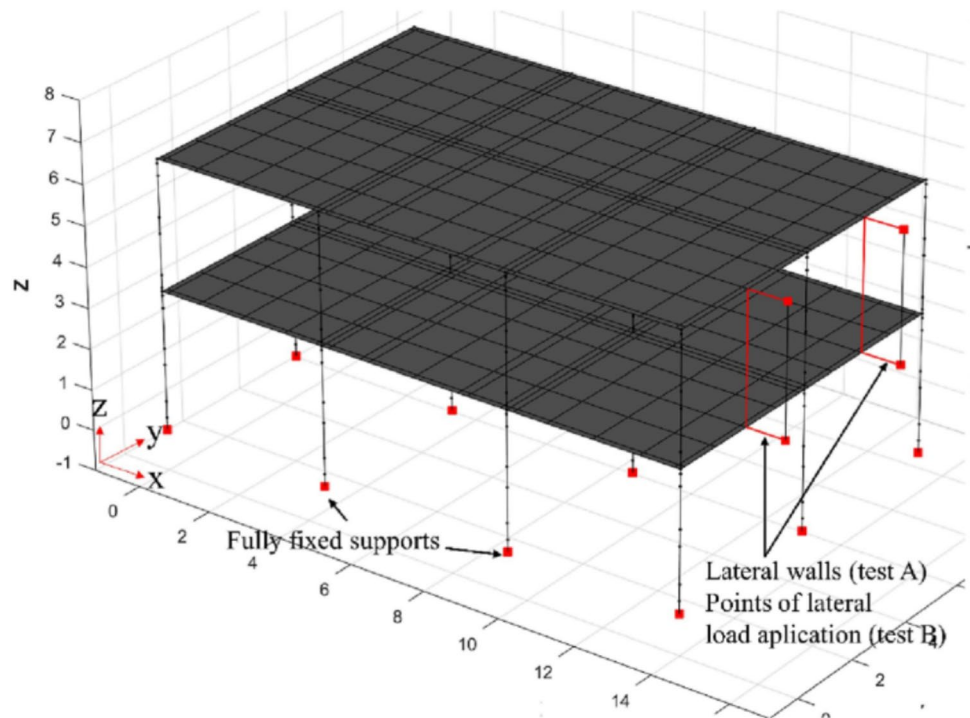


Fig. 5 FEM model of Group 1 and details of the adopted modeling strategy [108]

Fig. 6 FEM model of Group 2 [108]



Uniaxial material models for steel and concrete were assigned to the corresponding fibers, incorporating the hysteretic response of each material. External walls were modeled with linear spring elements.

Group 3 developed a detailed three-dimensional finite element model using ABAQUS, as shown in Fig. 7. The structure was modeled with its real geometry and size, with each reinforcing bar and stirrup explicitly represented. Eight-node 3D hexahedral elements were used for concrete, while truss elements modeled the reinforcement. The constitutive model for concrete provided by the software was calibrated with data from the literature, and bilinear behavior was assumed

for the steel. External walls were modeled with linear spring elements.

The models were analyzed and compared based on various factors, including the hysteretic response of the structure subjected to simulated seismic loading and lateral cyclic loading. In Groups 1 and 2, a slab and frame model was employed, yielding results that demonstrated good agreement with experimental data while offering greater simplicity in modeling. However, the analytical method known as Critical Shear Crack Theory (CSCT) [112] had to be utilized to accurately predict punching shear, which appears to be a typical requirement when

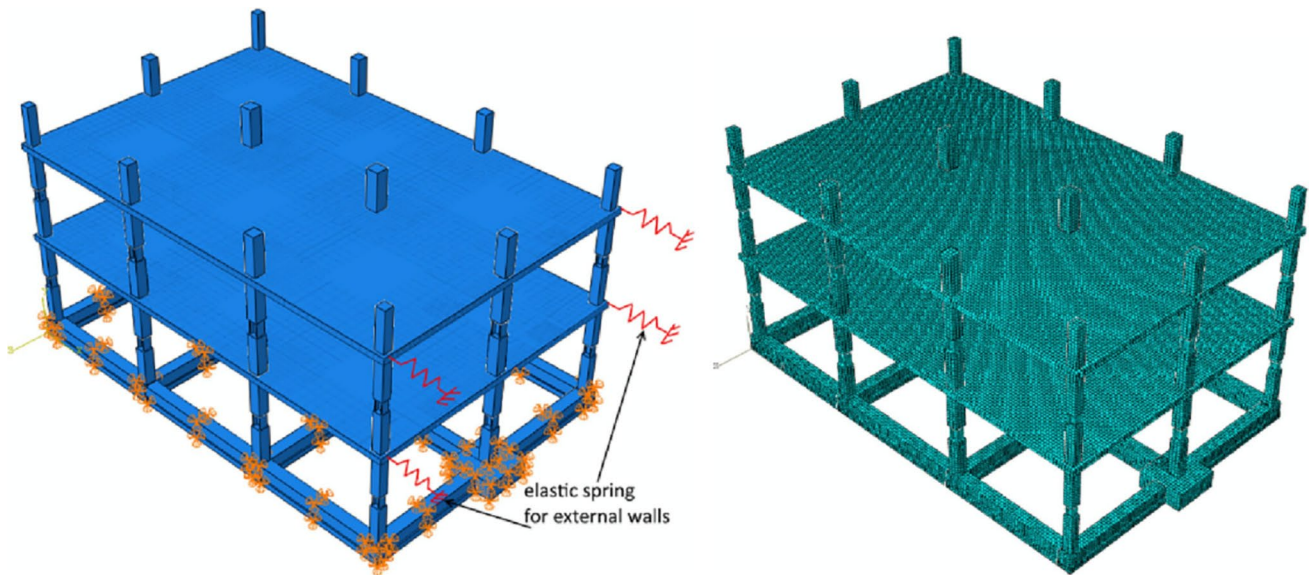


Fig. 7 Geometry, boundary conditions, and finite element mesh of the numerical model of Group 3 [108]

using currently available design analysis software [108]. Coronelli et al. [108] noted that the models could correctly estimate the drift level at which the first punching occurred; however, this was not associated with the correct location. Thus, while the prediction method is sufficiently conservative for design purposes, further research is necessary to improve capacity prediction and failure location accuracy.

Conversely, the 3D finite element model developed by Group 3 provided results comparable to those obtained by Groups 1 and 2 regarding code-based representations of the capacity curve and local results. This model enables a more in-depth investigation of local results. However, it is subject to a detailed description of geometry and reinforcement placement, requires a careful selection of material models, and incurs higher computational costs for post-processing results. Consequently, this method seems more attractive for research purposes than for routine design applications, particularly for large-scale analyses [108].

It is evident that numerical modeling of lateral cyclic and seismic loading is a significant topic due to its inherent complexity. Although numerous studies have explored modeling techniques for these structures, further research is required to deepen our understanding of how various parameters affect their behavior, particularly concerning punching shear. Additionally, the effects of strengthening techniques or alternative materials on punching shear resistance warrant further exploration. Finally, the study of seismic loads on large-scale structures and their impact on punching shear remains insufficiently addressed in existing literature.

2.4 Punching Shear in Slabs with Openings

In reinforced concrete structures, it is usual for buildings constructed with flat slabs to feature openings to accommodate electrical, telephone, gas, air conditioning, and plumbing installations [113]. However, the punching shear resistance can be significantly reduced when such openings are positioned near columns, as shown in Fig. 8. This is because the openings interfere with the formation of the punching shear cone [114]. Consequently, numerous experimental and numerical studies have sought to understand the influence

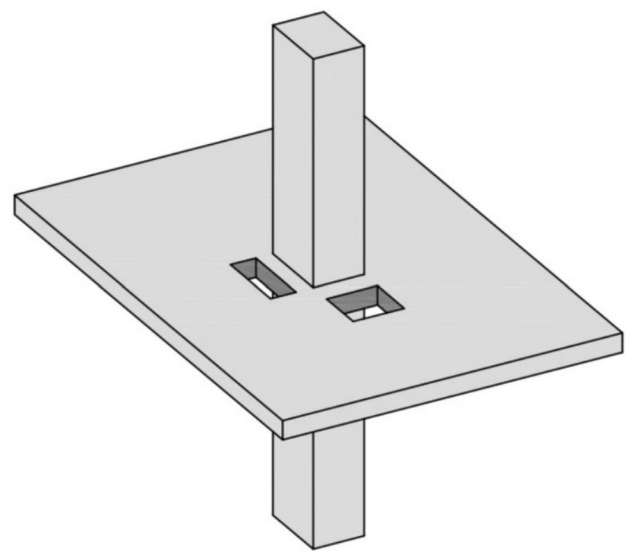


Fig. 8 Presence of openings near the column in flat slabs

of openings on punching shear failure, evaluating various configurations.

Most studies have focused on square and rectangular openings [28, 114–121], with only a limited number addressing circular openings [113, 122–124]. Additionally, several studies have examined the impact of the distance of openings from the column, ranging from adjacent to the slab-column connection to positions further away [44, 113, 114, 120, 125]. While some studies have focused on single openings [113–115, 126], others have analyzed configurations with two, three, or more openings [28, 117–120, 123, 124].

Research on slabs with shear reinforcement in the presence of openings remains limited [59, 117, 123], as do studies examining lateral loads [104, 125], unbalanced loads [44, 46, 114–116], or elevated temperatures [114]. Furthermore, only one study has investigated the effect of horizontal pipes crossing slabs under punching shear loads [127]. Some studies have explored reinforcement methods, such as FRP overlays or textile-reinforced mortar (TRM), to restore the load-carrying capacity of slabs with openings, enabling them to perform similarly to solid slabs of the same dimensions [117, 118, 126, 128].

Overall, studies have concluded that openings reduce load-carrying capacity, energy dissipation, initial stiffness, and first-crack load, significantly changing the crack patterns [28, 70, 114, 117, 118, 124, 129]. The load capacity decreases as the number and size of openings increase [113]. Critically, openings adjacent to the column have the most significant impact [117], whereas openings beyond a certain distance have an almost negligible effect on slab behavior [113]. For unbalanced loads, the reduction in load-carrying capacity is even more pronounced when the opening is positioned in the eccentric load direction [47, 115].

It is worth noting that experimental studies on slab openings and punching shear began in the last century [130]. However, this topic has only been explored more extensively in recent years through numerical simulations [44, 46, 47, 59, 70, 104, 116, 121, 122, 125, 128, 129, 131]. One primary objective of numerical studies has been to determine the critical distance beyond which openings no longer influence punching shear resistance. This distance is often expressed as a factor of the slab's effective depth [44, 121, 122, 131]. Numerical simulations typically employ commercial software and 3D hexahedral elements, enabling the visualization of crack patterns associated with punching shear failure. Modeling openings is generally a straightforward process involving the removal of a reinforced concrete volume from the slab, typically performed before meshing the finite element model.

Numerical studies often begin with validation against experimental results and are sometimes followed by parametric analyses that vary the number, size, and distance

of openings from the column. Most studies focus on slab-column connections without shear reinforcement, with only Polak [59] numerically analyzing slabs with transverse reinforcement. There is considerable variability regarding boundary conditions and loading. Studies have considered concentric, eccentric, and lateral loads applied to corner, edge, or interior columns. These details, the key findings, and modeling strategies from the evaluated studies are summarized in Table 3.

In this context, the studies highlight that openings near columns remain a current and relevant topic, with consistent conclusions emerging across different investigations. Common findings indicate that the openings' position, quantity, and dimensions influence the results. Another prevalent conclusion is that openings reduce punching shear capacity and stiffness, particularly when these openings are positioned adjacent to the column. There is also general agreement that increasing the distance between the openings and the columns enhances punching shear resistance. However, the precise distance at which openings cease to affect slab behavior remains a topic of debate. For instance, Živković et al. [44] suggest this distance should be $6d$, where d represents the effective depth of the slab, while other studies, such as Vieira et al. [122] and Genikomsou and Polak [121], propose a distance of $4d$. It is worth noting that design codes also present differing values for this parameter.

In the studies conducted by Genikomsou and Polak [121] and Balomenos et al. [131], which investigated edge and interior connections subjected to vertical or lateral loading, it was observed that the distance of the opening significantly influences crack distribution, as illustrated in Fig. 9. In this scenario, cracks in slabs with openings adjacent to the column propagate from the column and around the opening. Conversely, when openings are not adjacent to the column, secondary cracks form and begin to propagate from the corners of the openings, extending further away from the columns.

The same authors also conducted a parametric analysis by varying the dimensions of openings in slabs supported by interior columns under vertical loading. Genikomsou and Polak [121] demonstrated that the punching shear capacity of slabs with openings located $4d$ from the columns is equivalent to that of solid slabs. For smaller openings, between 50 and 70 mm, this critical distance decreases to $2d$. Thus, the size of the opening plays a significant role in reducing punching shear capacity, especially for large openings adjacent to the column.

From this perspective, it is noteworthy that the studies conducted on this topic generally employ robust numerical modeling approaches, utilizing the FEM appropriately and yielding results that predominantly converge with experimental data. The modeling of openings is typically straightforward, involving the removal of volumes before mesh

Table 3 Review of punching shear in slabs with openings

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Polak (2005) [59]	(Numerical analysis) FE Program	To investigate the impact of openings at the slab-column interface for corner and edge connections	Square openings at the slab-column interface were analyzed. Numerical data were compared with the authors' experimental results. Numerical modeling employed nine-node shell elements	The numerical model agreed well with experimental data regarding cracking and ultimate loads. The presence of openings reduced the punching shear strength
Abdulrahman et al. (2017) [126]	(Numerical analysis) FE Program/ABAQUS	To examine the impact of an opening near the column on the shear behavior of corner connections with FRP laminate	A square opening at the slab-column interface was modeled. The numerical simulation used eight-node 3D elements for the concrete and truss elements for the reinforcement. The opening was strengthened with carbon fiber-reinforced polymer (CFRP) laminates, and the results were compared with the authors' experimental data	The numerical model agreed well with experimental results for reinforcement strains and ultimate load, although the numerical curves were stiffer. The opening reduced punching shear strength, but the reinforcement mitigated the difference compared to a slab without openings
Genikomsou and Polak (2017) [121] and Balomenos et al. (2018) [131]	(Numerical analysis) FE Program/ABAQUS	To investigate the influence of column distance and opening size on the behavior of edge and interior connections	The numerical modeling was validated against experimental data from the literature, followed by a parametric analysis. Square openings, including those smaller, equal to, or larger than the column, were investigated. The numerical modeling employed eight-node 3D elements for the concrete and truss elements for the reinforcement. Crack propagation was also evaluated	Validation showed good agreement with experimental results. The parametric analysis concluded that openings positioned 4d from the column did not affect punching shear capacity. For openings near the column, failure cracks are initiated at the slab-column interface and around the opening
Yousef et al. (2019) [125]	(Numerical analysis) FE Program/ABAQUS	To evaluate the number, distance from the column, and size of openings on the punching shear of interior connections under cyclic lateral loading	The numerical model was validated against the authors' experimental data for ordinary and high-performance concrete slabs. Square openings smaller than the column were investigated. The modeling used eight-node 3D elements for the concrete and bar elements for the reinforcement	The position and size of openings significantly influenced the slab's punching shear resistance and the columns' top displacements. The numerical results closely matched the experimental data

Table 3 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Živković et al. (2019) [44]	(Numerical analysis) FE Program/ ABAQUS	To investigate the effect of opening distance on interior connections under symmetric and eccentric loading	The opening was square and equal in size to the column. For eccentric loading, the opening was positioned either on the same or the opposite side of the eccentricity. Respectively, concrete and reinforcement were modeled with solid eight-node 3D and truss elements	Openings influenced punching shear strength up to a distance of 6d from the column. For eccentric loading, openings on the same side as the eccentricity reduced punching shear strength by approximately 24% compared to those on the opposite side
Mostofinejad (2020) [116]	(Numerical analysis) FE Program/ ANSYS	To study the impact of column distance and opening size on edge or interior connections under unbalanced loads	The numerical model was validated against experimental data from the literature, and square openings equal to or smaller than the column were investigated. Hexahedral and line elements were used for the concrete and the reinforcement, respectively. Shear stress distribution across the slab was also analyzed	Maximum shear stresses occurred at a distance of $d/2$ from the column, regardless of openings. Placing an opening in this critical region increased stress concentration, significantly impacting larger openings. Internal columns experienced higher shear stress increases near openings compared to edge columns
Oukaili and Merie (2021) [128]	(Numerical analysis) FE Program/ ANSYS	To evaluate the influence of openings in BubbleDeck slabs (with spherical voids), both with and without CFRP sheets	The numerical modeling validated the experimental data provided by the authors. The analysis considered square openings of varying dimensions positioned at different distances from the column. Additionally, the study examined the effect of opening positions in slabs with different spherical void distributions. Tetrahedral elements were used for the concrete, while line elements represented the reinforcements	The numerical results closely aligned with experimental data for all cases. The pattern of void distribution had a more significant impact than the position and size of the openings. Slabs with openings exhibited more significant deflection under service loads
Akkaya et al. (2022) [129]	(Numerical analysis) FE Program/ ABAQUS	To investigate the structural behavior of slabs with large openings near columns	The numerical modeling validated the experimental data provided by the authors. The openings analyzed were larger than the column dimensions and were positioned in various configurations. The modeling employed 8-node 3D solid elements for the concrete and beam elements for the reinforcements	The numerical results were consistent with experimental observations. Large openings significantly reduced punching shear capacity, altered stiffness and impacted energy dissipation. The most critical case involved two large openings adjacent to and parallel to the column

Table 3 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
El-Mandouh and El-Hassan (2022) [104]	(Numerical analysis) FE Program/ ABAQUS	To investigate openings in edge connections with ordinary and lightweight concrete under cyclic lateral loading	The openings investigated were square, with dimensions smaller than the column, and positioned at various distances from the column. The modeling utilized 8-node 3D solid elements for the concrete and surface elements for the slab reinforcements. The numerical analysis aimed to validate experimental data from the authors	The validation of most models showed close agreement with experimental results regarding ultimate load. However, displacements at the top of the column just before failure were smaller than the experimental values
Alrousan and Alnemrawi (2022) [46] Alrousan and Alnemrawi (2022) [47]	(Numerical analysis) FE Program/ ABAQUS	To analyze openings in slabs reinforced with various materials, supported by interior columns, under eccentric loading	The validation used experimental data from the literature and a parametric study was performed. Openings were positioned either in the same direction or opposite the eccentricity, adjacent to the column, and examined under different eccentricities and reinforcement materials (steel, carbon fiber-reinforced polymer, or basalt fiber-reinforced polymer). The modeling employed 8-node 3D solid elements for the concrete and truss elements for the reinforcements	The numerical results closely matched the experimental data. Openings reduced plastic stiffness, energy absorption, and ductility capacity. For eccentrically loaded slabs, the most critical scenario occurred when the opening was positioned on the more heavily loaded side. An increase in concrete compressive strength enhances load capacity
Vieira et al. (2023) [122]	(Numerical analysis) FE Program/ DIANA	To investigate openings in slabs supported by interior columns	The research was validated with experimental data from the literature, and a parametric analysis was performed. Circular openings near the column were verified, varying in number and position. Numerical modeling employed 3D hexahedral elements for the concrete and individual isoparametric elements for the reinforcements	The model accurately represented experimental data from the literature. Openings positioned 4d from the column faces did not influence load capacity. When one or two openings were present, this distance was reduced to 3d. Increased compressive strength and higher flexural reinforcement ratios improved load capacity
Milligan and Polak (2023) [70]	(Numerical analysis) FE Program/ ABAQUS	To investigate openings in slabs supported by columns with various section shapes (L, T, cruciform)	A validated model was used for a parametric analysis investigating the effects of opening position, size, and quantity on the behavior of slabs supported by columns of different sectional shapes. The modeling employed 3D hexahedral elements for concrete and truss elements for reinforcements	Openings near columns reduced punching shear capacity and connection stiffness. When openings were positioned close to the column, they should be located as near as possible to the internal corners, typically a region of low shear stress

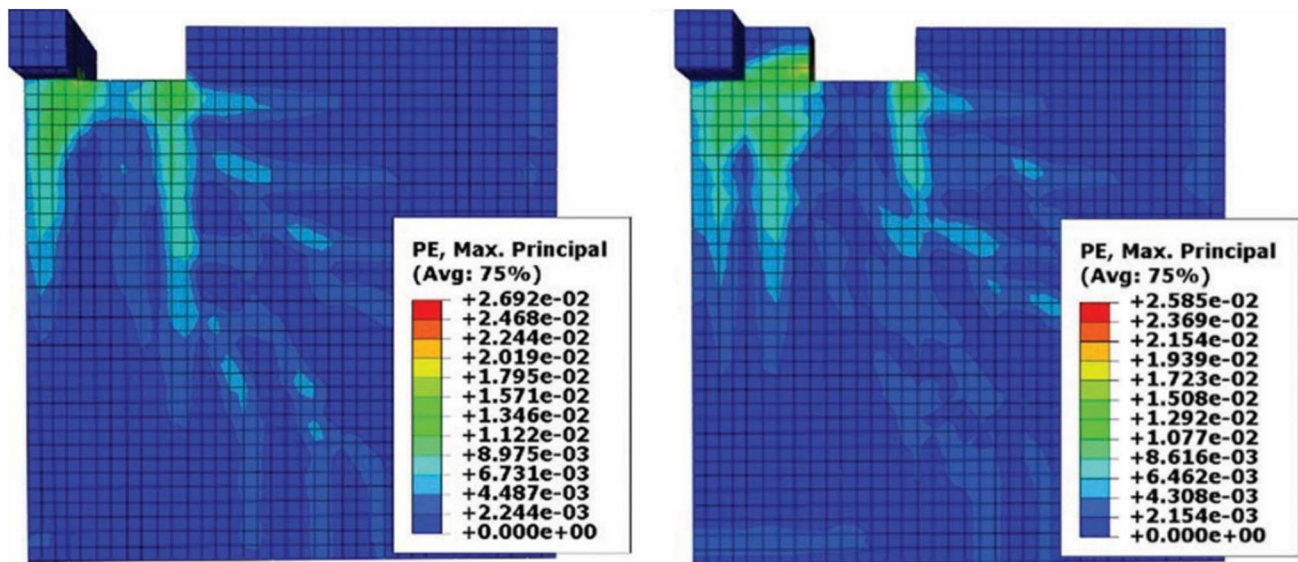


Fig. 9 Crack propagation in slabs with openings [131]

generation. In all the articles analyzed, numerical results are validated against experimental data, demonstrating that the models effectively capture real-world behavior. However, it is essential to emphasize that certain aspects, such as shear reinforcement or additional reinforcement around openings, have been insufficiently explored. Moreover, most studies focus on positioning openings away from columns to preserve load-bearing capacity. In practical applications, this may not always be feasible. Few investigations have explored strategies to maintain punching shear capacity while incorporating openings, such as structural reinforcements adjacent to the openings.

2.5 Punching Shear in Slabs with Non-Conventional Reinforced Concrete

The use of non-conventional reinforced concrete is primarily associated with the need to meet specific structural requirements. In punching shear, materials distinct from conventional concrete are employed for enhanced sustainability or improved structural performance in terms of load capacity, deformability, corrosion resistance, and lightness, among others. However, since the behavior of slabs under punching shear depends on the properties of concrete and steel, evaluating each new type of reinforced concrete used in structures becomes necessary.

From a sustainability perspective, notable options include coarse recycled concrete aggregate concrete (CRCAC) [132–140], geopolymers concrete (GPC) [141–145], Engineered Cementitious Composites (ECC) [38, 141, 146–151], and concretes incorporating alternative materials as partial substitutes for cement or aggregates [147, 152–155].

CRCAC utilizes construction waste as a substitute for natural aggregates, either partially or entirely, often resulting in lower mechanical strength and higher water absorption. Studies on its impact on punching shear show divergent results: some report significant reductions in load capacity [137], while others find CRCAC slabs to perform comparably to those made with ordinary concrete [132–135].

In GPC, conventional cement is replaced by silica-alumina-rich materials, eliminating the need for cement and reducing the carbon footprint [142]. GPC exhibits low shrinkage and creep strains, high resistance to chloride and sulfate attacks, superior fire and freeze resistance, and excellent mechanical and rheological performance [141]. However, due to the wide variety of materials and mix proportions, the final properties of GPC—and thus its punching shear behavior—can vary significantly [143]. ECC, also known as strain-hardening cementitious composites (SHCC) or ultra-high-toughness cementitious composites (UHTCC), demonstrates exceptional tensile strength and strain capacity due to the inclusion of fibers. This material is commonly used in structural repair due to its high ductility [141, 149]. When applied to slabs under punching shear, ECC enhances post-punching resistance, delays the initial cracking, and reduces the abruptness of failure [38, 141, 148, 156].

Another alternative is the use of high-performance and/or high-ductility concretes, such as high-strength concrete (HSC) [54, 125, 157–159], ultra-high-performance concrete (UHPC) [160–162], fiber-reinforced concrete (FRC) [163–165], steel-fiber-reinforced concrete (SFRC) [92, 166–173], steel-fiber recycled aggregate concrete (SFRAC) [174], high-performance fiber reinforced concrete (HPFRC) [175–177], ultra-high performance fiber reinforced concrete

(UHPRC) [39, 178–184], ultra-high performance self-compacting concrete (UHPRSCC) [185], steel fiber reinforced self-compacting concrete (SFRSCC) [186–188], and fibrous high strength self-compacted reinforced concrete (FHSS-CRC) [189].

High-strength concretes are increasingly becoming an attractive alternative in structural applications due to their ability to significantly enhance stiffness, flexural strength, and ductility [178]. This advantage stems from the fact that these materials exhibit higher durability, compressive strength, and tensile strength than ordinary concrete. However, the high cost associated with high-strength concrete often limits its application to specific regions of structural interest rather than being used throughout the entire structure. In the context of punching shear, high-performance concretes are commonly utilized in critical areas surrounding columns [39, 157, 160–162, 175, 176, 179], in layers within the tensioned or compressed regions, or at the slab-column connection zones [161, 181, 190]. Even in these localized applications, significant gains in the punching shear capacity of the structure are observed [39, 157, 175, 179, 190]. It is worth noting that high-performance concretes encompass a variety of formulations and properties. These materials can include ultra-high-performance characteristics, self-compacting properties, fiber reinforcement, and other specific attributes.

The inclusion of steel fibers primarily aims to enhance shear transfer capacity, thereby increasing punching shear strength and mitigating abrupt failure [166, 178, 191]. Gouveia et al. [169] emphasize that fibers significantly improve the tensile behavior of concrete by modifying its residual strength. Although steel fibers dominate, other types of fibers, such as synthetic [163, 164, 192–195], hybrid [155, 196], and amorphous metallic fibers [197], are gaining traction in experimental research. Fiber incorporation generally improves flexural and shear strength, deformation capacity, ductility, energy absorption, and altered crack patterns and density [92, 165, 167, 169, 170, 176, 180, 186, 187, 198–200].

Lightweight concrete represents a further alternative, principally aimed at reducing the structure's self-weight and, consequently, the requisite foundation size, as well as lowering seismic loads [141, 201]. This concrete uses aggregates with lower specific weights, such as expanded polystyrene balls [202], expanded clay [200, 201, 203, 204], and cenospheres [205], among others [192, 204, 206]. Lightweight concrete may exhibit reduced mechanical strength but offer better thermal and acoustic insulation [206]. For punching shear, lightweight concrete may decrease load capacity and result in more abrupt failure modes [202], necessitating further investigation.

It is worth mentioning that conventional reinforced concrete can be modified by altering the concrete mix or by

replacing the traditional steel reinforcement. In this context, the literature highlights several alternatives, such as the use of glass fiber reinforced polymer (GFRP) bars [30, 54, 63, 64, 88, 89, 159, 207–213], carbon fiber reinforced polymer (CFRP) bars [46, 105, 211, 214–218], and basalt fiber reinforced polymer (BFRP) bars [46, 219–221]. The primary motivation for choosing FRP bars lies in their high resistance to corrosion (one of the significant drawbacks of using steel), in addition to their high tensile strength, durability, and low density [43, 208, 218, 222]. However, Qian et al. [219] point out that one of the challenges in applying these materials is their low stiffness, which necessitates a higher reinforcement ratio to achieve stiffness levels comparable to those obtained with steel bars.

FRP bars can be employed throughout the slab in punching shear applications, serving as flexural and shear reinforcements [30, 88, 89, 210, 211] or exclusively in the shear reinforcements [209, 214, 218]. Additionally, numerous studies investigate the use of both alternative concretes and reinforcements to evaluate the experimental punching shear behavior [53, 54, 194, 208, 223] or conventional reinforced concrete slabs with some form of damage, such as corroded reinforcements, for instance [224, 225].

From a numerical perspective, the challenge of modeling punching shear in structures using alternative concretes lies in constitutive models that accurately represent these new materials. Generally, studies evaluate whether the proposed modeling approach can adequately describe the material's behavior. This evaluation often involves analyzing constitutive models, formulations, and uniaxial behavior curves provided by the authors, drawn from other studies in the literature, or based on design codes. When fibers are used, the post-cracking behavior of the concrete is modified by introducing residual tensile stresses [171, 172, 177, 188, 191, 192, 195]. For ultra-high-performance concretes, it is common to alter the material's uniaxial compression and/or tensile behavior [156, 182–184, 226]. For certain materials, such as HSC, CRCAC, and lightweight concrete, the standard concrete model is often employed, with parameter adjustments or distinct formulations used to predict tensile and compressive strengths and the modulus of elasticity [125, 138, 139, 182, 204]. The modeling of FRP reinforcements is more straightforward as they are generally represented with elastic behavior and abrupt failure, reflecting the behavior observed in experimental tests [46, 63, 64, 105, 212–216, 227].

In the early stages of research on this topic, simpler models were used, such as 2D models [171, 228] or elastic behavior assumptions for concrete [228]. However, more recent studies [143, 144, 182, 184, 191, 192, 229–232] predominantly employ 3D elements and/or complex nonlinear concrete models. These advanced models incorporate elastoplastic behavior, cracking, damage mechanics, viscoelastic

behavior, strain rate effects, and other relevant phenomena. Published studies aim to validate these models using experimental data provided by the authors or available in the literature—an essential step given the distinct behavior of alternative materials compared to conventional reinforced concrete. Subsequently, some studies propose formulations for predicting punching shear strength, comparing their findings with the load capacities estimated by principal design codes or conducting parametric analyses.

In numerical analyses, replacing conventional reinforced concrete entirely or partially has led to growing interest in FRP bars, particularly GFRP, and the application of ultra-high-performance concrete. High-performance concrete has demonstrated significant potential for enhancing punching shear capacity. When fibers are incorporated—whether in high-performance or conventional concretes—there is a notable increase in the structure's deformation capacity. FRP bars result in slabs with reduced stiffness but higher ultimate loads. However, it is essential to emphasize that these conclusions are specific to each analyzed study due to the diversity of materials and applications. Similarly, the findings regarding other alternative reinforced concrete discussed in Table 4 are limited to the scope of each investigation.

In most studies, numerical models do not include specific boundary or geometric conditions. Instead, the most commonly analyzed scenario is symmetric punching shear. These studies focus on accurately modeling new materials and evaluating their impact on crack patterns, ultimate load, and structural deformations using uniaxial behavior curves tailored to each material, developing precise mechanical strength formulations, or adjusting existing models. Nevertheless, some studies, such as those by Alrousan and Alnemrawi [46], investigated the effect of reinforcement materials in slabs subjected to asymmetric punching shear. In their modeling approach, truss elements were used to represent the reinforcement bars, as shown in Fig. 10, with only the uniaxial behavior curve modified to account for the steel, CFRP, and BFRP.

On the other hand, Huang et al. [226] examined an interior connection under symmetric loading by replacing ordinary concrete with UHTCC in the region near the column, as illustrated in Fig. 11. Similar to other studies mentioned in Table 4, the behavior of the alternative concrete was implemented using specific uniaxial curves that captured its material properties. In this investigation, 3D elements were employed for both types of concrete, while contact elements were employed to model the interface surfaces between the two materials.

Among the topics studied, high-performance concrete, steel fibers and FRP bars stand out not only because of their increasing use in existing structures but also because of the availability of data on the actual behavior of these materials. However, other areas remain underexplored, such as

GPC, CRCAC, synthetic fibers, and ECC, for which there is a limited number of publications regarding punching shear. Moreover, the analysis of punching shear in slabs made with these materials and alternatives like shear reinforcement, structural strengthening, or asymmetric loading conditions remains a relatively unexplored area in the current literature.

2.6 Strengthening of Slabs Subjected to Punching Shear

Using strengthening techniques in slabs subjected to punching shear is a contemporary approach aimed primarily at increasing the structure's load-bearing capacity [233–236]. The efficiency of such strengthening techniques depends on the superior characteristics of the strengthening materials, particularly tensile strength and ductility. These properties enable the interception of crack paths, delay the cracking load, and reduce the crack propagation speed [237–239]. The most common strengthening method involves the application of fiber-reinforced polymer (FRP). However, other materials such as steel plates, textile-reinforced mortar (TRM), and fabric-reinforced cementitious matrix (FRCM) are also employed. Despite their applications, research on these materials in punching shear scenarios is more limited than other structural members, such as columns, beams, shear walls, and brick infill walls [117, 236, 240].

Carbon fiber-reinforced polymer (CFRP) offers numerous advantages, including high mechanical strength, ease of application and installation, no alteration of the structure's dynamic characteristics, and excellent resistance to environmental effects [117, 241, 242]. Studies in literature have employed CFRP in various forms, such as sheets [33, 233–235, 243–249], strips [243, 250], fans [237], bars [140], or grids [251]. According to Torabian and Mostofinejad [237] and Saleh et al. [242], CFRP sheets are typically applied to the tension face of the slab to enhance its flexural capacity. Another approach is the installation of additional shear reinforcement, which generally improves ductility. Examples include FRP shear bolts or stirrup solutions employing FRP laminates, as investigated in several studies [241, 242]. Overall, the researchers compare slabs with and without reinforcement or examine parameters related to FRP application, such as material properties, positioning, and quantity used.

The application of CFRP typically involves bonding the composite material to the concrete, achieved through various techniques such as externally bonded reinforcement (EBR), near-surface mounted (NSM), or externally bonded reinforcement on grooves (EBROG). In some cases, CFRP sheets are used with prestressing, as demonstrated by Abdullah and Bailey [252], Abdullah et al. [253], and Koppitz et al. [250]. Another application involves using CFRP sheets

Table 4 Review of punching shear in slabs with non-conventional reinforced concrete

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Di Prisco and Felicetti (1997) [171]	(Numerical analysis) FE Program/ CASTEM	To validate the numerical modeling of slabs with SFRC	The authors modeled the slab using 2D elements. The fibers were incorporated into the model through modifications to the descending branch of the uniaxial tensile behavior, taking into account experimental results and theoretical formulations	The study revealed that the model could meticulously follow the loading path up to the peak strength. However, the model exhibited an inability to reproduce discrete radial cracks, making it more suitable to employ 3D modeling for better accuracy
Oliveira et al. (2004) [228]	(Numerical analysis) FE Program/ SAP	To examine the shear stresses in slabs made of high-performance concrete	The authors conducted experimental and numerical studies, utilizing computational modeling to determine the shear stresses around the columns. They analyzed slabs supported by rectangular columns of varying dimensions. Concrete was considered an elastic material, and shell elements were employed in the analysis	Based on the numerical study, the authors proposed a novel method for calculating punching shear strength, accounting for the rectangularity of the column
Xiao and Chin (2007) [195]	(Numerical analysis) FE Program/ ANSYS	To analyze the impact of polypropylene fibers on concrete	The authors compared numerical modeling with experimental data from the literature. The fibers were incorporated by using the experimental uniaxial behavior of fiber-reinforced concrete. Eight-node hexahedral elements were used to model the concrete, and link elements were used for the reinforcement. After, a parametric analysis was performed	The numerical model yielded results close to the experimental data for the ultimate load and the descending branch of the load-response curve. The study also found that increasing the slab thickness led to higher ultimate loads but could change the failure mode from flexural to punching shear
Mostafaei et al. (2011) [172]	(Numerical analysis) FE Program/ VecTor3	To validate numerical modeling of prestressed slabs with SFRC	Slabs made of conventional concrete and SFRC were analyzed. The fibers in the concrete were modeled by modifying the uniaxial tensile behavior. 3D hexahedral elements were employed, and the results were compared with experimental data	The numerical results were reasonably in agreement with the experimental findings. Including fibers rendered the behavior more ductile and mitigated the abrupt nature of the failure

Table 4 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Youn et al. (2014) [204]	(Numerical analysis) FE Program/ ABAQUS	To validate the numerical modeling of slabs made of lightweight concrete	3D hexahedral elements were used to model the concrete, while truss elements represented the reinforcement. Lightweight concrete was modeled based on experimental properties, with tensile strength adjustments according to formulations from the literature and model calibration considering different dilation angles	Once calibrated, the model demonstrated good agreement with experimental data regarding the failure mode. The authors also proposed an analytical equation for predicting punching shear strength in lightweight concrete slabs
Reis et al. (2015) [139]	(Numerical analysis) FE Program/ ATENA	To validate numerical modeling using experimental data from slabs with CRCAC	3D hexahedral elements were used for the concrete, and truss elements represented the reinforcements. A predefined concrete model available in the software was employed, and the authors analyzed various fracture energy values to calibrate it	The numerical model agreed well with the experimental results. The authors observed a significant dependency of the results on fracture energy. No substantial changes in slab behavior were observed when using CRCAC
Teixeira et al. (2015) [188]	(Numerical analysis) FE Program/ FEMIX	To investigate punching shear in slabs with SFRSCC through experimental and numerical analyses	Layered shell elements were employed, with one layer to model the reinforcement. The simulation used a constitutive model incorporating fracture modes. The fibers' behavior was introduced into the model by considering experimental data and calibrating the post-cracking curve. A parametric analysis varied the fracture energy and the second branch of the diagram that defines the crack-opening process	The calibrated model yielded excellent agreement with experimental results. An increase in the aspect ratio led to higher fracture energy and ultimate load capacity of the slab. Changes in the post-cracking branch significantly altered the structural behavior
Yousef et al. (2019) [125]	(Numerical analysis) FE Program/ ABAQUS	To examine HSC slabs with openings	The validation step used experimental data from the authors for slabs with normal and high-strength concrete subjected to cyclic loading. 3D solid and bar elements were used for the concrete and the reinforcement, respectively. Specific formulations for tensile strength and elastic modulus were considered for high-strength concrete	Numerical results closely matched experimental data. High-strength concrete increased the punching shear capacity by approximately 50% for slabs with and without openings

Table 4 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Hamoda and Hossain (2019) [215]	(Numerical analysis) FE Program/ ABAQUS	To investigate and compare shear reinforcement bars of CFRP and steel	3D elements were used to model the concrete, and truss elements represented the reinforcements. The study validated the model using experimental data from the authors and the literature. The numerical model was calibrated by varying material parameters. A parametric analysis was then conducted using different shear reinforcements made of CFRP and steel. A linear elastic model was employed to represent CFRP	The numerical model agreed well with the experimental results. CFRP shear reinforcement resulted in lower stiffness and cracking load compared to steel bars. In cases where CFRP bars were positioned within the punching shear cone, the slab's strength increased compared to scenarios without shear reinforcement
Wu et al. (2019) [183]	(Numerical analysis) FE Program/ ABAQUS	To investigate conventional concrete slabs with a layer of UHPFRC	Concrete and reinforcement were considered with 3D hexahedral and truss elements, respectively. An elastoplastic damage model was used for both conventional concrete and UHPFRC. Uniaxial elastic behavior in compression and a trilinear tensile curve were considered for UHPFRC. A validation was conducted with experimental data, followed by a parametric analysis	The results demonstrated that incorporating a UHPFRC layer increases punching shear resistance but reduces the deformation capacity. The compressive strength and thickness of the UHPFRC layer emerged as the predominant parameters influencing the structural response
Salama et al. (2020) [63] and Salama et al. (2020) [64]	(Numerical analysis) FE Program/ ANSYS	To examine slabs reinforced with GFRP bars supported on edge columns	The study validated the simulation with experimental data and provided a parametric analysis. Hexahedral elements were used for the concrete and bar elements for the reinforcement, with GFRP bars employed for flexural and shear reinforcements (closed stirrups). GFRP was modeled as a linear elastic material	The numerical results agreed well with the experimental data. The use of GFRP stirrups increased the ultimate load capacity. The higher flexural GFRP bar stiffness correlated with increased punching shear resistance

Table 4 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Said et al. (2020) [212]	(Numerical analysis) FE Program/ ANSYS	To analyze lightweight concrete slabs reinforced with shear bars made of steel or GFRP and bolts	The modeling adopted 3D elements for the concrete and link elements for the reinforcement. The post-peak tensile behavior was disregarded, assuming zero tension after reaching the maximum tensile strength. GFRP was modeled as a linear elastic material. Different shear reinforcement configurations, varying material, anchorage, and position, were tested. A validation step was performed using the experimental authors' data	The results showed excellent agreement with experimental data. Shear reinforcement significantly improved load capacity, more for steel bars than for GFRP. Bolts fixed to slabs using steel plates altered the failure mode from punching shear to flexural-punching
Al-Rousan et al. (2020) [216]	(Numerical analysis) FE Program/ ANSYS	To analyze bridge decks reinforced with steel, CFRP, and GFRP bars	Hexahedral elements were used for the concrete, and link elements for the bars. The validation step considered experimental data from the literature. CFRP and GFRP bars were modeled as linear elastic materials in tension, while steel bars were represented as perfectly elastoplastic. Parametric analysis varied concrete strength, reinforcement ratio, and material type	The numerical results closely matched the experimental data. CFRP and GFRP bars enhanced load capacity compared to steel bars, with CFRP bars yielding superior performance to GFRP
Jafarian et al. (2020) [214]	(Numerical analysis) FE Program/ ABAQUS	To investigate slabs reinforced with CFRP and GFRP grids	The numerical model was validated with experimental data from the literature. The FRP grids, used as shear reinforcement, were modeled as linear elastic materials. The modeling employed 3D elements for the concrete and truss elements for the reinforcement. Parametric analysis explored variations in grid spacing, position, and material	The results agreed well with the experimental data. Grids increased load capacity and energy dissipation, with CFRP grids showing superior performance compared to GFRP

Table 4 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
El-Gendy and El-Salakawy (2021) [105]	(Numerical analysis) FE Program/ ATENA	To investigate slabs reinforced with CFRP and GFRP bars	A validation step and a parametric analysis were performed. The study investigated slabs under cyclic lateral loading. Steel bars were used in the column, while FRP bars were employed in the slabs. Tetrahedral elements modeled the concrete, and truss elements were used for the reinforcement. Steel was modeled as an elastoplastic material, while FRP exhibited linear elastic behavior	Numerical validation showed excellent agreement with experimental data. CFRP provided higher load capacity and lower total drift ratio than GFRP
Sousa et al. (2021) [184]	(Numerical analysis) FE Program/ ABAQUS	To analyze hybrid slabs made of conventional concrete and UHP-FRC	The numerical model, validated against experimental data, employed 3D elements for the concrete and truss elements for the reinforcement. The UHPFRC model incorporated specific parameters and uniaxial tension and compression curves from the literature. A parametric analysis investigated the rational use of UHPFRC	Validation results closely matched most experimental data. Parametric analysis indicated that using a UHP-FRC layer in the compressed zone was more efficient than in the tensile zone. Slabs with UHPFRC showed higher deformation capacity in the critical shear zone
Kadhim et al. (2021) [182]	(Numerical analysis) FE Program/ ABAQUS	To investigate hybrid slabs made of conventional concrete and UHPFRC	The modeling, validated with experimental data, employed 3D elements for the concrete and truss elements for the reinforcement. The UHPFRC model incorporated specific tension and compression curves from the literature, and other parameters were calibrated. Parametric analysis was performed	The numerical model showed good agreement with experimental data. Increasing the UHPFRC area in the slab enhanced load capacity. The compressive strength ratio between the two concretes significantly impacted the structural response

Table 4 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Kitano et al. (2021) [192]	(Numerical analysis) FE Program	To validate a numerical model for lightweight concrete slabs with fibers	The modeling utilized 3D elements, simulating lightweight concrete reinforced with PVA fibers to specific uniaxial curves from the literature. For the tensile curve, residual stresses after cracking due to the fibers were considered. The elastic modulus was reduced to account for the lightweight aggregate. The results were compared against experimental data. The slabs also included prestressing, which was numerically modeled	The use of lightweight aggregates reduces punching shear resistance. However, the introduction of pre-stressing and the addition of short fibers enhances the punching shear resistance
Madkour et al. (2022) [229]	(Numerical analysis) FE Program/ ABAQUS	To investigate slabs reinforced with GFRP bars	The modeling employed 3D elements for the concrete and truss elements for the reinforcement. The concrete constitutive model parameters were calibrated. The simulation was validated against experimental data from the literature, followed by a parametric analysis	The model agreed well with the experimental results. The ductility index developed for slab-column connections reinforced with GFRP bars was generally lower than connections reinforced with conventional steel bars
Alrousan and Alnemrawi (2022) [46]	(Numerical analysis) FE Program/ ABAQUS	To investigate slabs reinforced with steel, CFRP, or BFRP bars	The study validated experimental data from the literature and conducted a parametric analysis. Slabs were evaluated with varying eccentricities and openings. The modeling utilized 8-node 3D elements for the concrete and truss elements for the reinforcement. Steel was modeled as an elastoplastic material, while CFRP and BFRP were modeled as linear elastic materials	Compared to steel reinforcement, CFRP and BFRP bars reduced elastic stiffness, cracking load, energy absorption, post-cracking behavior, and ductility. BFRP reinforcement was the most critical situation

Table 4 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
El-Kassas et al. (2022) [189]	(Numerical analysis) FE Program/ ANSYS	To investigate FHSSRC slabs	Experimental and numerical analyses were performed on slabs with varying percentages and types of fibers (basalt or steel), with and without shear reinforcement. The modeling employed 3D elements for the concrete and link elements for the reinforcement. Uniaxial curves obtained experimentally were used to model the materials. Subsequently, a parametric analysis was conducted	The validation stage yielded excellent approximations of the failure load. The addition of fibers made the structural behavior more ductile without altering the punching shear failure surface
Sanabria Díaz et al. (2022) [177]	(Numerical analysis) FE Program/ DIANA	To investigate hybrid slabs made with conventional concrete and HPFRC	The modeling was validated against experimental data from the literature. It employed 3D elements for the concrete and truss elements for the reinforcement. The HPFRC was modeled based on experimental data, with modifications to the post-cracking behavior in the uniaxial tensile diagram. After validation, the model was used for a parametric analysis	The validation results closely matched the experimental data. The analysis demonstrated that even small regions of HPFRC significantly enhance the ductility of the structural response. Larger areas of HPFRC may even alter the failure mode from punching shear to flexural failure
Zhang et al. (2023) [213]	(Numerical analysis) FE Program/ ABAQUS	To investigate slabs reinforced with GFRP bars	The model was validated against experimental data using 3D elements. The study analyzed slabs subjected to impact loading and high temperatures. Scenarios with fire followed by cooling were also investigated. The elastic-brittle constitutive model was adopted for the GFRP bars	Under high temperature and impact conditions, the behavior of slabs with GFRP bars is similar to that of RC slabs. However, high temperatures affect displacement and impact force more significantly when using GFRP
Huang et al. (2023) [226]	(Numerical analysis) FE Program/ ABAQUS	To investigate hybrid slabs with a UHTCC region	UHTCC was considered using specific uniaxial curves derived from equations in the literature. The modeling employed 3D solid and contact elements between the UHTCC and conventional concrete. Validation was performed using experimental data from the literature	The results agreed well with the experimental findings. A UHTCC region increased the failure load, enhanced ductility, and altered the location of the primary crack

Table 4 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Chen et al. (2023) [143] and Chen et al. (2025) [144]	(Numerical analysis) FE Program/LS-DYNA	To investigate GPC slabs subjected to impact loading	The modeling utilized 3D elements and contact elements to simulate the impact load. A specific constitutive model was employed, incorporating modifications to certain constants. One study focused on GPC slabs, while another investigated fiber-reinforced geopolymer concrete slabs (FR-GPC). The strain rate effect was incorporated into the model. Validation was performed using experimental data, followed by a parametric analysis	GPC slabs exhibit typical punching shear failure. Formulations used for conventional concrete can be applied to estimate the resistance of GPC slabs under static conditions. FR-GPC slabs can withstand higher impact forces and absorb more energy than GPC
Elsamak et al. (2023) [156]	(Numerical analysis) FE Program/ABAQUS	To investigate slabs made of conventional concrete, HSC, SHCC, and UHPFRC	The study conducted experimental and numerical analyses on slabs with different types of concrete, with and without shear reinforcement. The shear reinforcement included high-strength steel bolts fixed on steel plates and steel tubes filled with SHCC. The modeling employed 3D elements for the shear reinforcement and concrete and truss elements for the longitudinal bars. Specific uniaxial curves were used for each type of concrete. A case study was also performed	The numerical validation results closely matched the experimental findings. Steel assemblies increased the punching shear capacity of slabs made with regular concrete, although punching shear failure was not entirely prevented. Internally unreinforced slabs made of SHCC and UHPFRC avoided punching shear failure and exhibited ductile bending collapse
Yehia et al. (2023) [230]	(Numerical analysis) FE Program/ABAQUS	To investigate UHPC slabs with micro steel fibers	The modeling employed 3D elements for the concrete and truss elements for the reinforcement. The software's constitutive model was calibrated using parameter values recommended in the literature. Validation was performed with proprietary experimental data, followed by a parametric analysis	Modeling results agreed well with the experimental data. The use of UHPC altered the failure mode from pure brittle shear failure to slightly ductile failure, increased the load capacity, and modified the inclination of the failure surface

Table 4 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
McGinnis et al. (2024) [138]	(Numerical analysis) FE Program/ABAQUS	To model hollow-core slabs with CRCAC	The study utilized 3D elements for concrete and truss elements for reinforcement. The authors presented experimental and numerical analyses of hollow-core slabs subjected to flexure, shear, and punching shear. Recycled aggregate concrete was modeled similarly to conventional concrete	The punching shear results obtained showed some variation compared to experimental data. The stress pattern corresponded to the expected distribution for the punching shear load configuration
Mostafa et al. (2024) [231]	(Numerical analysis) FE Program/ANSYS	To investigate conventional reinforced concrete slabs with GFRP gratings	The modeling employed 3D elements for the concrete and link elements for the reinforcement. The elastic-brittle constitutive model was applied to the GFRP bars. The study validated the simulation using proprietary experimental data and conducted a parametric analysis	The validation results closely matched the experimental data concerning the failure load. The specimens with GFRP gratings displayed greater punching shear perimeters and failure loads
Alkhattabi et al. (2024) [227]	(Numerical analysis) FE Program/V3	To investigate slabs reinforced with GFRP bars	The elastic-brittle constitutive model was employed for the GFRP bars, accounting for slippage. The modeling used 3D tetrahedral elements for the concrete. Validation was performed against proprietary experimental results, followed by a parametric analysis varying the span-to-depth and column-aspect ratios	The validation results were consistent with the experimental data. The parametric analysis revealed that increasing the column-aspect ratio reduces punching shear stresses while increasing the span-to-depth ratio decreases shear stresses
Ors et al. (2024) [232]	(Numerical analysis) FE Program/ABAQUS	To investigate UHPC slabs with steel fibers	The numerical model was used as a data generation source for machine learning studies to refine normative expressions for prestressed UHPC slabs. The modeling employed 3D elements for the concrete and truss elements for the reinforcement. Validation was conducted using experimental data, followed by a parametric analysis	The results closely matched the experimental findings. The proposed correction factors primarily depend on the concrete strength, the aspect ratio of the fibers, and the strength of the shear reinforcement steel

Table 4 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Barros et al. (2024) [173]	(Numerical analysis) FE Program/ Various software	To investigate punching shear in SFRC slabs through a blind competition	The study involved researchers from various countries through a blind competition. Following an experimental program of SFRC slabs subjected to punching shear, participants submitted predictions regarding failure loads, displacements, and deformations. Numerical results were obtained using smeared crack models, a concrete damage plasticity model, and discrete crack models, among others	The numerical models demonstrated that finite element method could adequately represent punching shear in SFRC slabs, provided the material behavior is correctly implemented. Some approaches yielded results that closely matched the experimental data
Lin et al. (2025) [191]	(Numerical analysis) FE Program/ DIANA	To investigate prestressed slabs with SFRC	The study validated numerical data against experimental results from the literature. A set of slabs and columns was modeled. Prestressing was investigated in slabs with either passive reinforcement or SFRC. The numerical analysis employed 3D elements for the concrete and embedded bar elements for the reinforcement. Steel fibers were incorporated into the modeling using a theoretical tensile diagram that accounted for crack opening in this type of concrete	Applying the constitutive model for the fibers demonstrated numerical results that closely matched experimental findings, with good agreement in crack patterns. Compared to conventional reinforcement solutions, steel fibers resulted in better crack control and higher punching shear capacity at slab-column connections
Afshani and Rofooei (2025) [200]	(Numerical analysis) FE Program/ LS-DYNA	To investigate slabs with lightweight concrete, prestressing, and steel fiber	The study presented an experimental and numerical analysis of slabs subjected to impact loading, made with lightweight concrete, investigating the effects of prestressing and steel fibers. In the modeling, 3D solid elements were used for concrete, while beam elements were used for reinforcement bars. Contact elements were implemented, and strain rate effects were considered in the analysis	The numerical results were close to the experimental ones regarding maximum load and impulse-time response. The numerically obtained failure mechanism closely resembled that observed in the experimental tests

Table 4 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Su et al. (2025) [151]	(Numerical analysis) FE Program/ABAQUS	To investigate slabs locally enhanced using ECC	The study conducted an experimental analysis employing ECC in the region near the column. The numerical analysis was validated with experimental data, and a parametric study was carried out. Specific uniaxial stress–strain curves were used to model ECC. The analysis employed 3D elements for concrete and truss elements for reinforcement	Excellent results were obtained in terms of the load–displacement response. Using ECC could modify the failure mode from punching shear to a combined punching shear–flexural failure, increase load capacity and ductility

around slab openings to increase localized resistance [117, 126, 254].

An alternative strengthening technique is the TRM method, in which a cement-based inorganic mortar material behaves as a binder. This composite material compounds steel, basalt, carbon, or glass textiles within the mortar layer [117]. Mercimek et al. [240] highlight that, compared to FRP, TRM offers advantages such as low cost, resistance to high temperatures, suitability for wet surfaces, low thermal conductivity, and high load-bearing capacity. Although relatively few studies have explored this technique for punching shear failure [117, 240, 255, 256], there is a consensus on its potential, as it significantly enhances the load-bearing capacity compared to traditional solid slabs. Strain-hardening cementitious composites (SHCC) or ultra-high-performance engineered cementitious composites (UHPECC) are also common in structural reinforcement analyses. In such cases, a material layer is typically applied to the bottom region, forming a kind of drop panel with properties specifically tailored to improve punching shear resistance. This approach often increases ultimate load capacity and enhances ductility [257, 258].

The use of steel sheets as a method of strengthening slabs subjected to punching shear has also been investigated. In this approach, the reinforcement is attached to the concrete using epoxy and bolts, which can also function as shear reinforcement. Although this technique is less commonly applied to address punching shear, some studies have utilized aluminum sheets [238, 239] or steel plates [259–261]. In all cases, an increase in both the ultimate load capacity and the stiffness of the structure was observed. From this perspective, Mansour et al. [238] and Ghalla et al. [239] utilized aluminum sheets sourced from waste generated by factories and large workshops, a measure aligned with sustainability in civil construction. Additionally, it is worth noting that steel sheets are commonly applied externally to the top surface of slabs. However, Ghalla et al. [239] experimentally and numerically investigated the application of this strengthening technique within the structure before slab casting, resembling shear reinforcement.

From a numerical perspective, it is noteworthy that shell elements are frequently utilized for modeling the sheets [33, 126, 239, 247, 249, 260, 262]. However, 3D elements can also be used [262]. In the case of TRM meshes, Mercimek et al. [240] adopted truss elements. It is also common to use contact elements, particularly in scenarios involving FRP composites [33, 126, 246, 249]. Concerning material constitutive models, the modeling of FRP as a linear-elastic material up to failure and steel as an elastoplastic material, with or without hardening, poses no significant challenges. However, the performance and proper functioning of the reinforcements in numerical models depend on the accurate simulation of surface bonding and the potential for

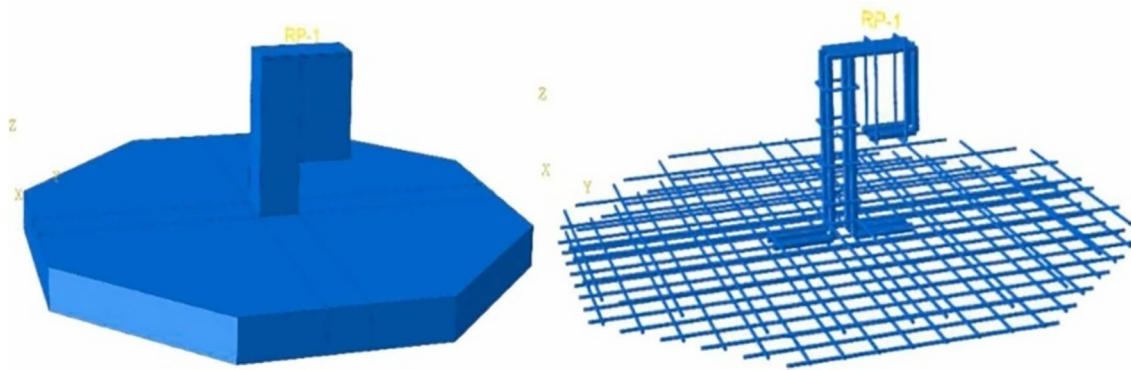


Fig. 10 Example of modeling the solid geometry and lines of BFRP, CFRP, and steel reinforcement bars [46]

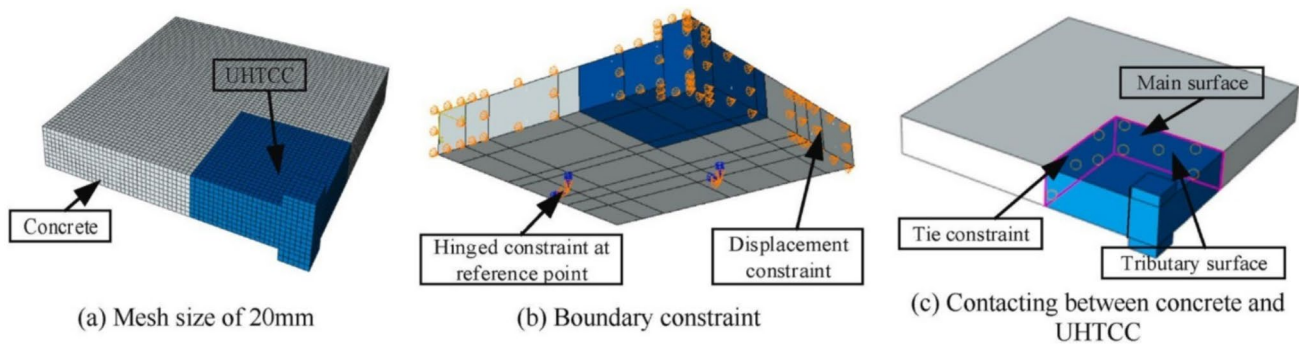


Fig. 11 Example of modeling the geometry (solid and mesh) of ordinary concrete, UHTCC, and their interface contact [226]

slippage between the surfaces. In this regard, some studies have applied slip laws derived from literature or experimental tests, achieving numerical validation results that closely matched experimental data [33, 246].

Researchers have concluded that strengthening techniques increase the ultimate load capacity [126, 128, 239, 240, 246, 249, 252, 256, 260]. When FRP sheets were employed, reduced deformation capacity of the structure was observed [128, 246, 252, 262]. Conversely, the use of steel plates demonstrated improved ductility [260]. Furthermore, the primary objective of numerical studies has been to evaluate the effect of reinforcement, often validating the results with experimental data from the literature. As presented in Table 5, many studies have also presented conclusions regarding the position of reinforcement elements, sheet thickness, and the effects of overlapping layers.

In this context, the review of numerical studies conducted to date on strengthening techniques highlights that, although alternatives such as SHCC, aluminum sheets, UHPECC, and TRM have been utilized, most studies focus on FRP sheets. Among these studies, many employ 3D elements for modeling concrete and shell elements to represent the FRP sheets, as illustrated in Fig. 12, adapted from Hamoda et al. [257]. In this case, the authors also used a combination of

a bonded UHPECC layer, CFRP sheets, and threaded steel bolts. Figure 12 further compares the experimentally and numerically obtained concrete crack patterns, demonstrating a strong correlation between the results.

When modeling strengthening techniques, several aspects have drawn the attention of researchers, such as contact elements to represent the bond between materials and the implementation of constitutive models for the strengthening materials, which adds complexity to the numerical models. In this regard, knowledge gaps persist in this area, as evidenced by the limited number of published articles. These gaps include studies on alternatives to FRP sheets and analyses of strengthening techniques applied to previously damaged slabs or those subjected to specific conditions, such as asymmetric punching shear, for example.

2.7 Use and Modeling of Shear Reinforcement in Slabs Subjected to Punching Shear

Although there are various techniques to enhance punching shear resistance, as discussed in previous sections, shear reinforcement offers several advantages that make it one of the most commonly employed strategies. Unlike alternatives like column capitals or increasing slab thickness, shear

Table 5 Numerical review of strengthening techniques in punching shear

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Agbossou et al. (2008) [262]	(Numerical analysis) FE Program/ ANSYS	To investigate slabs strengthened with externally bonded FRP strips	The study conducted experimental and numerical analyses of slabs and beams strengthened with CFRP strips. The numerical model was validated using experimental data followed by a parametric analysis. The modeling employed 3D hexahedral elements for concrete, link elements for steel reinforcement, or layered shell elements. CFRP was modeled using 3D hexahedral or shell elements. The study assessed the influence of the low mechanical quality of the concrete between steel bars and CFRP	The validation phase yielded accurate approximations of experimental results for beams. Composites significantly reduced the overall rotation angle of the slab around the loading column. The mechanical quality of the concrete between the steel and the composite was a limiting factor in externally bonded strip composites
Farghaly and Ueda (2010) [246]	(Numerical analysis) FE Program/ 3D CAMUI	To analyze slabs strengthened externally with FRP sheets	The study validated its numerical modeling with experimental data from the literature. It used 3D elements for concrete, plate elements for FRP, and contact elements for material interfaces. The FRP-concrete interface mechanical behavior was modeled through the relationship between local shear stress and slip	The validation phase achieved accurate approximations of experimental results, and the slip model was deemed appropriate. FRP increased the stiffness and punching shear load capacity of the slabs, and the overlap of FRP layers reduced their slippage
Ebead and Saeed (2010) [33]	(Numerical analysis) FE Program/ ADINA	To investigate strengthening slabs using steel or FRP plates secured with steel bolts	The concrete was modeled using 3D hexahedral and steel reinforcement with truss elements. Steel and FRP plates were modeled using shell elements with interface elements to simulate bonding. Steel was considered a bilinear material, while FRP was linear elastic. The study investigated experimental slabs subjected to concentric or eccentric loading	Validation results showed an excellent approximation of the ultimate load capacity and the corresponding displacements. The numerical model successfully captured the slippage between the plates and the concrete, with the highest slippage observed near the column where the plates overlapped

Table 5 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Radnić et al. (2015) [247]	(Numerical analysis) FE Program	To analyze slabs strengthened with CFRP laminates subjected to impact loading	The modeling used layered shell elements, with steel reinforcement and CFRP treated as specific layers. The strain rate effect was included in the material modeling. The numerical model was validated against experimental data from the authors. CFRP was applied to the slab's tensioned bottom surface	The numerical model closely matched the experimental results. Slabs strengthened with CFRP showed only slightly greater ultimate bearing capacity under static and impact loading than conventional reinforced slabs
Barros et al. (2017) [248]	(Numerical analysis) FE Program/ FEMIX	To analyze slabs strengthened with U-shaped CFRP laminates	The authors conducted experimental and numerical analyses of slabs and beams using CFRP laminates anchored within the structure. 3D solid and cable elements were used for concrete and steel reinforcement or CFRP, respectively. CFRP was modeled with linear behavior up to its ultimate tensile strength. After validation, the model was used for parametric analysis	The numerical results closely matched the experimental data. The use of CFRP strengthening increased the punching shear load capacity. The slab's ultimate load capacity increased with the tensile strength of the CFRP laminates. However, the slab's stiffness remained unaffected by this property
Abdulrahman et al. (2017) [126]	(Numerical analysis) FE Program/ ABAQUS	To investigate the corner connections with openings strengthened with CFRP laminates	The numerical model utilized 8-node 3D elements for concrete, shell elements for CFRP laminates, and cohesive elements to simulate material bonding. The laminates were placed around the column with extended ends, and a parametric analysis evaluated various arrangements	The numerical model agreed well with the experimental data. The strengthening technique increased ultimate load capacity, albeit marginally. Changes in laminate arrangement or quantity did not significantly improve load capacity
Abdullah and Bailey (2018) [252]	(Numerical analysis) FE Program/ ABAQUS	To verify strengthening slabs with prestressed or non-prestressed CFRP laminates	3D hexahedral elements were used for concrete, truss elements for steel reinforcement, layered shell elements for FRP, and connector elements between CFRP and concrete. CFRP was modeled as linear elastic, and the interface was modeled with elastic-damage behavior. The model was validated against experimental data from the authors' previous studies	The results demonstrated excellent approximation of experimental data. The use of FRP strengthening increased punching shear capacity but reduced deformation capacity. Prestressed FRP plates significantly reduced the deformation capacity

Table 5 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Taresh et al. (2021) [260]	(Numerical analysis) FE Program/ABAQUS	To evaluate slabs using steel angle plates anchored by steel bolts as a strengthening technique	Experimental and numerical analyses were conducted on slabs strengthened with steel plates anchored by steel bolts at the slab-column connection. Hexahedral elements were used for the concrete, shell elements for the steel plates, and truss elements for the reinforcement and steel bolts. After calibration and validation, the model was employed in a parametric analysis. The shear bolts and angle plates were modeled with a bilinear strain-hardening behavior	The validation results showed excellent agreement with the experimental data. Steel angle plates prevented abrupt failure and increased the failure load, deformation capacity, stiffness, and energy absorption. The confinement pressure on the concrete near the column expanded the critical section
Silva et al. (2021) [249]	(Numerical analysis) FE Program/ABAQUS	To investigate severely damaged reinforced concrete slabs strengthened with CFRP laminates	The study presents an experimental and numerical analysis of slabs strengthened after severe damage. The damaged concrete was removed and replaced, followed by implementing CFRP laminate in various positions and configurations. The modeling used tetrahedral elements for the concrete, shell elements for the CFRP, and cohesive elements for the interface between materials	The validation step demonstrated that the numerical results were close to the experimental ones. Radial arrangements of CFRP laminates contributed more significantly to load capacity than orthogonal configurations. Providing CFRP in multiple layers proved more effective than placing the same area of CFRP in a single layer on the tension surface
Oukaili and Merie (2021) [128]	(Numerical analysis) FE Program/ANSYS	To investigate the use of CFRP laminates in Bubbledeck slabs with openings	The modeling was validated with experimental data provided by the authors. Tetrahedral elements were used for the concrete, line elements for the reinforcement, and shell elements for the CFRP laminates, assuming perfect adhesion between materials	In all cases, numerical results closely matched experimental data. The strengthening technique increased post-cracking stiffness and the ultimate load capacity in several models
Mercimek et al. (2022) [240]	(Numerical analysis) FE Program/ABAQUS	To analyze slabs strengthened with shear studs, TRM, or CFRP laminates	The study evaluated slabs with various strengthening techniques, both experimentally and numerically. Hexahedral elements were used for the concrete, beam elements for the reinforcement, truss elements for the carbon mesh in TRM, thin shell elements for CFRP laminates, and tetrahedral elements for the shear studs	Numerical results agreed well with the experimental data. Regarding load-carrying capacity, diagonal strips were more effective before punching shear, while orthogonal strips yielded better results after punching shear. Strengthening techniques analyzed enhanced slab capacity, with CFRP and TRM showing the most significant improvements

Table 5 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Ammash et al. (2022) [261]	(Numerical analysis) FE Program/ABAQUS	To evaluate slabs strengthened with steel plates	Numerical modeling was validated with experimental data and employed in parametric analysis. After the slabs reached approximately half of their ultimate load, strengthening was applied, and the width and length of the steel plates were varied. Calibration was based on experimental results	The simulation results agreed well with the experimental data. Increased steel stiffeners expanded the critical punching shear perimeter and enhanced the ultimate load capacity. Higher concrete compressive strength and flexural reinforcement ratio also increased load capacity
Le-Nguyen et al. (2023) [256]	(Numerical analysis) FE Program/Cast3M	To investigate slabs strengthened with FRCM layers	The modeling employed 3D elements to validate experimental data and conduct parametric analysis. The FRCM behavior was modeled using constitutive laws based on experimental material data, incorporating damage evolution and uniaxial tension behavior. Material parameters were calibrated through an iterative model	Calibration achieved excellent numerical validation results. The use of FRCM significantly enhanced the ultimate load capacity and altered the crack patterns. The strengthening technique activated the dowel effect mechanism, increasing bearing capacity and potentially providing ductility
Ghalla et al. (2024) [239]	(Numerical analysis) FE Program/ABAQUS	To investigate slabs strengthened with internally embedded aluminum sheets	The study included experimental and numerical analysis of slabs strengthened with internal aluminum sheets. Aluminum sheets and reinforcement were modeled as materials with perfect plasticity. Solid 3D elements were used for the concrete, truss elements for the reinforcement, and membrane strain shell elements for the aluminum sheets. The model was validated against experimental data. Subsequent parametric analyses explored the configuration of the strengthening sections and the thickness of the reinforced aluminum sections	Validation results closely matched experimental data for all models. Aluminum sheets increased ultimate load capacity, delayed crack formation, restricted crack propagation speed, and prevented concrete cover splitting prior to complete failure. The use of small interspaces between the strengthening sections significantly increased the absorbed energy

Table 5 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Hamoda et al. (2024) [257]	(Numerical analysis) FE Program/ ABAQUS	To investigate the combination of bonded UHPECC layer, CFRP sheets, and threaded steel bolts	Experimental and numerical analyses were performed, with the numerical validation employing 3D finite elements for concrete and truss elements for reinforcement. Shell elements were used for the CFRP sheets. The UHPECC overlay was applied at and around the interior column, while CFRP sheets were positioned outside the overlay boundary. Threaded steel bolts were used as shear reinforcement	Numerical results agreed well with experimental data. The parametric study revealed that increasing the width of the UHPECC layer and/or the column diameter enhanced the punching shear capacity. Strengthening the slab with the combined UHPECC overlay and post-installed bolts increased elastic stiffness, higher cracking, and ultimate loads
Ghalla et al. (2025) [258]	(Numerical analysis) FE Program/ ABAQUS	To investigate punching shear in slabs with SHCC drop panels	The authors conducted an experimental and numerical study on slabs with SHCC drop panels. The research also examined GFRP or galvanized steel meshes and bolts within the drop panel. The numerical analysis employed 3D elements for the concrete and truss elements for the reinforcements to validate the modeling against experimental data. A parametric study varied the number and position of steel bolts	The numerical results closely aligned with experimental findings. Compared to the control slab, all strengthened configurations showed substantial improvements in cracking load, ultimate load, and energy absorption. Increasing the number of anchors led to higher load capacities. The cracking and failure loads varied depending on the strengthening method

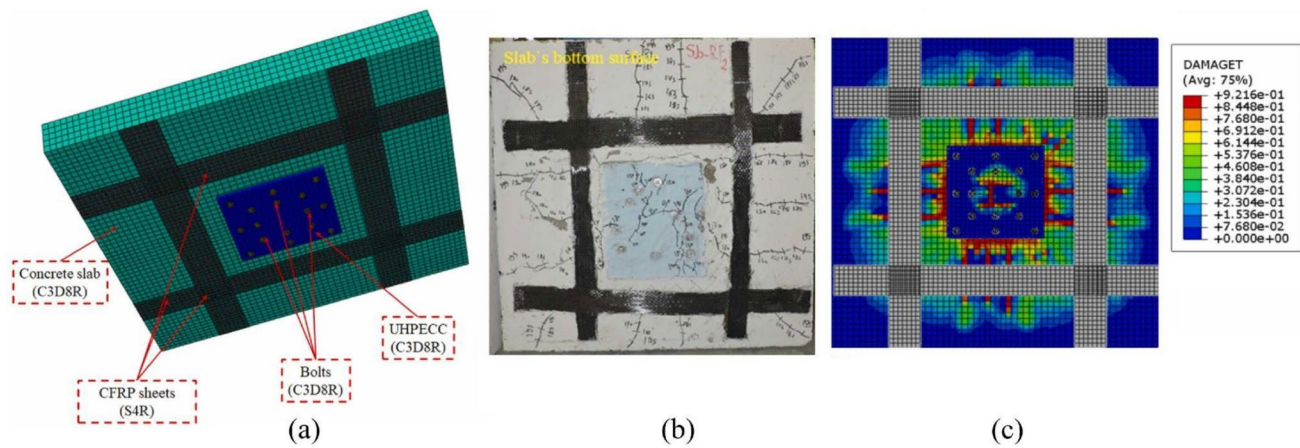


Fig. 12 Example of (a) a numerical model used for the strengthening technique employing CFRP sheets and the corresponding (b) experimental and (c) numerical crack patterns [Adapted from 257]

reinforcement does not alter the final geometry of the structure or significantly impact its total weight. Additionally, in contrast to alternative concretes, certain types of shear reinforcement do not require specific materials that may be difficult to source, and depending on the selected type, shear reinforcement can be straightforward and cost-effective.

The concept behind shear reinforcement is relatively simple: a stronger material, such as steel, is placed in the regions where principal shear cracks tend to form, ultimately creating the punching shear cone. This approach ensures that the slab retains significant strength even as the concrete develops cracks. Consequently, this method generally increases the load-bearing capacity and ductility of the structure [31, 42, 215, 263–267]. Literature and practical design exhibit various types of shear reinforcement used as an option for flat slabs, some of which are illustrated in Fig. 13.

Recent experimental studies have increasingly focused on alternative forms of shear reinforcement, such as shearheads [42, 102, 103], star-shaped steel plates [268], welded shear with chords and web [267], rectangular-shaped spiral stirrup arrangements [269], inverted T-shaped steel [71], grids [214], steel sheets [96, 270], hook reinforcement [271], bent-up bars [215, 272, 273], pyramids of swimmer bars [274], among others [215, 272, 275]. These alternatives differ from conventional solutions like stirrups and studs. Nevertheless, many recent investigations examine traditional shear reinforcement, particularly variations in bar positioning, diameter, and attachment methods [34, 189, 276–281]. Studies also explore the use of shear reinforcement under distinct loading conditions, such as lateral cyclic loads and seismic actions [84, 86, 93], asymmetric loading [29, 52], or shear reinforcement produced from materials other than steel [85, 88–90].

Additionally, post-installed shear reinforcement aims to enhance punching shear resistance after the slab has already

been loaded. Examples include shear bolts [31, 81, 97, 264, 265, 282–284] and concrete screws [285–287]. However, it is crucial that post-installed reinforcements are effectively anchored and that their installation does not damage existing flexural reinforcement within the structure.

In addition to improving load-bearing capacity and ductility, the insertion of shear reinforcement can alter the location of the punching shear cone and the slab's cracking pattern [64, 279, 288]. This behavior is influenced not just by the type of reinforcement used but by parameters such as positioning, distance from the column, reinforcement ratio, and anchorage, among others. From this perspective, punching shear failure may occur within the reinforced zone (failure crossing one or more rows of shear reinforcement), outside the reinforced zone (failure in the unreinforced region), or directly at the maximum punching shear capacity (failure within the compression zone near the column face) [288], as illustrated in Fig. 14.

Numerical analyses involving shear reinforcement present several challenges in modeling and result accuracy. First, it is necessary to determine whether the shear reinforcement will be modeled using truss, line, shell, or beam elements. This choice depends on the complexity of the model, the desired precision, and, importantly, the type of reinforcement employed. For studs, stirrups, bolts, and bent-up bars, it is common to use truss and line elements [42, 275, 283, 284] or beam elements [264, 266, 273]. Conversely, shell elements have been employed for steel sheets and inverted T-shaped steel reinforcements [71, 270]. Additionally, concrete can be modeled using plate elements [31, 289], a combination of 3D, joint, and plate elements [280], or solid 3D elements [42, 264–267, 270, 274, 279, 281, 283–286].

Another critical factor to consider in some cases is the bond between the steel and concrete in the reinforcement. While perfect bond assumptions may be adequate in some

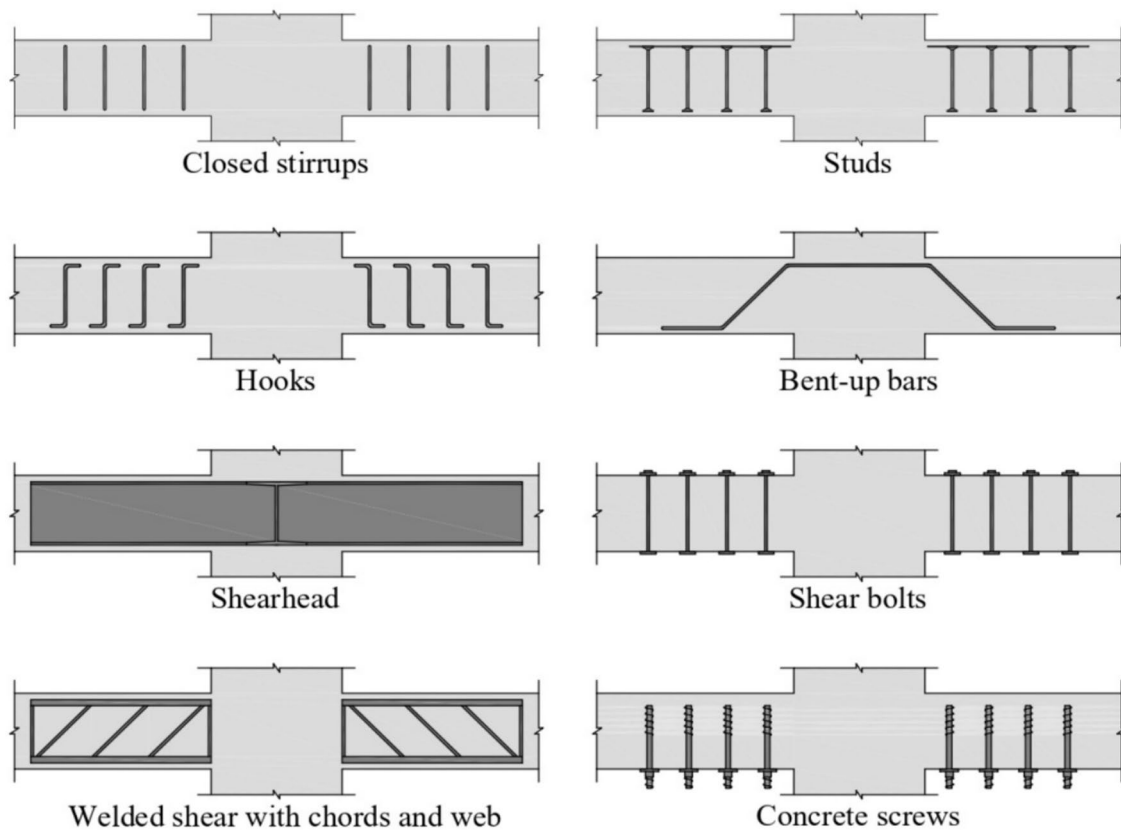


Fig. 13 Examples of shear reinforcement used in slab-column connections subjected to punching shear loads

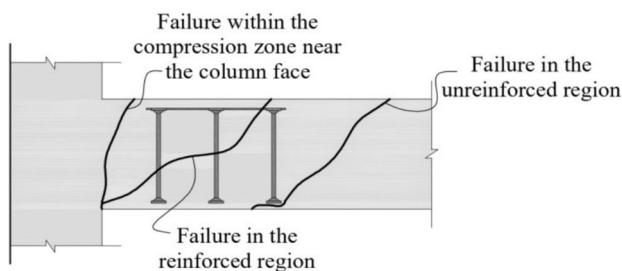


Fig. 14 Punching shear failure locations in slab-column connections with shear reinforcement

scenarios, Spiegl et al. [286] emphasize that, for concrete screws, for example, a bond-slip rule defining the steel–concrete interaction must be incorporated. This consideration increases processing time and introduces a new challenge: defining and implementing the governing law for reinforcement slip behavior.

Furthermore, another difficulty in modeling experimental data relates to the anchorage of reinforcement during testing, a feature that becomes more complex to simulate numerically. Experimentally, Oliveira et al. [290] and Lima et al. [291] highlight that the anchorage method of closed

stirrups, for example, can significantly influence the behavior. In numerical models, this issue can result in discrepancies with experimental data, particularly regarding the location of the principal shear crack. From the perspective of constitutive models, significant difficulties are generally not encountered, as most reinforcement options are steel-based, a material commonly used in flexural reinforcement. However, studies have investigated materials such as CFRP and GFRP [64, 212, 214, 225], which necessitate adjustments to the constitutive behavior.

Consequently, it is standard practice for numerical studies to validate their findings against experimental data, either from their experiments or the literature, often including a parametric analysis. Overall, the studies summarized in Table 6 achieved good agreement in validation results, particularly concerning failure load. However, the location of the shear crack did not always match experimental observations. Regarding the results, there is a consensus that shear reinforcement increases failure load and ductility. Still, direct comparisons of conclusions are challenging due to variations in reinforcement type, position, ratio, material, and other characteristics across studies.

The numerical modeling of shear reinforcement allows not just the assessment of their impact on the punching shear

Table 6 Numerical review of the punching shear in slabs with shear reinforcement

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Polak (2005) [59]	(Numerical analysis) FE Program	To numerically model shear bolts as shear reinforcement	The numerical modeling was developed using nine-node shell elements with varying numbers of layers. The reinforcement was modeled by modifying the properties of a concrete layer in the transverse direction, adding stiffness in that direction, and restricting crack opening in the transverse plane	The numerical model accurately represented experimental data regarding ultimate load and ductility. The presence of shear reinforcement contributed significantly to the increase in ultimate load capacity and ductility
Wörle (2014) [285]	(Numerical analysis) FE Program/ABAQUS	To evaluate the impact of post-installed concrete screws as shear reinforcement	An experimental and numerical study used screws of different diameters and installation methods. In all cases, the screws were arranged radially. The numerical model used 3D elements for the concrete and beam elements for the flexural reinforcement. The screws were modeled as elastoplastic materials with kinematic couplings connected with springs	The validation demonstrated good agreement with experimental results. The use of post-installed screws significantly increased the ultimate load capacity, mainly when epoxy adhesive was employed. Ensuring a proper connection between the slab's tension and compression zones with sufficient anchorage depth is imperative
Balomenos et al. (2015) [283]	(Numerical analysis) FE Program/ABAQUS	To investigate the impact of shear bolts as shear reinforcement	Numerical modeling was employed to validate experimental data and to perform a probabilistic analysis. Hexahedral 3D elements were used for the concrete, while truss elements represented the reinforcement, assuming perfect bond conditions. The shear bolts were arranged orthogonally	The results demonstrated that the numerical model successfully predicted experimental behavior regarding ultimate load and displacements. Shear bolts significantly increased the slab's ultimate load capacity and ductility
Baig et al. (2016) [264]	(Numerical analysis) FE Program/ANSYS	To study the impact of external steel shear bolts as shear reinforcement	Experimental and numerical analyses were conducted, comparing slabs with and without external steel shear bolts, while varying the concrete's compressive strength. The numerical simulation used 3D elements for the concrete and line elements for the flexural reinforcement. Shear reinforcement was positioned orthogonally and modeled with beam elements by removing concrete volumes	The results showed good agreement between experimental and numerical outcomes. External steel shear bolts significantly increased ultimate load capacity, ductility, and the critical failure perimeter while altering the punching shear failure mode to flexural failure. However, placing the shear reinforcement far from the column had minimal impact on the slab's behavior

Table 6 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Genikomsou and Polak (2016) [284]	(Numerical analysis) FE Program/ABAQUS	To examine the impact of shear bolts as shear reinforcement	The numerical modeling investigated various methods of modeling shear reinforcement, comparing results with experimental data. The concrete was modeled with solid 3D elements, while shear bolts were modeled with truss, shell, or solid. The study also analyzed the impact of the number of reinforcement layers on the structure's behavior	The numerical results were in close agreement with experimental data. The most accurate predictions for ultimate load and ductility were obtained when shear bolts were modeled as beam elements. All slabs failed in regions without shear reinforcement. The concrete behavior could be modeled similarly for slabs with and without shear reinforcement
Saleh et al. (2018) [265]	(Numerical analysis) FE Program/ATENA	To evaluate the impact of using post-installed steel bolts as shear reinforcement	Experimental and numerical analyses were performed. The numerical model employed tetrahedral 3D elements for the concrete and line elements for the flexural reinforcement. The shear bolts were modeled as cables embedded within the concrete elements. Following validation, a parametric analysis was performed	The use of shear bolts improved punching shear resistance and ductility. Their application shifted the failure mode from punching shear to flexural failure. Increasing the compressive strength of the concrete enhances load capacity, particularly in slabs without shear reinforcement. The numerical results closely approximated experimental data concerning ultimate load capacity
Papadopoulos et al. (2018) [275]	(Numerical analysis) FE Program/ABAQUS	To explore J-bars and stirrups as shear reinforcement in bridge decks	Bridge decks with shear reinforcement and extended column bars were experimentally tested and validated in numerical modeling. The column reinforcement was modeled with beam elements, while other reinforcements were modeled with truss elements	The results showed good agreement with experimental data, indicating punching shear of the concrete in the slab's top layer. The study concluded that shear reinforcement in the first row significantly influenced slab behavior
Al-Hamd et al. (2019) [42]	(Numerical analysis) FE Program/ABAQUS	To evaluate the impact of using shearheads as shear reinforcement	Eight-node 3D elements were used for the concrete, while truss elements represented the reinforcement. The study also examined the effects of fire on the structure. The shear reinforcement consisted of a steel grid fabricated from steel profiles, with two distinct shearhead configurations analyzed	Shearheads increased punching shear resistance by expanding the control perimeter and had an effect similar to increasing the column section. The employed shearheads demonstrated resistance to fire effects. Numerical models closely matched experimental data for all configurations analyzed

Table 6 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Hamoda and Hossain (2019) [215]	(Numerical analysis) FE Program/ABAQUS	To investigate and compare CFRP and steel shear reinforcement	The concrete was modeled with 3D elements, while truss elements represented the reinforcement. The numerical model was validated with experimental data from the authors and the literature. Calibration was performed by adjusting material parameters, followed by a parametric analysis using bottom bars, bent-up bars, vertical or inclined bars, and closed stirrups	The numerical results agreed well with the experimental results. The most effective approach to enhancing load capacity was the straight or vertical bars at locations intersecting the expected path of punching shear cracks. Both steel and CFRP shear reinforcements improved ductility and punching shear resistance
Al-Gasham et al. (2019) [270]	(Numerical analysis) FE Program/ABAQUS	This study assessed the impact of using steel sheets as shear reinforcement in slabs with spherical voids	Experimental and numerical analyses were conducted on slabs with spherical voids and shear reinforcement. The numerical model was validated, followed by a parametric study varying the number and thickness of steel sheets. Solid 3D elements were used for the concrete, truss elements for the reinforcement, and shell elements for the steel sheets, assuming perfect bonding between steel and concrete	The results closely matched experimental data. The use of steel sheets resulted in a more ductile failure mode. While spherical voids reduced punching shear capacity, combining them with steel sheets increased the ultimate load beyond that of a solid slab of similar dimensions. Increasing the thickness enhanced the steel sheets' activation
Yankelevsky et al. (2020) [273]	(Numerical analysis) FE Program/ANSYS AUTODYN	To evaluate bent-up bars as shear reinforcement in progressive collapse scenarios caused by punching shear	Numerical modeling aimed to validate experimental data related to impact loading. Once validated, the model was used to analyze progressive collapse scenarios. Among various analyses, the use of shear reinforcement was examined. Solid elements were used for the concrete, and beam elements were used for the reinforcement	The validation phase agreed well with the experimental data. In the progressive collapse analysis, shear reinforcement reduced maximum and permanent deflections. Increasing the number of bars prevented punching shear failure. The maximum deformation supported by the steel influenced the results, potentially avoiding total collapse
Navarro et al. (2020) [266]	(Numerical analysis) FE Program/ABAQUS	To assess the impact of bolts as shear reinforcement on punching shear resistance	The numerical modeling was validated against experimental data from the literature. Subsequently, a parametric analysis investigated the bolts' position, number, spacing, and diameter. Solid elements were used for the concrete, truss elements for flexural reinforcement, and beam elements for modeling the bolts	The numerical model agreed well with the experimental results. The parametric analysis concluded that increasing the number of bolts enhanced the ultimate load capacity. Bolts positioned radially achieved the best results, increasing both ultimate load and ductility

Table 6 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Salama et al. (2020) [64]	(Numerical analysis) FE Program/ ANSYS	To evaluate the influence of GFRP closed stirrups on punching shear	The numerical modeling validated experimental data. Hexahedral 8-node elements were used for the concrete, and bar elements were used for the reinforcement. Flexural and shear reinforcements were made of GFRP and modeled as a linear elastic material	The results showed good agreement with experimental data. GFRP stirrups increased the ultimate load capacity, reduced crack widths, and improved the distribution of shear stresses
Lewiński and Więch (2020) [289]	(Numerical analysis) FE Program	To analyze the impact of double-headed studs as shear reinforcement on punching shear resistance	The numerical modeling was validated using the authors' experimental data. Triangular shell elements were used for the concrete. The study focused on the implementation and modeling of constitutive models. The study also provided several comparisons with other numerical studies in the literature	The results showed satisfactory agreement with the experimental data regarding ultimate load capacity. The use of planar elements reduced computational costs
Said et al. (2020) [279]	(Numerical analysis) FE Program/ ANSYS	To investigate the use of studs as shear reinforcement in slabs	The numerical modeling was validated with the authors' experimental data. Slabs with and without studs were analyzed, varying their position (orthogonal and radial) and reinforcement ratio. Solid elements were used for the concrete, and bar elements were employed for the reinforcement	The results closely matched the experimental data. Studs increased the punching shear perimeter, ultimate load capacity, and ductility. This reinforcement also resulted in less abrupt failure and altered the cracking pattern. Increasing the flexural reinforcement ratio further improved ductility
Jafarian et al. (2020) [214]	(Numerical analysis) FE Program/ ABAQUS	To analyze slabs reinforced with CFRP and GFRP grids as shear reinforcement	The model was validated using experimental data from the literature. The FRP grids were modeled as linear elastic materials. Solid 3D and truss elements were used for the concrete and the reinforcement bars, respectively. After validation, a parametric analysis investigated the shear reinforcement's spacing, position, and material	The results agreed well with the experimental data. The use of grids increased the punching shear resistance, with a more pronounced improvement when CFRP was used compared to GFRP. Radial positioning of the grids yielded higher load capacities than orthogonal positioning

Table 6 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Said et al. (2020) [212]	(Numerical analysis) FE Program/ANSYS	To investigate lightweight concrete slabs reinforced with steel, GFRP bars, or bolts as shear reinforcement	Solid 3D and link elements were used for the concrete and the reinforcement, respectively. The post-peak tensile behavior was disregarded; thus, the tensile stress was set to zero after reaching its maximum strength. GFRP was modeled as a linear elastic material. Various shear reinforcement configurations were tested, varying the material, fixation method, and position. The numerical model was validated using the authors' experimental data	The numerical results showed excellent agreement with the experimental data. Shear reinforcement significantly improved load capacity, with steel bars showing the most substantial increase compared to GFRP. Bolts fixed to the slabs with steel plates altered the punching shear failure mode to a flexural punching failure
Najmi et al. (2021) [274]	(Numerical analysis) FE Program/ABAQUS	To evaluate the pyramids of swimmer bars as shear reinforcement for punching shear resistance	The numerical model was validated using the authors' experimental data, which included slabs with various dimensions, thicknesses, and swimmer bar diameters. Solid 3D elements were used for the concrete, and truss elements were employed for the reinforcement. Once validated, the model was used for parametric analysis, altering the number and position of the swimmer bars	The validation showed satisfactory results. The use of swimmer bars significantly increased the ultimate load capacity and ductility. The first pyramid of the swimmer bar assembly contributed most to the punching shear resistance. Larger diameter corner bars further improved the capacity
Spiegl et al. (2021) [286]	(Numerical analysis) FE Program/ATENA	To evaluate the impact of using screws as shear reinforcement on punching shear behavior	The numerical modeling was validated against experimental data from the literature. The study examined slabs with varying numbers, diameters, and positions of screws. Hexahedral elements were used for the concrete, while truss elements represented the flexural reinforcement. The screws were modeled as truss elements with a bond-slip relationship between the screw and the concrete	The numerical results closely matched the experimental data, particularly regarding punching shear capacity. The presented curves indicated slightly higher cracking loads and stiffness in the numerical model compared to experimental results. The simulated shear crack patterns were consistent with those observed experimentally

Table 6 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Setiawan et al. (2021) [280]	(Numerical analysis) FE Program/ADAPTIC and ATENA	To analyze different modeling approaches for slabs with headed studs as shear reinforcement	The modeling was validated using experimental data from the literature. Two approaches were developed: one employing 3D solid elements and the other combining 3D joint elements with nonlinear shell elements. The reinforcement was modeled using line elements. All slabs were equipped with headed stud shear reinforcement, with variations in reinforcement ratios	Both modeling approaches agreed well with experimental results regarding punching shear capacity. The use of solid elements realistically simulates internal crack patterns along radial saw-cuts and the influence of the shear reinforcement ratio
Jia and Chiang (2022) [34]	(Numerical analysis) FE Program/ABAQUS	To analyze slabs with shear studs subjected to unbalanced loading	Experimental data from the literature validated the numerical model. The study investigated slabs with varying numbers and diameters of shear studs and different moment-to-shear ratios. Hexahedral elements were used for the concrete, while truss elements represented the reinforcement	The numerical model demonstrated good agreement with experimental results. The number of shear studs influenced punching shear resistance, while the diameter of the bars had a less pronounced effect
El-Kassas et al. (2022) [189]	(Numerical analysis) FE Program/ANSYS	To investigate FHSSRC slabs with and without closed stirrups as shear reinforcement	Experimental and numerical analyses were conducted on slabs with variations in fiber type (basalt or steel) and percentage, as well as the use of shear reinforcement. The modeling utilized 3D solid elements for the concrete and link elements for the reinforcement. A parametric analysis was also performed	Excellent agreement was achieved between the numerical and experimental punching shear capacity results during validation. For slabs without fibers, shear reinforcement significantly improved load capacity. However, fibers proved more efficient than shear reinforcement in enhancing punching shear resistance
Xue et al. (2022) [71]	(Numerical analysis) FE Program/ABAQUS	To investigate slabs supported on L-shaped columns reinforced with capitals or inverted T-shaped steel for shear enhancement	The study conducted both experimental and numerical analyses. The numerical model utilized 3D solid elements for concrete, truss elements for steel reinforcement, and shell elements for T-shaped steel. The impact of slab thickness, the presence of capitals, and the use of inverted T-shaped steel were examined. Following calibration and validation, the model was applied to a parametric analysis	The numerical validation demonstrated excellent agreement with experimental results. Capitals, increase in slab thickness, or inverted T-shaped steel significantly improved punching shear capacity. Moreover, increasing the width of the inverted T-shaped steel web altered the slab's failure mode from punching shear to flexural failure

Table 6 (continued)

Study	Analysis method	Objectives and/or scope	Main points	Main findings
Elsamak et al. (2023) [156]	(Numerical analysis) FE Program/ABAQUS	To investigate slabs reinforced with high-strength steel bolts fixed on steel plates and steel tubes filled with SHCC	The study examined slabs with and without shear reinforcement using different concrete types: ordinary concrete, HSC, SHCC, and UHPFRC. The numerical model employed 3D elements for concrete and shear reinforcements and truss elements for longitudinal reinforcements	The numerical results agreed well with the experimental data. Steel assemblies increased the punching shear capacity of slabs with regular concrete but did not eliminate punching shear failure. Additionally, the application of steel assemblies altered the location of the failure cone, moving the failure surface away from the column
Jiang and Shi (2024) [267]	(Numerical analysis) FE Program/ABAQUS	To analyze slabs with welded shear reinforcement	The numerical model was validated against experimental data from the literature. The study focused on slabs with orthogonally arranged welded shear reinforcement. The modeling used 3D solid elements for concrete, beam elements for welded shear reinforcement, and truss elements for flexural reinforcement. After validation, the model was employed in a parametric analysis	The numerical results agreed well with the experimental data. Internal welded shear reinforcement significantly increased punching shear capacity and ductility. Furthermore, increasing the web shear reinforcement ratio enhanced the peak strength and ultimate displacement. However, the chords of welded shear reinforcement had minimal influence on punching shear
Ewees et al. (2024) [281]	(Numerical analysis) FE Program/ANSYS	To assess the impact of stirrups and tensile and compressive reinforcement on punching shear	The study included both experimental and numerical analyses. The numerical model utilized 3D solid elements for concrete and link elements for reinforcements. A parametric analysis was also conducted	The results agreed well with the experimental data for ultimate load and final displacements. The use of stirrups and an increase in concrete compressive strength enhanced ultimate load capacity and energy absorption, with a slight effect on the crack pattern

strength of the connection but also an understanding of how much they are stressed, their deformations, and their influence on the crack pattern and the formation of the punching shear cone. In this regard, it is worth noting that there are various ways to model reinforcement, and the choice depends on the type of reinforcement, the adopted numerical model, the available finite elements, and the desired level of accuracy. Regarding the finite elements, studies such as those by Jia and Chiang [34] and Genikomsou and Polak [284] investigated the differences in the response of slabs with shear reinforcement, such as shear bolts and studs, using different types of elements, including beam, truss, shell, and 3D solid elements. Conversely, as shown in Fig. 15, studies like those by Al-Gasham et al. [270] evaluated the stress levels in shear reinforcement, such as shearheads and steel sheets, respectively. In this context, the potential of numerical tools in analyzing shear reinforcement becomes evident. They enable the identification of stresses and deformations at any desired point, even in locations that would be challenging to measure experimentally.

Finally, it is evident that shear reinforcement as an alternative to enhance punching shear capacity and ductility is an efficient and well-established measure. Numerical simulations introduce some challenges when modeling slabs with such shear reinforcement. However, the considerations presented in previously published works generally allow for good approximations of available experimental data. A recent shift in numerical studies on shear reinforcement appears to be toward alternatives to conventional

options (e.g. studs and stirrups). These studies focus not only on modeling aspects but also on the structural effects of employing these new elements. Moreover, a significant number of studies have investigated the use of post-installed shear bolts as a means of retrofitting existing structures and improving their resistance capacity. Given the emergence of new materials in the construction sector, alongside evolving demands for flat slab construction, research on shear reinforcement appears promising. This is especially true for cases of punching shear that deviate from symmetric loading conditions in slabs made of conventional reinforced concrete.

2.8 Other Cases of Punching Shear

Although there are various relevant cases of punching shear, most studies focus on the scenarios discussed in the previous sections, which were detailed accordingly. In the subsections below, other cases of punching shear will be briefly presented to provide a general understanding of how numerical research in this field has developed without delving deeply into each study. Topics include punching shear in alternative systems such as biaxial hollow-core slabs, hollow slabs, waffle slabs, one-way slabs, sandwich slabs, and mixed (steel–concrete) connections. The impact of drop panels, studies conducted on bridge decks, and punching shear under different loading conditions, such as explosions, impacts, and high temperatures, will also be addressed. Finally, the study of progressive collapse in flat slabs, a

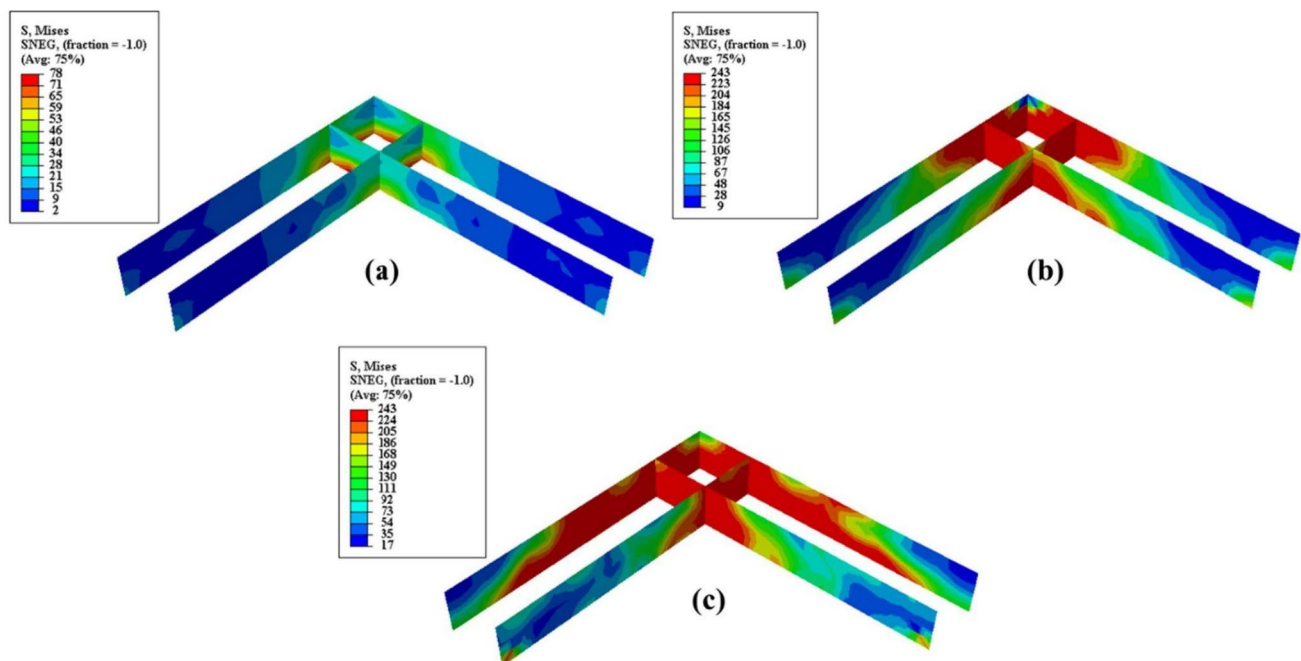


Fig. 15 Von Mises stress distribution in steel sheets used as shear reinforcement [270]

phenomenon typically associated with punching shear failure and post-punching behavior in slab-column connections, will be discussed.

2.8.1 Punching Shear in Hollow, Biaxial Hollow, and Waffle Slabs

While solid flat slabs offer advantages over conventional slab systems supported by beams, their geometry can be modified to incorporate other desirable characteristics. For instance, to reduce the self-weight of the structure, geometric voids can be introduced, either unidirectionally (e.g. hollow-core slabs) or bidirectionally (e.g. biaxial hollow slabs). These voids can be created using removable plastic forms (as in waffle slabs) or forms that remain embedded within the structure (as in biaxial hollow slabs). Moreover, incorporating drop panels or capitals can significantly enhance punching shear capacity. Lastly, it is essential to analyze one-way slabs, where punching shear must be evaluated in conjunction with unidirectional shear behavior.

In biaxial hollow slabs, the introduction of plastic voids reduces the volume of concrete, the slab's self-weight, and, consequently, the loads applied to the foundations. Experimental studies have sought to determine how the positioning of voids influences structural behavior, as voids placed near columns significantly reduce punching shear capacity, especially when they intersect the failure punching shear cone [278, 292]. These studies have also evaluated the use of shear reinforcement [270, 271, 278], precast slabs [278], openings [128, 293], reinforcement [128], asymmetric loading [45], and various void shapes, such as spheres [45, 128, 278, 293–295], ellipses [271], cuboids [294], and others [292, 296, 297].

Numerically, modeling these structures' geometry and finite element mesh is more complex than solid flat slabs.

Some types of voids require highly refined meshes with elements of varying shapes. Numerical studies have focused on determining the optimal distance of voids from columns to minimize the impact on punching shear capacity [37, 296] or quantifying the load reduction caused by voids in critical regions [36, 270, 295]. These studies typically include a validation phase [36, 37, 45, 128, 270, 295, 296], often complemented by parametric analyses [37, 296] or comparisons with analytical predictions [295, 296]. Key topics investigated include shear reinforcement [36, 37, 270], void dimensions and positioning [37, 45, 128, 295, 296], and asymmetric punching shear [36, 37, 45]. Notably, all cited studies employed 3D hexahedral or tetrahedral elements, modeling concrete as nonlinear material.

In this context, Schimanowski et al. [37] conducted a numerical analysis using 3D tetrahedral solid elements to evaluate slabs with spherical voids. As shown in Fig. 16, the authors concluded that the distance of the spheres from the column reduces their interference, both in terms of cracking and regarding the ultimate load and displacements of the slab. Consequently, when the spheres are positioned farther away, as in Case C, the structural behavior is similar to that of a solid slab. However, when the spheres are closer to the column, as in Case A, the punching shear cone forms within the first layer of spheres, significantly reducing the punching shear strength of the connection.

In hollow-core slabs, voids are positioned continuously in one direction, making punching shear failures uncommon, as these slabs are not typically supported directly on columns. However, punching shear failures may still occur under concentrated loads. Han et al. [298] experimentally investigated a novel prefabricated concrete hollow-core composite floor slab with a steel bar truss skeleton. Similarly, McGinnis et al. [138] examined traditional hollow-core slabs through experimental and numerical analyses, focusing

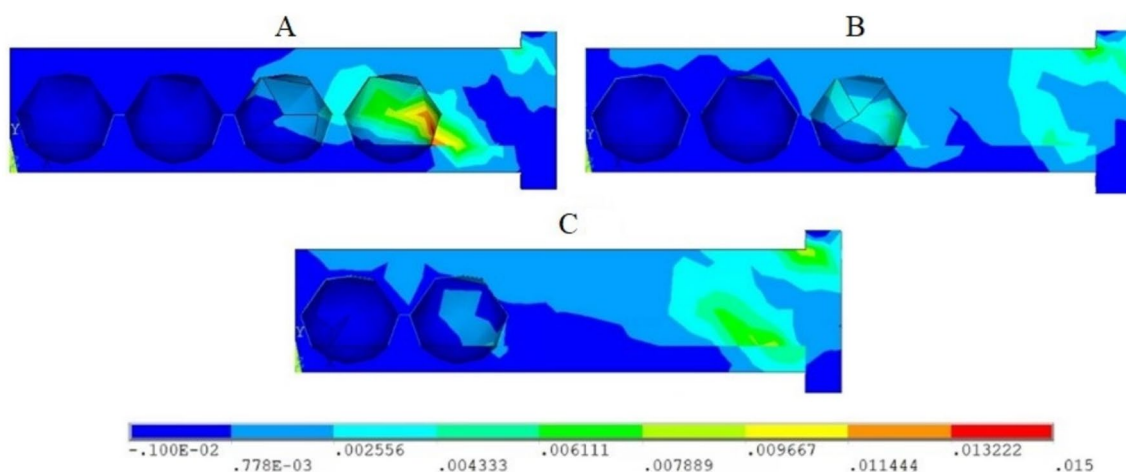


Fig. 16 Principal tensile strain in the concrete of a biaxial hollow core slab [37]

on punching shear, flexure, and shear, and validating the numerical model.

Waffle slabs represent another system that reduces self-weight by using voids. These voids are created by removing forms after the concrete hardens, leaving an underside different from that of solid flat slabs. Like biaxial hollow slabs, waffle slabs may exhibit reduced punching shear resistance because the potential failure surface extends into the waffle section. Research to date has focused on experimental analyses, examining solid regions around columns [40, 56, 299, 300], shear reinforcement [299], and asymmetric punching shear [40, 56, 301].

2.8.2 Slabs with Drop Panels or Capitals and One-Way Slabs

In some cases, it is desirable for slab-column connections to exhibit higher punching shear resistance, which is possible through shear reinforcement, additional reinforcement, alternative reinforced concrete, or by increasing the flexural reinforcement ratio, concrete compressive strength, column dimensions, or slab thickness. While increasing slab thickness is an efficient solution for enhancing load capacity, it significantly increases self-weight and reduces the floor-to-ceiling height of a building. A more efficient alternative is the use of drop panels (Fig. 17a) or capitals (Fig. 17b), which generally consist of localized increases in slab thickness around columns [48, 61, 258, 302–306]. These can also be implemented as hidden panels made of ultra-high-performance fiber-reinforced concrete (UHPFRC) [179] (Fig. 17c) or internal steel drop panels [69, 307] (Fig. 17d).

In general, drop panels or capitals significantly enhance the load capacity of slab-column connections, affecting post-punching behavior and potentially altering the failure cone

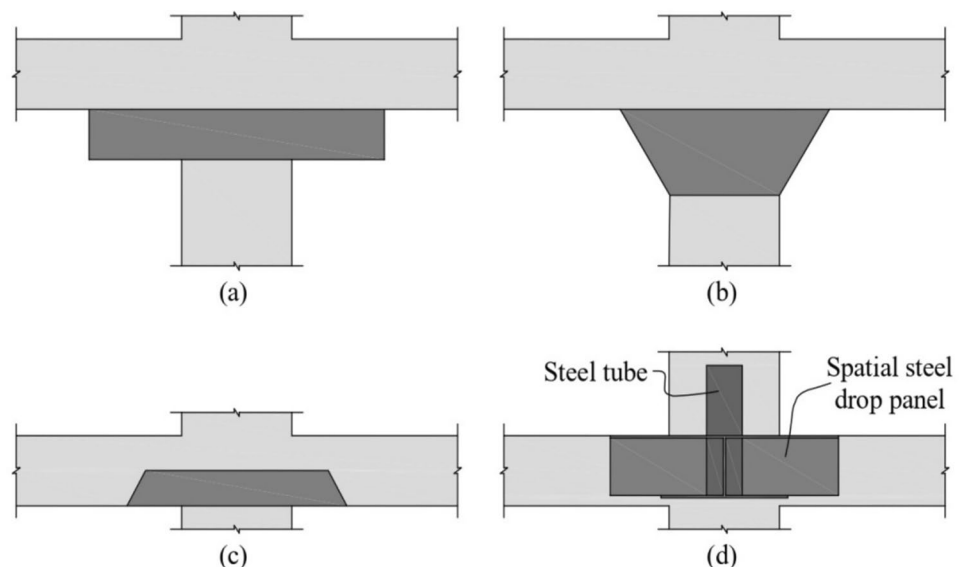
location. Experimental studies on drop panels and capitals typically examine their dimensions [302, 303, 307], interactions with columns of different geometries [69], steel columns [69, 307], progressive collapse [304], and shear reinforcement [179, 307]. Numerical simulations facilitate the modeling of this alternative by altering slab geometry and/or material properties. These simulations often use 3D solid elements [48, 258, 305] or layered shell elements [61, 306] to model the connections, with objectives including model validation [48, 61, 258], parametric analyses [258, 305], and comparisons with analytical predictions [48, 306]. Drop panels have been investigated with progressive collapse [48, 305], asymmetric punching shear [48], and alternative concrete [258].

Finally, another failure mode associated with slabs of distinct geometries is punching shear in one-way slabs. Sousa et al. [308] noted that these slabs are found in bridge decks, industrial floors, and residential buildings. When concentrated loads are applied away from supports, these structures may fail due to unidirectional shear, such as beams or punching shear. Experimental analyses [308–311] have focused on detailing cracking patterns, displacements, and deformations under varying load application points, comparing punching shear resistance with analytical predictions. Numerical studies have aimed to develop validated models capable of accurately describing the behavior of these structures [308].

2.8.3 Punching Shear in Prestressed Flat Slabs

Prestressed concrete has become a staple in civil and industrial construction, finding application in buildings, bridges, and viaducts. Generally, prestressing in concrete structures reduces deflections and the propagation of cracks. In the case of punching shear, prestressing in

Fig. 17 Slabs with (a) drop panel, (b) capital, (c) UHPFRC hidden capital, and (d) internal steel drop panel



flat slabs significantly enhances load-bearing capacity, controls deflections, and limits cracking in serviceability limit states while reducing slab thickness. The efficacy of prestressing in concrete structures is attributed to the generation of in-plane compressive stresses, the decompression moments, and the vertical component of inclined tendons near the column [312, 313]. Although experimental studies on prestressed flat slabs are limited compared to the number of experimental tests conducted on reinforced concrete slabs, existing research has demonstrated that prestressing can significantly improve load-bearing capacity [55, 69, 95, 138, 172, 191, 192, 313–316].

The study of prestressed slabs under punching shear loads has also been limited in numerical modeling. This is partly due to the scarcity of experimental data and partly due to the complexity of modeling the interaction between concrete and prestressing tendons. In the existing studies, 3D solid elements have been used to model concrete, while beam elements [316–318], bar elements [191, 192], or truss elements [138, 312] have been used to model tendons. These analyses require incorporating specific bond relationships between steel and concrete [312, 316] and implementing techniques to apply prestressing. While most studies applied prestressing through tendons with initial tension [69, 138, 172, 312, 316–318], Amir et al. [315] employed prestressing via stresses in end steel beams on a bridge deck, and Kitano et al. [192] applied volume strain to prestressed concrete steel.

Key variables analyzed in simulations include tendon position [312], tendon eccentricity [69, 312], applied prestressing force [172, 192, 317], and investigations of prestressed systems in bridges [315], hollow-core slabs [138], alternative concrete materials [138, 172, 192], impact loading scenarios [317, 318], and connections involving steel columns or steel drop panels [69]. These studies typically incorporate a numerical validation phase [69, 172, 191, 192, 312, 315, 316], sometimes followed by parametric analyses and comparisons with analytical predictions [312, 316]. Among the primary conclusions, it has been noted that prestressing increases load-bearing capacity due to in-plane compressive forces, tendon eccentricities, and deviation forces [172, 192, 312, 316, 317]. Additionally, Amir et al. [315] determined that the level of transverse prestressing also affects cracking behavior; Chen et al. [69] further observed that the application of prestressing effectively increases the cracking load of specimens, delays crack development on slab surfaces, and reduce crack width; Jiao et al. [316] investigated an increase in post-punching resistance when employing prestressing.

2.8.4 Punching Shear in Bridge Decks

Although the study of punching shear in bridge decks is less extensive than in flat slabs, this type of failure is a significant concern due to the application of concentrated loads from vehicle tires. The most common focus in numerical analyses has been on bridge decks supported by steel girders [216, 315, 319–321]. However, studies on decks with concrete beams [322, 323], cantilever bridge decks [324], slab-column joints of slab bridges [275], and skew composite steel–concrete bridges [325] have also been developed. Unlike simulations in flat slabs, where supports are often modeled as rigid plates, bridge deck studies typically involve the entire modeling of supporting girders to account for support effects. Regarding deck modeling, studies have employed various approaches, including 3D solid elements [216, 275, 315, 319], shell elements [321, 322, 324, 325], layered shell elements [323], and plane stress elements [322].

Generally, all studies include a numerical validation phase. Many also present parametric analyses [216, 324, 325] and analytical studies [319, 321, 322, 324, 325], comparing the obtained loads with code predictions or proposing new formulations. Setiawan et al. [324] additionally provide a literature review, identifying open scientific questions. These studies encompass various types of bridges, differing in materials, geometry, load applications, and modeling parameters. Consequently, no consensus can be drawn about punching shear in bridge decks. Among the frequently explored topics is the use of FRP bars [216, 319], aimed at mitigating environmental impacts (such as corrosion) on reinforcements; the study of prestressed bridges [315, 323, 326]; and the effect of compressive membrane action [320, 322, 323, 326]. Belletti et al. [323] state that it is widely recognized that compressive membrane action, also referred to as arching action, increases both the bending and punching shear capacities of reinforced concrete structures. To observe compressive membrane action, the slab or deck must be subjected to some form of horizontal translational restraint, and the net tensile strain along a longitudinal fiber must be non-zero if there is no horizontal restraint [322]. Experimental studies on this topic are limited due to the high costs of testing full-scale bridge decks. The most addressed experimental topics include punching shear in decks with alternative reinforced concrete [220, 327, 328], prestressing [329, 330], and fatigue analysis [327, 329, 330].

2.8.5 Punching Shear in Slabs with Steel Tubular Columns and Sandwich Slabs

The most common and extensively studied occurrence of punching shear concerns failures in flat reinforced concrete slabs. Some alternatives that modify the original

characteristics of the structure can be adopted to address this issue. One such option involves steel tubular columns, which may be filled or unfilled with concrete, characterizing the slab-column connection as a hybrid system or mixed structure. Steel tubular columns offer many advantages over reinforced concrete columns, including rapid construction, a smaller footprint, and superior structural performance [66]. Furthermore, Yan and Wang [331] and Luu et al. [332] highlight advantages in terms of structural efficiency, fire resistance, seismic performance, load-bearing capacity, stiffness, ductility, and energy absorption. However, the smooth finish of the external surface of steel tubes can reduce punching shear resistance due to the lack of adhesion between the steel column and the concrete slab. This issue necessitates mechanical systems, typically consisting of a steel insert welded to the column and integrated into the flat slab as shearheads [91, 102, 103, 331, 333–336]. Additionally, other solutions include welded shear studs [337], welded shear studs combined with steel plates and bent-up rebars [66, 67], and horizontal bearing plates with vertical ribs [332].

Recent analyses aim to develop mechanical systems for the ends of steel tubular columns while also evaluating specific geometric and loading conditions, such as corner connections [66], lateral cyclic loading [91, 102, 103], or prestressed slabs [332]. Numerical studies typically involve validation stages [66, 67, 331, 333, 334, 337], followed by parametric analyses [67, 331, 333, 337] and analytical studies [67, 103, 334]. The most common modeling approach employs 3D solid elements for the entire structure [66, 67, 333, 337] or a combination of 3D elements for the slab and shell elements for the column [102, 103, 331, 333]. Generally, these studies concluded that the flat slab–steel tube shear connection system behaves as an enlarged column in terms of punching shear resistance [66, 67].

Another type of composite structure, less explored in terms of punching shear, involves I-shaped steel columns connected to reinforced concrete slabs. Hamoda et al. [150] investigated these connections, employing ECC and UHPECC at the slab-column interface to improve the punching shear resistance of the connections, leveraging their high strain-hardening performance and superior tensile and compressive strengths. The authors conducted experimental and numerical studies, demonstrating good agreement between the results.

Another structural system susceptible to punching shear failure is the sandwich slab system, consisting of a concrete layer combined with one or two steel plates, referred to as steel–concrete (SC) or steel–concrete–steel (SCS) sandwich slabs, respectively. In such systems, punching shear failure arises from localized loads, leading to significant deformation of the steel plates and typical cracking in the concrete, which assumes a truncated cone shape. In this context, the steel plate can prevent the propagation of concrete cracks.

According to Wang et al. [338], this system exhibits higher ductility due to the steel layers, a property enhanced by shear connectors that transfer forces across interfaces, such as tie bars and headed studs. In sandwich systems, the formwork, composed of steel plates and shear connectors, is prefabricated at the factory, eliminating the need for demolding before concrete casting [339]. This method improves construction quality and efficiency, albeit at a significantly higher cost than conventional flat slabs [338].

Compared to traditional reinforced concrete structures, SCS composite structures demonstrate excellent mechanical performance under static, seismic, impact, or blast loads [339]. Applications include power containment structures, marine oil platforms, protective layers for submarine tunnels, ice-resistant walls for offshore platforms, and other major engineering facilities exposed to harsh environments as external protective layers [338, 339]. Numerical analyses of these systems commonly employ 3D finite elements, incorporating contact elements between the steel plates and the concrete layer [338–342]. Studies have evaluated SC [341, 342] and SCS [338–340] sandwich slabs with connectors such as headed studs [341, 342] or a combination of headed studs and tie bars [338, 339]. All studies included a numerical validation stage, with some extending into parametric analyses [338, 339, 342]. Variations explored in these studies include the use of alternative concrete types, such as UHPC and SFRC [338–340], the behavior of sandwich slabs under impact loads [338], and the effects of varying the shapes of loading heads [340].

2.8.6 Punching Shear in Slabs Subjected to High Temperatures, Explosions, and Impacts

The study of punching shear in slabs subjected to adverse conditions, such as explosions, high temperatures, and fire, has been sparsely explored in the literature, particularly in numerical simulations. For high temperatures and fire scenarios, experimental studies have aimed to identify how different materials behave and how these behaviors influence the slab's response [343–345]. In numerical studies, Al-Hamd et al. [42] investigated the use of shearheads in slabs exposed to fire, concluding that the structural behavior can be adequately modeled using 3D finite elements while accounting for temperature variations in the model. Similarly, Zhang et al. [346] observed that the numerical model exhibited premature punching shear failure when high temperature was considered. Moreover, Zhang et al. [213] concluded that exposure to high temperatures significantly degrades the impact resistance of GFRP-reinforced concrete slabs, leading to more severe punching shear failures. In the case of explosions, resulting from terrorist attacks or accidental detonations, some studies have sought to understand how this type of loading affects

structural behavior and whether it can induce punching shear failure due to the application of concentrated dynamic loads [347–349].

Conversely, the study of slabs subjected to impact scenarios—a subset of dynamic punching shear—has gained significant attention in recent years. However, the number of studies involving dynamic analyses remains significantly lower than in static cases due to the complexity of solving such problems [350]. In these cases, punching shear may not result directly from stress concentrations around the column but rather from failure induced by a concentrated impact load. In this context, Yankelevski et al. [273] point out that punching shear is the primary failure mode of slabs subjected to concentrated loads, and these slabs exhibit lower deformations at failure compared to static cases. Consequently, studies in this field simulate impact loads that may result from traffic collisions, falling rocks, landslides, floods, tsunamis, aircraft crashes, intentional aggression during service life, or airborne debris during cyclonic events [338, 351, 352].

Goswami et al. [351] conducted an analytical study with a comprehensive literature review and classified impacts based on the rigidity of the impactor as either hard or soft impacts and based on velocity as low-velocity (1–10 m/s), medium-velocity (10–100 m/s), or high-velocity (100–1000 m/s) impacts. In this scenario, it is worth noting that low- and medium-velocity impact tests are generally conducted using a drop hammer. In contrast, materials subjected to high-velocity loading are typically tested using the Hopkinson bar test [247]. Furthermore, the response of a slab subjected to impact can be governed by either global response, local response, or a combination of both. When the structural behavior is characterized by a global response, energy dissipation mainly occurs through overall member deformation, and flexural failure may be observed on the concrete's backside due to bending cracks [318, 351]. In contrast, when local failure dominates, the local response primarily governs the slab behavior, and energy dissipation is restricted to damage and deformations around the impacted area, leading to annular shear cracks. Wang et al. [318] mention that various failure modes can be observed in this scenario,

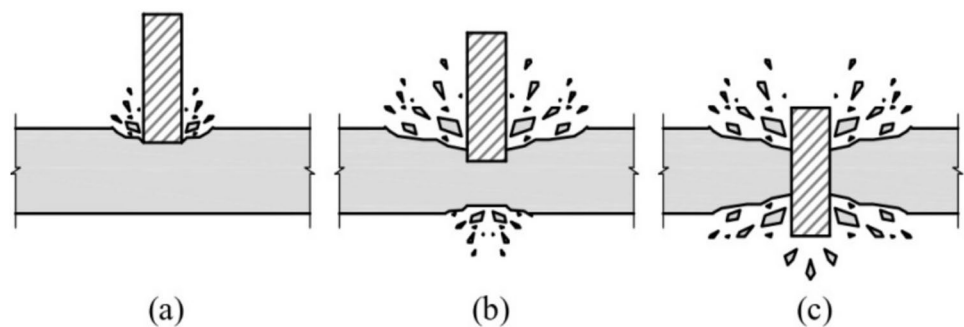
including penetration, spalling, scabbing, and perforation, as illustrated in Fig. 18.

Thus, in the case of a hard impact, the impactor behaves as a rigid body, and its energy dissipation is negligible. Conversely, the impactor undergoes significant deformation in a soft impact, dissipating considerable impact energy [351]. Regarding low-velocity impacts, slabs exhibit both global and localized responses, a phenomenon that has been increasingly studied due to its occurrence in reinforced concrete structures. Afshani and Rofooei [200] highlight several effective strategies for controlling slab deflection, minimizing crack formation, and mitigating damage, including reducing reinforcement spacing, increasing slab thickness, using shear stirrups, employing high-strength or fiber-reinforced concrete, retrofitting with FRP sheets, and implementing prestressed concrete.

Several factors influence the response of slabs under impact loading, such as the mechanical properties of constituent materials, impact energy, contact area, reinforcement ratio, boundary conditions, and mesh sensitivity [351, 353]. In this context, it is evident that material behavior is significantly affected in punching shear scenarios under impact loading, requiring experimental, analytical, and numerical studies to account for strain-rate effects. The mechanical properties of concrete and steel under high-rate loading conditions, such as impact and blast, differ significantly from those under quasi-static loading, with higher strain rates leading to increased compressive and tensile strength [247, 352].

This phenomenon, known as the strain-rate effect, has been extensively studied in the literature and must be incorporated into numerical analyses involving impact loading. This can be achieved using analytical formulations from design codes and existing research, which correlate static properties, such as tensile and compressive strength and elastic modulus, with their dynamic behavior. These formulations typically consider the evolution of strain rate during loading, allowing for real-time updates of material properties. Advanced software capable of solving nonlinear dynamic problems is commonly used for numerical investigations of impact scenarios. These simulations typically

Fig. 18 Punching shear damage caused by impact: (a) penetration and spalling, (b) scabbing, and (c) perforation



employ 3D solid elements, which offer constitutive models that accurately capture plastic behavior, material damage, and strain-rate effects. Contact elements are also frequently included in modeling interfaces between the impactor and the slab.

In this context, studies have sought to validate numerical simulations using data from drop-weight tests [200, 213, 247, 318, 338], pendulum impact tests [143, 144], or short-duration impulsive loads applied to entire slabs [354]. Other research efforts have investigated how impact velocity influences punching shear behavior [355–357] or have analyzed missile impacts on flat slabs and plates in greater detail [317, 358–362]. Numerical analyses predominantly rely on 3D models [200, 213, 318, 338, 351, 354–356], requiring careful consideration of key aspects such as dynamic increase factors for material properties, inertial effects, and stress wave propagation. Typically, these studies include a validation phase, followed by parametric analyses [143, 144, 213, 338, 355, 356] or the development of failure load prediction models [143, 351].

Recent research has explored how different parameters influence slab response under impact, including reinforcement ratio, concrete properties, shear reinforcement, and retrofitting techniques [232, 350, 353, 363–365]. More recent numerical studies on punching shear under impact have focused on evaluating this loading condition in slabs made of alternative materials [143, 144, 200], sandwich slabs [366], and the effect of impactor geometry on structural behavior [352]. Ultimately, findings indicate that results strongly depend on accurately considering strain-rate effects, inertial forces, and key modeling aspects such as element meshing and boundary conditions [134, 353].

Additionally, punching shear failure may result from slab-column connection failures during impulse pressure events [354] or slab-on-slab impacts [273, 305], which are typical actions in progressive collapse scenarios, further elaborated in the next section. It is important to note that in impact loading, a high-intensity force is applied over a short period, and the impacted slab experiences interaction pressure from the impacting slabs, which depends on the masses and velocities of the slabs. In contrast, during impulsive pressure events, the pressure–time history is predefined, applied over the slab area, and independent of slab properties [354].

2.8.7 Post-Punching Shear Behavior and Progressive Collapse in Flat Slabs

As previously discussed, punching shear is a type of failure that can result in the partial or complete detachment of the slab, making the analysis of post-punching behavior essential. In this context, when localized failure occurs, subsequent punching shear failure may initiate at adjacent joints on the same floor due to load redistribution, potentially

leading to large-scale failure propagation, which can trigger progressive horizontal collapse. In the case of complete slab detachment, the impact of the detached structure on the lower floor can cause punching shear failure in the lower slabs. In such scenarios, the entire slab volume around the column may experience severe damage, and connection failure occurs along the column core circumference, ultimately leading to the vertical progressive collapse of the structure.

From this perspective, Yang et al. [367] cite several real-world cases of progressive collapse, noting that it can occur due to improper structural modifications, overloading of structures with insufficient material strength, exposure to high temperatures during fires, and column removal (caused by explosions or vehicular collisions, for example). Under this perspective, Pérez-Díaz et al. [368] conducted a case study of a real-life scenario involving progressive collapse. The investigation identified that the collapse was caused by excessive loading and faults in the cross-brace area, including concrete disintegration, cavities, reduction in bar fixings, and excessive grouping. Additionally, the importance of studying the post-punching mechanism lies in evaluating the resistance of slab-column joints to mitigate the progressive collapse of flat plate systems [369]. It is worth mentioning that some studies have conducted numerical analyses emphasizing post-punching behavior and achieved excellent approximations to experimental results [369, 370].

Regarding progressive collapse, Yankelevsky et al. [371] highlight that such scenarios involve dynamic punching shear, where failure is a time-dependent event requiring consideration of specific effects such as damping, strain rate, and inertial effects. The authors state that dynamic punching shear can be categorized into two groups. The first group is associated with low velocities and includes column removal, as illustrated in Fig. 19a, or localized blast or impact loading. The second group, illustrated in Fig. 19b, refers to impact-induced failures, which occur when a slab, detached from its supports due to either static or dynamic failure, falls at a considerably higher velocity and impacts the slab below, generating high-magnitude dynamic interaction pressures between the impacting slabs, thereby leading to progressive collapse [371]. In this context, Yang et al. [367] note that various studies have addressed column removal and horizontal collapse through numerical and experimental analyses. However, few researchers have focused on studying vertical progressive collapse caused by slab-column joint failures.

In the case of horizontal progressive collapse analysis, which occurs on the same floor, the punching shear failure of one connection transfers forces to adjacent connections, triggering a progressive collapse. This may result from a connection's isolated punching shear failure or a column's loss. Experimental studies utilized reduced-scale models to create a group of slabs and columns connected together, simulating an entire floor at a smaller scale [314, 367, 372–375].

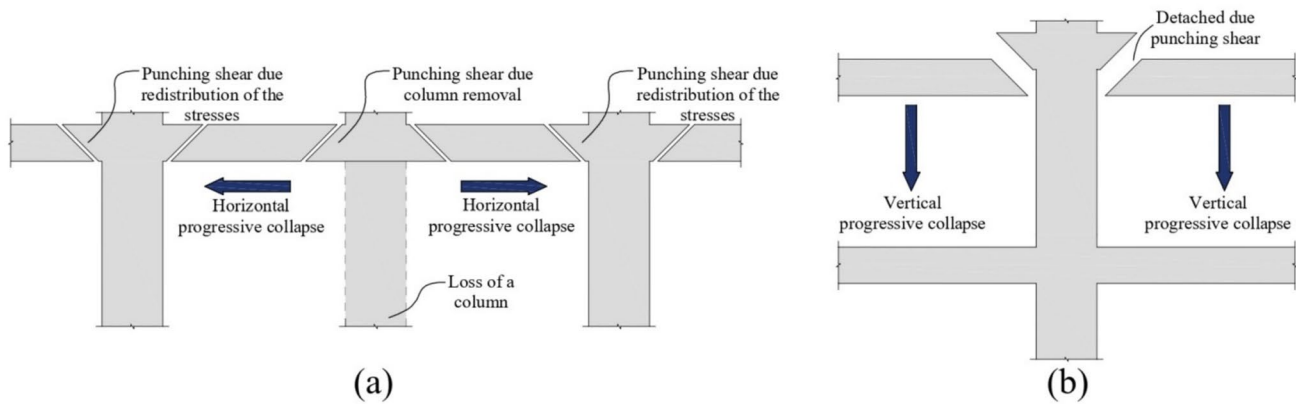


Fig. 19 Examples of (a) horizontal progressive collapse due to column removal scenario and (b) vertical progressive collapse due to slab detachment

For punching shear failure caused by loading, the load was applied to the slabs at various points until failure occurred [367]. In the case of column removal, an initial load was applied to the set of slabs, followed by the abrupt removal of a column [372].

In numerical analyses of punching shear due to column removal, which constitutes the majority of studies on progressive collapse, a validation stage was conducted [376–380], using either original results or those from the literature, followed by proposals for analytical formulations to predict loads [376, 380, 381]. In these models, a set of slabs and columns was constructed to simulate an entire floor. A specific load was applied to the structure, after which a column was removed, and the redistribution of stresses in the remaining connections (including internal, edge, and corner connections), cracking, structural damage, and the occurrence of horizontal progressive collapse were observed. These numerical analyses employed concrete modeled with 3D elements [372, 376–379], 3D elements combined with shell elements [381], shell and beam elements [306, 382, 383], or layered shell elements [380, 384]. Overall, the numerical tools used were able to reproduce behavior closely matching experimental results, indicating that FEM provides satisfactory results even for complex cases. Additionally, several studies highlight the significant role of compressive membrane action in these analyses for the progressive collapse resistance of flat plate systems [382, 383, 385, 386].

In the case of slab impact on another slab, which may lead to vertical progressive collapse, the structural response is entirely different from the column removal scenario. In the impact case, shear failure is distributed over a volume of the concrete slab surrounding the column, differently from the shear failure surfaces resulting from static and dynamic conditions [361]. Here, the implicit pressure–time action on the impacting slabs depends on factors such as their contact, their masses, and the impact velocity [361]. For numerical

analyses of progressive collapse caused by slab impact, no experimental data are available for model validation due to the lack of laboratory results. Thus, numerical models were developed either to analyze scenarios proposed by the authors or real situations [361]. In this scenario, specific geometric and boundary conditions or experimental data from low-velocity drop-weight impacts [273] were used in the validation step. In the numerical simulation, 3D modeling was employed to evaluate a slab detached from its supporting columns due to static punching shear failure and falling under gravity toward the slab below, which is initially undamaged [273, 305, 371]. In this context, the impact on progressive collapse was assessed by incorporating drop panels [305], shear reinforcement [273], or examining the effects on the structure’s columns [371], rather than solely on the slabs, as is typically the case in punching shear studies.

3 Discussion of the State-of-the-Art Review in Punching Shear

Previous reviews on punching shear have primarily focused on experimental tests involving alternative materials or lateral loads [82, 387, 388]. Ramos et al. [388] investigated slabs made with coarse recycled concrete aggregate, aiming to identify the experimental studies conducted in this field and assess the applicability of design code expressions for such slabs. Their results highlighted the need for further research in this area. Isufi and Ramos [387] sought to identify and highlight gaps in the existing literature by reviewing tests on slabs made with high- and ultra-high-performance concrete, as well as fiber-reinforced concrete, in connections subjected to both non-seismic and seismic loading. Similarly, Coronelli et al. [82] presented a critical

review of the state of the art of experimental research on the seismic response of reinforced concrete flat slabs.

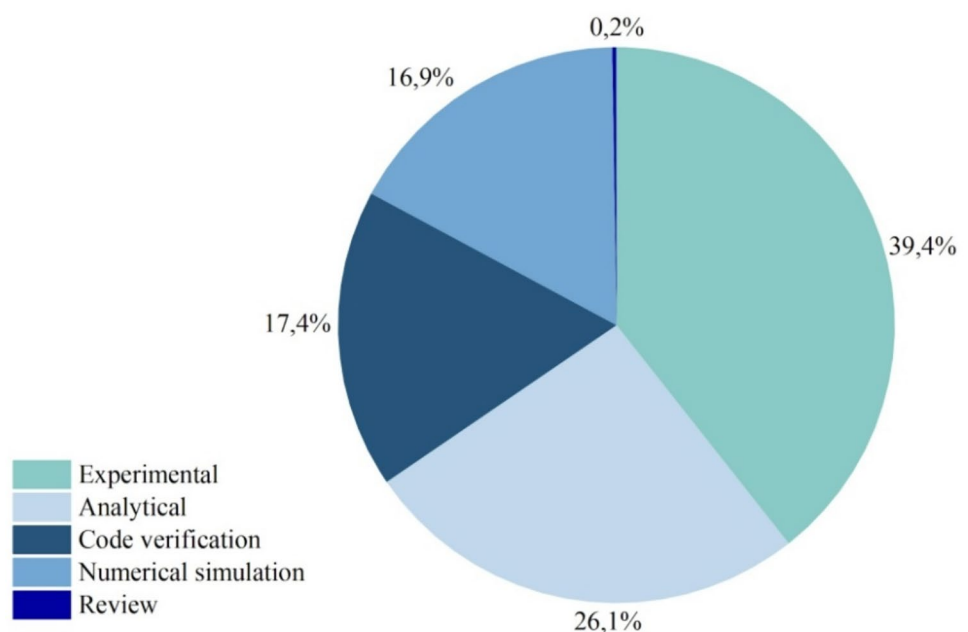
In contrast, the review presented in this study aims to provide a broader perspective on the punching shear cases investigated in the literature, focusing on a different aspect than previously published reviews. This work emphasizes numerical analyses of punching shear, highlighting the models used in the literature, relevant studies in the field, their contributions, and the various scenarios in which punching shear has been studied. To the best of the authors' knowledge, no other review article has addressed numerical studies on punching shear with the same level of detail as the present research.

As identified in this review, numerical analyses have increasingly gained prominence in punching shear studies, although most research remains concentrated on experimental investigations. For comparison purposes, articles published in reputable journals were collected and classified into five types of analyses: numerical analysis (detailed in this article), experimental studies, analytical studies (including reliability studies, development of methods or equations for load prediction), verification of normative formulations, and literature reviews. Figure 20 presents the percentage of each type of analysis considering these articles. It is worth noting that the limited number of review studies in this area is evident in the presented graph, which indicates that 0.2% of all research on punching shear is comprised exclusively of literature reviews. This highlights a significant gap in synthesizing and critically analyzing existing knowledge, emphasizing the need for comprehensive reviews to identify trends, challenges, and opportunities in the field.

From this perspective, it is worth mentioning that studies addressing more than one type of analysis (numerical, experimental, analytical, and code verification) were counted in each relevant classification. This reveals a clear predominance of experimental analyses in research on punching shear, accounting for approximately 39% of the studies. In these cases, a wide range of alternative materials used in slabs subjected to punching shear can be observed, along with variations in geometry and reinforcement, tests on slabs with shear reinforcement, asymmetric punching, strengthening techniques, different loading types, and construction systems, conducted at different scales and under various testing conditions.

Following this, around 26% of the research focuses on analytical studies, many of which propose formulations capable of predicting punching shear capacity under various scenarios. Notably, about 17% of the analytical studies employ reliability techniques or Machine Learning, which have seen significant growth over the last years due to advances in the computational field. Panahi and Genikomsou [19] point out that these techniques have become advantageous in structural engineering due to their high speed and accuracy in obtaining behavioral predictions. In punching shear studies, available data from the literature, both experimental and numerical, are generally used to get load predictions, especially in complex simulations involving strengthening techniques or alternative materials, which are often not covered in detail by design codes [389–393]. Specifically in numerical analyses, this tool presents excellent potential as it can be utilized in two main ways. One approach involves obtaining numerical data from a properly calibrated model using actual data, which will then serve as

Fig. 20 Types of analyses in articles on punching shear in slabs



input for artificial intelligence to develop analytical models [232]. The other possibility concerns using Machine Learning techniques for calibrating parameters applied in numerical simulations, such as constitutive model parameters [19].

Moreover, a substantial portion of the studies published about punching shear aims to present, either exclusively or alongside other analyses, comparisons of results with code predictions, which comprise 18% of the total. In this regard, numerical modeling makes a significant contribution, as it enables the analysis of punching shear under various conditions, evaluating materials or techniques that are not always addressed in design codes. Additionally, numerical analyses are also used to verify whether existing expressions can be applied in critical situations, accounting for factors not always considered in design standards. Thus, these comparisons and verifications make numerical simulations valuable tools for updating normative expressions.

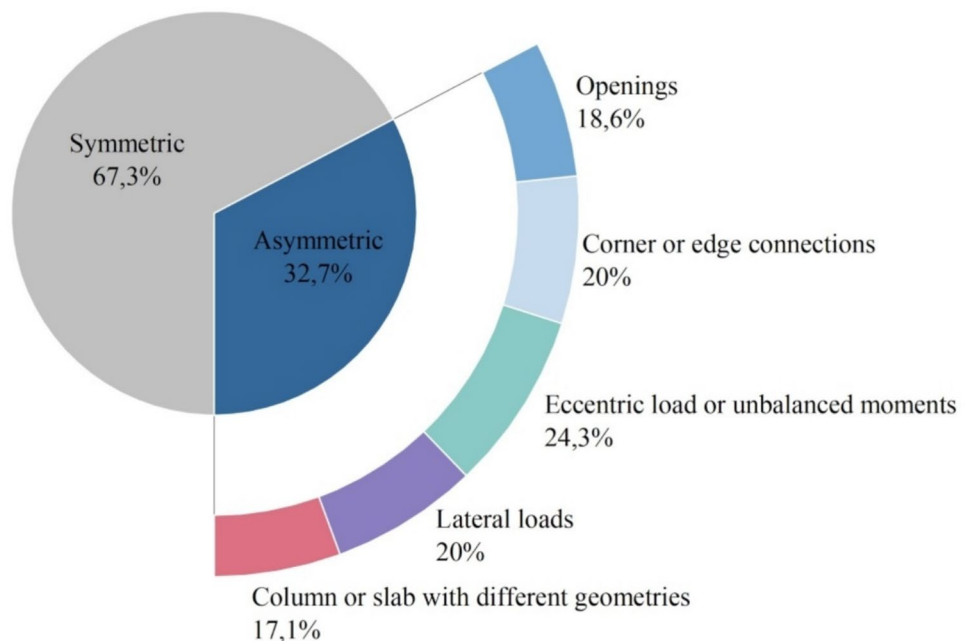
Finally, numerical studies represent only 17% of the total analyses on this topic, indicating an area with potential in this field. Thus, this review of numerical simulations identified the research profile of published studies. Regarding the type of punching shear analyzed, numerical studies focused primarily on symmetric punching shear, i.e., internal connections in slabs and columns with symmetric geometry and loading, resulting in uniform stresses in the critical punching shear section. As shown in Fig. 21, approximately 33% of the published articles address asymmetric punching shear. In this case, studies focused on eccentric loads and/or unbalanced moments (24.3%), followed by lateral load applications (20%), corner or edge connections (20%), openings near the column in slabs (18.6%), and connections involving

slabs and/or columns with distinct geometries (17.1%). However, it is relevant to highlight that in practical design scenarios, symmetric punching shear represents a rare condition regarding loading and geometry, with asymmetric punching shear being far more prevalent. Moreover, when considering all types of analyses (review, numerical, experimental, code verification, and analytical) from published papers, symmetric punching shear accounted for approximately 74%, a slightly higher number than that observed in numerical analyses (67.3%).

Regarding alternative materials, there is a discrepancy between all published articles and those focusing solely on numerical analyses. In the former case, around 29% of the articles employed alternative materials to conventional reinforced concrete. However, this percentage dropped to 19.4% in numerical simulations, as shown in Fig. 22. These researches concentrate on high-performance and ultra-high-performance concrete or slabs with CFRP, GFRP, or BFRP bars, which partially or fully replace traditional steel reinforcement. However, the challenges in developing and implementing constitutive models that adequately represent alternative materials contribute to the focus on conventional reinforced concrete. In this regard, it becomes evident that topics of significant relevance to the construction industry, particularly those aimed at sustainability, such as geopolymers concrete and concrete with coarse recycled concrete aggregate, constitute only a slight fraction of numerical punching shear studies.

When analyzing the use of shear reinforcement, 25% of numerical studies investigated the behavior of slabs employing this alternative. The most common solutions include

Fig. 21 Percentage of articles with numerical analyses on symmetric and asymmetric punching shear



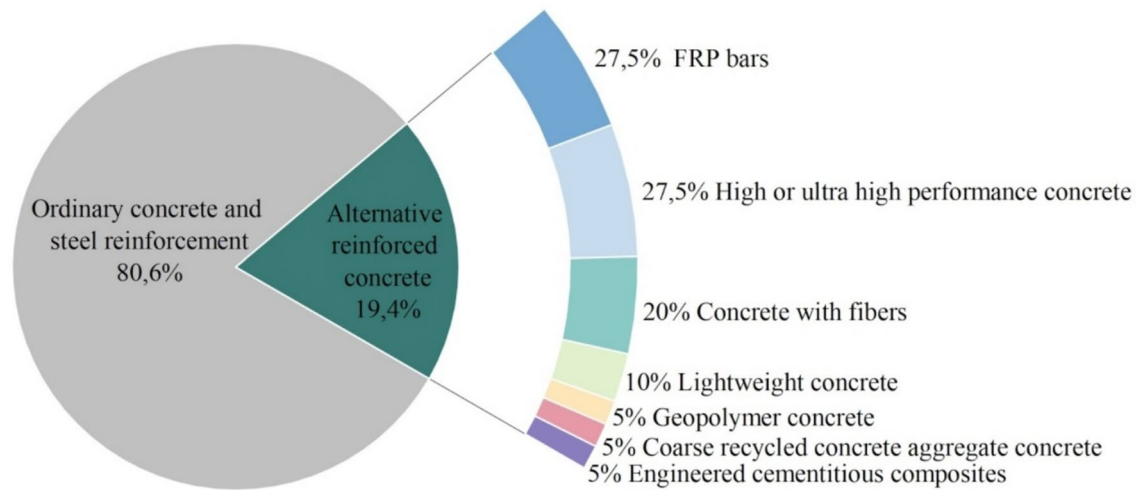


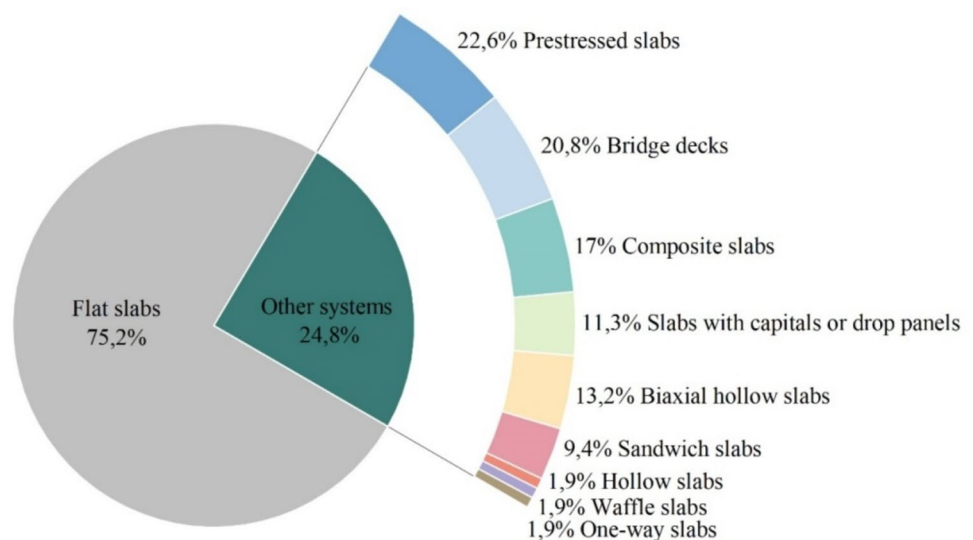
Fig. 22 Percentage of articles with numerical analyses on punching shear in slabs made of conventional and alternative reinforced concrete

shear bolts, closed stirrups, and studs, which constitute the majority of numerical studies on shear reinforcement. Other approaches include bent-up bars, shearheads, and grids, among others, exploring different solutions for structural behavior. Regarding load capacity and ductility, shear reinforcement can significantly enhance performance, offering a simple and, in most cases, cost-effective solution for structures. From a numerical perspective, modeling such reinforcements is generally straightforward, especially for studs and stirrups, but their inclusion can increase the model's complexity due to additional factors such as reinforcement deformations and the location of the main crack, which are influenced by aspects like anchorage and experimental bond behavior.

Regarding the types of slabs in which punching shear is evaluated through numerical modeling, there is a clear

predominance of analysis in flat slabs, as presented in Fig. 23. A significantly lower number of studies have been identified for other systems, representing only about 25%. Within the study of different systems, the most researched topics are prestressed slabs (23%) and bridge decks (21%), followed by composite slabs (17%), slabs with capitals or drop panels (11%), biaxial hollow slabs (13%), sandwich slabs (9%), hollow slabs (2%), waffle slabs (2%), and one-way slabs (2%). Among the reasons that explain the limited research studies on these topics are the scarcity of available experimental studies, which complicates the numerical validation process, the challenges associated with modeling complex geometries such as the voids of different shapes in biaxial hollow slabs, and the difficulties in considering pre-stressing. Additionally, another factor impacting the reduced number of studies is the fact that these systems are less

Fig. 23 Percentage of articles with numerical analyses on punching shear in different slab types



commonly used in design practice, whether due to the need for specific elements, such as forms and materials, the lack of specialized labor, associated costs, or lower demand, as seen in the case of bridge decks. It is also worth mentioning that other areas of punching shear studies represent smaller portions of the total published numerical articles but significantly contribute to the field, particularly in understanding punching shear related to strengthening techniques (8%), impact on flat slabs (7%), and progressive collapse (7%).

Finally, numerical studies on punching shear have demonstrated that existing tools can accurately reproduce the behavior of slabs subjected to this type of failure, capturing many experimental behaviors with precision. Each topic analyzed revealed unique characteristics, whether through simplifications, the choice of specific element types, or boundary conditions applied in the simulations. Thus, from a comprehensive review of published studies, it is possible to identify which topics have been extensively studied and which still require further attention.

4 Conclusions

This article comprises a comprehensive literature review on numerical simulations of punching shear in slabs, presenting high-quality numerical papers published in renowned journals. Various scenarios examined asymmetric punching shear, slabs with shear reinforcement, structural strengthening for punching shear, slabs made with alternative materials, and other relevant topics. Among the main conclusions drawn, the following can be highlighted:

1. Nearly all numerical studies include a validation phase, which may be accompanied by a parametric study or an analytical analysis proposing formulations for load prediction or comparing results with code-based predictions.
2. There is a predominant use of 3D finite elements for modeling concrete and truss or bar elements for reinforcement, although the literature presents other alternatives.
3. Numerical studies can show satisfactory comparisons with experimental results, demonstrating good approximation for rupture loads, stiffness, first-crack loads, and crack patterns.
4. The simplest punching shear scenario—considering symmetric loading and geometry in conventional reinforced concrete slabs—is still under investigation. However, such studies focus on constitutive models or modeling techniques.
5. Asymmetric punching shear may arise due to boundary conditions and geometry. Verifying these structures is crucial for design practices and has gained significant interest among researchers in recent years. However, only one-third of numerical analyses address this issue, with the study of symmetric punching shear cases remaining predominant.
6. Only a few studies focus on corner connections, continuous slabs, irregular or rectangular slabs, and columns with varying geometries. Furthermore, few investigations combine asymmetric punching shear scenarios with other variables, such as alternative concrete materials or structural strengthening. Additionally, an extremely low number of studies evaluate critical asymmetry scenarios, such as combined loading and geometric asymmetry.
7. Research on lateral cyclic loading has gained traction in recent years, yielding significant conclusions for the field of punching shear. Future studies in this area will focus on slabs made with alternative concretes or applying dynamic loading to simulate seismic or wind-induced effects.
8. The study of seismic loads on large-scale structures and impact loads remains insufficiently explored in current literature. This gap is due to the increased complexity of dynamic punching shear problems, which must consider strain rate effects, inertial forces, and material damage. The use of multiscale coupling simulations could enhance these analyses by considering material effects at the micro-scale, capturing their combined behavior at the meso-scale, and applying them in large-scale 3D finite element models at the macro-scale.
9. Numerical studies on the presence of slab openings primarily focus on positioning them away from columns to preserve load-bearing capacity. In practice, however, this is not always feasible. Few investigations have explored strategies to maintain punching shear capacity with openings, such as structural strengthening or additional reinforcement adjacent to the openings.
10. Alternative materials in slabs account for only about 20% of all numerical punching shear analyses despite their potential contributions to reducing self-weight, increasing punching shear resistance or ductility, enhancing structural durability, and improving sustainability in structural engineering.
11. The use of alternative materials introduces greater complexity to numerical analyses, as these materials often require advanced constitutive models for accurate representation. Research in this field mainly focuses on using FRP bars as alternatives to steel reinforcement or high-performance concrete as an alternative to conventional concrete. A promising research avenue involves developing constitutive models that accurately represent the behavior of these new materials, possibly employing homogenization theory.

12. The use of alternative concretes, such as geopolymers, concrete, lightweight concrete, coarse recycled aggregate concrete, and strain-hardening cementitious composites, has been minimally explored in numerical analyses despite their potential advantages in certain aspects of punching shear. The study of these materials represents a significant gap in punching shear analysis and is identified as a promising future research direction, particularly for materials aligned with sustainability goals.
13. Shear reinforcement is an efficient and widely modeled alternative in numerical analyses; however, only 25% of studies incorporate such reinforcements. Current research focuses on evaluating parameters for traditional reinforcements or determining the behavior of structures employing novel shear reinforcement types.
14. Punching shear under specific loading conditions, such as impacts, explosions, and high temperatures, as well as punching shear in composite structures or sandwich slabs, has received limited research attention.
15. Numerical studies on punching shear analyze predominantly flat slabs, representing approximately 75% of publications in the field. Systems like ribbed slabs and voided slabs offer advantages such as weight reduction but require further exploration, as they represent critical cases for punching shear, mainly when openings are near columns.
16. The study of progressive collapse has become essential to prevent such catastrophic failures. However, most research has focused on horizontal progressive collapse, with only a few numerical studies addressing vertical progressive collapse.
17. Machine Learning and artificial intelligence tools appear particularly promising when combined with numerical analyses, either by utilizing data from simulations or calibrating parameters used in numerical models.

Several topics related to punching shear remain particularly promising and warrant further numerical investigation. These include asymmetric punching shear, dynamic punching shear, and punching shear in slabs with alternative reinforced concrete. On the other hand, some topics are less promising, as consolidated conclusions have already been extensively established. Examples include the specific analysis of symmetric punching shear in conventional reinforced concrete slabs and the study of traditional shear reinforcement systems.

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Author Contribution E.R.Z.S. conceptualized the topic, wrote the main text, prepared the figures, collected the data, and developed the tables. J.P.T., A.C.F. and P.M.L. conceptualized the topic, supervised the study, and reviewed the manuscript.

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Declarations

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References

1. Zhang C, Gholipour G, Mousavi AA (2021) State-of-the-art review on responses of RC structures subjected to lateral impact loads. *Arch Comput Methods Eng* 28:2477–2507. <https://doi.org/10.1007/s11831-020-09467-5>
2. Miranda MP, Tamayo JP, Morsch IB (2022) Benchmark examples for structural system changes: analytical and numerical approaches. *Arch Comput Methods Eng* 29:3609–3637. <https://doi.org/10.1007/s11831-022-09709-8>
3. Spessatto RP, Campos Filho A, Lazzari BM, Lazzari PM, Pacheco AR (2024) Study via the finite element method of the behavior of deck joints in a bridge composed of precast concrete segments. *Struct Concr* 25(5):3164–3187. <https://doi.org/10.1002/suco.202300180>
4. He D, Li P (2025) Shaking table test and numerical simulation of rocking wall frame structure considering dynamic soil-structure interaction. *Eng Struct* 322:119156. <https://doi.org/10.1016/j.engstruct.2024.119156>
5. Machado GG, Campos Filho A, Lazzari PM, Lazzari BM, Pacheco AR (2023) Numerical simulation by the finite element method of the constructive steps of a precast prestressed segmental bridge. *Struct Eng Mech* 85(2):163–177. <https://doi.org/10.12989/sem.2023.85.2.163>
6. Sánchez BS, Tamayo JP, Morsch IB, Miranda MP (2021) A non-linear geometric model for pre-stressed steel-concrete composite beams. *J Braz Soc Mech Sci Eng* 43:233. <https://doi.org/10.1007/s40430-021-02938-1>
7. Lazzari PM, Campos Filho A, Lazzari BM, Pacheco AR, Gomes RRS (2019) Numerical simulation of the constructive steps of a cable-stayed bridge using ANSYS. *Struct Eng Mech* 69(3):269–281. <https://doi.org/10.12989/sem.2019.69.3.269>
8. Quevedo FPM, Bernaud D, Campos Filho A (2022) Numerical analysis of deep tunnels in viscoplastic rock mass considering the creep and shrinkage of the concrete lining. *Int J Geomech*. [https://doi.org/10.1061/\(ASCE\)GM.1943-5622.0002282](https://doi.org/10.1061/(ASCE)GM.1943-5622.0002282)
9. Tamayo JLP, Awruch AM (2016) On the validation of a numerical model for the analysis of soil-structure interaction problems. *Lat Am J Solids Struct* 13(8):1545–1575. <https://doi.org/10.1590/1679-78252450>
10. Ojha S, Pal P, Mehta PK (2025) Computational analysis of curved prestressed concrete box-girder bridges using finite element method. *Sci Rep* 15:7050. <https://doi.org/10.1038/s41598-025-91172-z>

11. Xiao RY, O'Flaherty T (2000) Finite-element analysis of tested concrete connections. *Comput Struct* 78(1–3):247–255. [https://doi.org/10.1016/S0045-7949\(00\)00075-4](https://doi.org/10.1016/S0045-7949(00)00075-4)
12. Shu J, Plos M, Zandi K, Johansson M, Nilenius F (2016) Prediction of punching behaviour of RC slabs using continuum non linear FE analysis. *Eng Struct* 125:15–25. <https://doi.org/10.1016/j.engstruct.2016.06.044>
13. Shu J, Belletti B, Muttoni A, Solari M, Plos M (2017) Internal force distribution in RC slabs subjected to punching shear. *Eng Struct* 153:766–781. <https://doi.org/10.1016/j.engstruct.2017.10.005>
14. Navarro M, Ivorra S, Varona FB (2018) Parametric computational analysis for punching shear in RC slabs. *Eng Struct* 165:254–263. <https://doi.org/10.1016/j.engstruct.2018.03.035>
15. Goh CYM, Hrynyk TD (2018) Numerical investigation of the punching resistance of reinforced concrete flat plates. *J Struct Eng* 144(10):04018166. [https://doi.org/10.1061/\(ASCE\)ST.1943-541X.0002142](https://doi.org/10.1061/(ASCE)ST.1943-541X.0002142)
16. Wosatko A, Winnicki A, Polak MA, Pamin J (2018) Role of dilatancy angle in plasticity-based models of concrete. *Arch Civil Mech Eng* 19:1268–1283. <https://doi.org/10.1016/j.acme.2019.07.003>
17. Faria R, Marreiros R, Ramos AP, Jesus C (2020) Influence of the top reinforcement detailing in the behaviour of flat slabs. *Structures* 23:718–730. <https://doi.org/10.1016/j.istruc.2019.12.007>
18. Lapi M, Secci L, Teoni E, Ramos AP, Orlando M (2020) A hybrid method for the calibration of finite element models of punching-shear in R/C flat slabs. *Comput Struct* 238:106323. <https://doi.org/10.1016/j.compstruc.2020.106323>
19. Panahi H, Genikomsou AS (2022) Comparative evaluation of concrete constitutive models in non-linear finite element simulations of slabs with different flexural reinforcement ratios. *Eng Struct* 252:113617. <https://doi.org/10.1016/j.engstruct.2021.113617>
20. Palhares RA, Rossignoli FW, Melo GS, Lima HJN (2022) A numerical model capable of accurately simulating the punching shear behavior of a reinforced concrete slab. *Struct Concr* 23:1134–1150. <https://doi.org/10.1002/suco.202100773>
21. Abbasi MSA, Baluch MH, Azad AK, Rahman HHA (1992) Non-linear finite element modelling of failure modes in RC slabs. *Comput Struct* 42(5):815–823. [https://doi.org/10.1016/0045-7949\(92\)90192-3](https://doi.org/10.1016/0045-7949(92)90192-3)
22. Polak MA (1998) Modeling punching shear of reinforced concrete slabs using layered finite elements. *ACI Struct J* 95(1):71–80. <https://doi.org/10.14359/528>
23. Goh CYM, Hrynyk TD (2020) Nonlinear finite element analysis of reinforced concrete flat plate punching using a thick-shell modelling approach. *Eng Struct* 224:111250. <https://doi.org/10.1016/j.engstruct.2020.111250>
24. Menétrey P, Walther R, Zimmermann T, Willam KJ, Regan PE (1997) Simulation of punching failure in reinforced-concrete structures. *J Struct Eng*. [https://doi.org/10.1061/\(ASCE\)0733-9445\(1997\)123:5\(652\)](https://doi.org/10.1061/(ASCE)0733-9445(1997)123:5(652))
25. Vidosa FG, Kotsovos MD, Pavlovic MN (1988) Symmetrical punching of reinforced concrete slabs: an analytical investigation based on nonlinear finite element modeling. *ACI Struct J* 85(3):241–250. <https://doi.org/10.14359/2609>
26. Erdogan H (2023) Strain-based grid beam model for predicting punching failure of slab–column connections. *Arab J Sci Eng* 49:6007–6026. <https://doi.org/10.1007/s13369-023-08420-5>
27. Menétrey P (2002) Synthesis of punching failure in reinforced concrete. *Cement Concr Compos* 24:497–507. [https://doi.org/10.1016/S0958-9465\(01\)00066-X](https://doi.org/10.1016/S0958-9465(01)00066-X)
28. Santos JB, Souza RM, Melo GD, Gomes RB (2024) Investigating punching shear in slabs with unbalanced moments and openings. *Eng Struct* 299:117105. <https://doi.org/10.1016/j.engstruct.2023.117105>
29. Ferreira MP, Oliveira MH, Melo GSSA (2019) Tests on the punching resistance of flat slabs with unbalanced moments. *Eng Struct* 196:109311. <https://doi.org/10.1016/j.engstruct.2019.109311>
30. Gouda A, El-Salakawy E (2016) Behavior of GFRP-RC interior slab-column connections with shear studs and high-moment transfer. *J Compos Constr* 20(4):04016005. [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0000663](https://doi.org/10.1061/(ASCE)CC.1943-5614.0000663)
31. Park H, Choi K (2006) Improved strength model for interior flat plate-column connections subject to unbalanced moment. *J Struct Eng* 132(5):694. [https://doi.org/10.1061/\(ASCE\)0733-9445\(2006\)132:5\(694\)](https://doi.org/10.1061/(ASCE)0733-9445(2006)132:5(694))
32. Coronelli D (2010) Grid model for flat-slab structures. *ACI Struct J* 107(6):645–653. <https://doi.org/10.14359/51664012>
33. Ebead U, Saeed H (2010) Modeling of reinforced concrete slabs strengthened with fiber-reinforced polymer or steel plates. *ACI Struct J* 107(2):218–227. <https://doi.org/10.14359/51663538>
34. Jia Y, Chiang JCL (2022) Finite element analysis of punching shear of reinforced concrete slab–column connections with shear reinforcement. *Appl Sci* 12(19):9584. <https://doi.org/10.3390/app12199584>
35. Zheng B, Zheng W, Cao B, Zhang Y (2023) Nonlinear finite element analysis of non-symmetrical punching shear of rectangular flat slabs supported on square columns. *Eng Struct*. <https://doi.org/10.1016/j.engstruct.2022.115451>
36. Schimanowski ERZ, Campos Filho A, Lazzari BM, Lazzari PM (2024) Numerical simulation of punching shear in flat slabs with spherical voids subjected to symmetric and asymmetric loadings. *IBRACON Struct Mater J* 17(3):e17316. <https://doi.org/10.1590/S1983-41952024000300016>
37. Schimanowski ERZ, Campos Filho A, Lazzari BM, Lazzari PM (2024) An investigation of punching shear in flat slabs with spherical voids using the finite element method. *Struct Concr*. <https://doi.org/10.1002/suco.202400489>
38. Amiri M, Esfahani MR (2023) Effect of using Engineered Cementitious Composites (ECC) on failure behavior of flat slab-column connections. *Structures* 47:2397–2407. <https://doi.org/10.1016/j.istruc.2022.12.058>
39. Sine A, Pimentel M, Nunes S (2022) Experimental investigation on punching shear behaviour of RC-(R)UHPFRC composite flat slabs without transverse reinforcement. *Eng Struct* 255:113951. <https://doi.org/10.1016/j.engstruct.2022.113951>
40. Al-Bayati AF, Leong LT, Clark LA (2018) Eccentric punching shear of waffle slab. *ACI Struct J* 115(1):163–173. <https://doi.org/10.14359/51701090>
41. Farzam M, Fouad NA, Grünberg J, Fard MY (2010) Punching of RC slabs under eccentric loads. *Struct Concr* 11(4):211–220. <https://doi.org/10.1680/stco.2010.11.4.211>
42. Al-Hamd RKS, Gillie M, Cunningham LS, Warren H, Albostami AS (2019) Novel shearhead reinforcement for slab-column connections subject to eccentric load and fire. *Arch Civil Mech Eng* 19(2):503–524. <https://doi.org/10.1016/j.acme.2018.12.011>
43. Setiawan A, Vollum RL, Macorini L, Izzuddin B (2019) Efficient 3-D modelling of punching shear failure at slab-column connections by means of nonlinear joint elements. *Eng Struct*. <https://doi.org/10.1016/j.engstruct.2019.109372>
44. Živković S, Bešević M, Purčar MV, Kozarić L (2019) Nonlinear finite element analysis of punching shear strength of eccentrically loaded RC flat slabs with opening. *KSCE J Civ Eng* 23(11):4771–4780. <https://doi.org/10.1007/s12205-019-0075-5>
45. Oukaili NK, Hussein LF (2020) Experimental and analytical study on punching shear strength of Bubbledecks. *ACI Struct J* 117(2):17–31. <https://doi.org/10.14359/51721360>

46. Alrousan RZ, Alnemrawi BR (2022) Punching shear behavior of FRP reinforced concrete slabs under different opening configurations and loading conditions. *Case Stud Constr Mater* 17:e01508. <https://doi.org/10.1016/j.cscm.2022.e01508>
47. Alrousan RZ, Alnemrawi BR (2022) The influence of concrete compressive strength on the punching shear capacity of reinforced concrete flat slabs under different opening configurations and loading conditions. *Structures* 44:101–119. <https://doi.org/10.1016/j.istruc.2022.07.091>
48. Wang J, Li Y, Guan H, Yang Y, Diao M, Xu W, Chen Y (2023) Punching and post-punching shear failure behaviors of laterally-restrained and eccentrically loaded flat plate slab-column joints with drop panels. *Eng Struct*. <https://doi.org/10.1016/j.engstruct.2022.115585>
49. Sanabria Díaz R, de Albuquerque NGB, de Almeida LC, Trautwein LM, Sales de Melo G (2024) Flat slab-edge column connections subjected to outward and inward eccentricity: Numerical modeling and comparison with codes. *Struct Concr* 25(5):3211–3231. <https://doi.org/10.1002/suco.202301070>
50. Mendes RP, Mesquita LC, Ferreira MP, Trautwein LM, Marvila MT, Marques MG (2024) Numerical evaluation of the punching shear strength of flat slabs subjected to balanced and unbalanced moments. *Buildings*. <https://doi.org/10.3390/buildings14040985>
51. Vargas D, Lantsoght EOL, Genikomsou AS (2022) Flat slabs in eccentric punching shear: experimental database and code analysis. *Buildings* 12:2092. <https://doi.org/10.3390/buildings12122092>
52. Hernández Fraile D, Setiawan A, Santos JB, Muttoni A (2023) Punching tests on edge slab-column connections with refined measurements. *Eng Struct* 288:116166. <https://doi.org/10.1016/j.engstruct.2023.116166>
53. Salama AE, Hassan M, Benmokrane B (2021) Punching-shear behavior of glass fiber-reinforced polymer-reinforced concrete edge column-slab connections: experimental and analytical investigations. *ACI Struct J*. <https://doi.org/10.14359/51730525>
54. Salama AE, Hassan M, Benmokrane B, Ferrier E (2020) Modified strip model for punching-shear strength of FRP-reinforced concrete edge-column slab connections. *Eng Struct* 206:110769. <https://doi.org/10.1016/j.engstruct.2020.110769>
55. Barbán VV, Melo GS (2020) Carvalho AL (2020) Punching shear tests of post tensioned flat slabs supported on edge columns. *Struct Concr* 21:247–261. <https://doi.org/10.1002/suco.20190160>
56. Al-Bayati AF, Leong LT, Clark LA (2019) Edge punching shear of waffle slab with moment transfer perpendicular to slab's free edge. *ACI Struct J* 116(4):243–252. <https://doi.org/10.14359/51715630>
57. Salama AE, Hassan M, Benmokrane B (2019) Effectiveness of glass fiber-reinforced polymer stirrups as shear reinforcement in glass fiber-reinforced polymer reinforced concrete edge Slab-Column Connections. *ACI Struct J* 116(5):97–112. <https://doi.org/10.14359/51716757>
58. Megally S, Ghali A (2000) Punching of concrete slabs due to column moment transfer. *J Struct Eng*. [https://doi.org/10.1061/\(ASCE\)0733-9445\(2000\)126:2\(180\)](https://doi.org/10.1061/(ASCE)0733-9445(2000)126:2(180))
59. Polak MA (2005) Ductility of reinforced concrete flat slab-column connections. *Comput-Aided Civil Infrastruct Eng* 20:184–193. <https://doi.org/10.1111/j.1467-8667.2005.00386.x>
60. Marinković SB, Alendar VH (2008) Punching failure mechanism at edge columns of post-tensioned lift slabs. *Eng Struct* 30(10):2752–2761. <https://doi.org/10.1016/j.engstruct.2008.03.009>
61. Wang W, Teng S (2008) Finite-element analysis of reinforced concrete flat plate structures by layered shell element. *J Struct Eng*. [https://doi.org/10.1061/\(ASCE\)0733-9445\(2008\)134:12\(1862\)](https://doi.org/10.1061/(ASCE)0733-9445(2008)134:12(1862))
62. Soares LFS, Vollum RL (2016) Influence of continuity on punching resistance at edge columns. *Mag Concr Res* 68(23):1225–1239. <https://doi.org/10.1680/jmacr.16.00052>
63. Salama AE, Hassan M, Benmokrane B (2020) Nonlinear finite-element analysis of glass fiber-reinforced polymer-reinforced concrete slab-column edge connections. *ACI Struct J* 117(5):217–231. <https://doi.org/10.14359/51724682>
64. Salama AE, Hassan AM, Benmokrane AB (2020) Effect of GFRP shear stirrups on strength of two-way GFRP RC edge slabs: experimental and finite-element investigations. *J Struct Eng*. [https://doi.org/10.1061/\(ASCE\)ST.1943-541X.0002593](https://doi.org/10.1061/(ASCE)ST.1943-541X.0002593)
65. Matseevich T (2023) Finite element analysis of the bearing capacity of beamless floor slabs under punching, taking into account the design parameters of the contacting elements. *Buildings* 13(5):1221. <https://doi.org/10.3390/buildings13051221>
66. Yu JL, Wang YC (2020) Punching shear behavior of an innovative connection between steel-tubular column and flat-concrete slab. *J Struct Eng*. [https://doi.org/10.1061/\(asce\)st.1943-541x.0002724](https://doi.org/10.1061/(asce)st.1943-541x.0002724)
67. Yu JL, Wang YC (2020) Modelling and design method for static resistance of a new connection between steel tubular column and flat concrete slab. *J Construct Steel Res*. <https://doi.org/10.1016/j.jcsr.2020.106254>
68. Siqueira JPA, Albuquerque EJP, Ruiz MF, Melo GS (2024) Punching shear in flat slabs with re-entrant corner columns and shear reinforcement. *Eng Struct* 310:118098. <https://doi.org/10.1016/j.engstruct.2024.118098>
69. Chen J, Zhang R, Li A, Che W, Wu S, Deng X (2024) Punching shear performance of prestressed steel-concrete composite slab-column joints with spatial steel drop panels. *Structures* 62:106334. <https://doi.org/10.1016/j.istruc.2024.106334>
70. Milligan GJ, Polak MA (2023) Influence of openings on the punching shear behaviour of concrete slabs supported on special-shaped columns. *Eng Struct* 284:115968. <https://doi.org/10.1016/j.engstruct.2023.115968>
71. Xue J, Zhang W, Xu J, Yuan Z, Zhao F, Zhang C (2022) Investigation of the punching shear behaviors of reinforced concrete slab-steel reinforced concrete L-shaped column connections. *Adv Struct Eng* 25(14):2906–2921. <https://doi.org/10.1177/13694332221109035>
72. Milligan GJ, Polak MA (2021) Impact of column rectangularity on punching shear strength: code predictions versus finite element analysis. *J Struct Eng* 147(2):04020331. [https://doi.org/10.1061/\(ASCE\)ST.1943-541X.0002889](https://doi.org/10.1061/(ASCE)ST.1943-541X.0002889)
73. Milligan GJ, Polak MA, Zurell C (2022) Finite element analysis of punching shear behavior of reinforced concrete slabs supported on walls. *Struct Concr* 23:1118–1133. <https://doi.org/10.1002/suco.202100692>
74. Setiawan A, Vollum RL, Macorini L, Izzuddin BA (2020) Punching shear design of RC flat slabs supported on wall corners. *Struct Concr* 21:859–874. <https://doi.org/10.1002/suco.201900514>
75. Setiawan A, Vollum RL, Macorini L, Izzuddin BA (2020) Punching of RC slabs without transverse reinforcement supported on elongated columns. *Structures* 27:2048–2068. <https://doi.org/10.1016/j.istruc.2020.08.017>
76. Zheng B, Zheng W, Zhang Y, Wang G (2025) Punching failure and analytical model for rectangular flat slabs with equal flexural capacities in orthogonal directions. *Eng Struct* 323:119305. <https://doi.org/10.1016/j.engstruct.2024.119305>
77. Milligan GJ, Polak MA, Zurell C (2020) Finite element analysis of punching shear behaviour of concrete slabs supported on rectangular columns. *Eng Struct* 224:111189. <https://doi.org/10.1016/j.engstruct.2020.111189>

78. Loo YC, Guan H (1997) Cracking and punching shear failure analysis of RC flat plates. *J Struct Eng* 123(10):1321. [https://doi.org/10.1061/\(ASCE\)0733-9445\(1997\)123:10\(1321\)](https://doi.org/10.1061/(ASCE)0733-9445(1997)123:10(1321))
79. Abolhelm R, Hrynyk TD (2024) Thick-shell finite element analysis of reinforced concrete flat plates with non-uniform connection stresses. *Eng Struct*. <https://doi.org/10.1016/j.engstruct.2024.117634>
80. Genikomsou AS (2024) Seismic damage assessment of reinforced concrete slab-column connections—review of test data, code provisions and analytical models. *Buildings* 14:465. <https://doi.org/10.3390/buildings14020465>
81. Almeida AFO, Ramos AP, Lúcio V, Marreiros R (2020) Behavior of RC flat slabs with shear bolts under reversed horizontal cyclic loading. *Struct Concr* 21:501–516. <https://doi.org/10.1002/suco.201900128>
82. Coronelli D, Muttoni A, Pascu IR, Ramos AP, Netti T (2020) A state of the art of flat-slab frame tests for gravity and lateral loading. *Struct Concr* 21:2764–2781. <https://doi.org/10.1002/suco.202000305>
83. Ramos AP, Isufi B, Marreiros R, Bolešová M, Gajdošová K (2022) Rational use of HPFRC in slab-column connections under reversed horizontal cyclic loading. *Eng Struct* 270:114903. <https://doi.org/10.1016/j.engstruct.2022.114903>
84. Isufi B, Rossi M, Ramos AP (2021) Influence of flexural reinforcement on the seismic performance of flat slab-column connections. *Eng Struct* 242:112583. <https://doi.org/10.1016/j.engstruct.2021.112583>
85. Eladawy M, Hassan M, Benmokrane B, Ferrier E (2020) Lateral cyclic behavior of interior two-way concrete slab-column connections reinforced with GFRP bars. *Eng Struct* 209:109978. <https://doi.org/10.1016/j.engstruct.2019.109978>
86. Almeida AFO, Alcobia B, Ornelas M, Marreiros R, Ramos AP (2020) Behaviour of reinforced-concrete flat slabs with stirrups under reversed horizontal cyclic loading. *Mag Concr Res* 72(7):339–356. <https://doi.org/10.1680/jmacr.18.00209>
87. Inácio M, Isufi B, Lapi M, Ramos AP (2020) Rational use of high-strength concrete in flat slab-column connections under seismic loading. *ACI Struct J* 117(6):297–310. <https://doi.org/10.14359/51728080>
88. El-Gendy MG, El-Salakawy EF (2020) Gravity load effect on seismic response of glass fiber-reinforced polymer-reinforced concrete slab-column edge connections. *ACI Struct J* 117(5):41–52. <https://doi.org/10.14359/51724665>
89. El-Gendy MG, El-Salakawy EF (2020) GFRP Shear Reinforcement for slab-column edge connections subjected to reversed cyclic lateral load. *J Compos Constr* 24(2):04020003. [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0001000](https://doi.org/10.1061/(ASCE)CC.1943-5614.0001000)
90. El-Gendy M, El-Salakawy E (2019) Effect of flexural reinforcement type and ratio on the punching behavior of RC slab-column edge connections subjected to reversed-cyclic lateral loads. *Eng Struct* 200:109703. <https://doi.org/10.1016/j.engstruct.2019.109703>
91. Lee C-H, Kang THK, Kim J-W, Song J-K, Kim S (2019) Seismic performance of concrete-filled tube column-reinforced concrete slab connections with shearhead keys. *ACI Struct J* 116(2):233–244. <https://doi.org/10.14359/51713316>
92. Gouveia ND, Faria DMV, Ramos AP (2019) Assessment of SFRC flat slab punching behaviour – part II: reversed horizontal cyclic loading. *Mag Concr Res* 71(1):26–42. <https://doi.org/10.1680/jmacr.17.00344>
93. Isufi B, Ramos AP, Lúcio V (2019) Reversed horizontal cyclic loading tests of flat slab specimens with studs as shear reinforcement. *Struct Concr* 20:330–347. <https://doi.org/10.1002/suco.201800128>
94. Almeida AFO, Inácio MMG, Lúcio VJG, Ramos AP (2016) Punching behaviour of RC flat slabs under reversed horizontal cyclic loading. *Eng Struct* 117:204–219. <https://doi.org/10.1016/j.engstruct.2016.03.007>
95. Himawan A, Teng S (2014) Cyclic behavior of post-tensioned slab-rectangular column connections. *ACI Struct J* 111(1):177–188. <https://doi.org/10.14359/51686516>
96. Xia C, Shi Q, Rong C, Wang B (2024) Punching and lateral cyclic behavior of concrete slab-column connections reinforced with steel sheets. *Structures* 67:106928. <https://doi.org/10.1016/j.istruc.2024.106928>
97. Ramos AP, Isufi B, Marreiros R, Coronelli D, Netti T, Tornaghi ML, Tsionis G, Muttoni A (2023) Seismic performance of strengthened slab-column connections in a full-scale test. *J Earthquake Eng* 27(9):2299–2318. <https://doi.org/10.1080/13632469.2022.2112320>
98. Muttoni A, Coronelli D, Tornaghi ML, Martinelli L, Pascu IR, Ramos AP, Tsionis G, Bamonte P, Isufi B, Setiawan A (2022) Deformation capacity evaluation for flat slab seismic design. *Bull Earthq Eng* 20:1619–1654. <https://doi.org/10.1007/s10518-021-01302-x>
99. Genikomsou AS, Polak MA (2015) Finite element analysis of punching shear of concrete slabs using damaged plasticity model in ABAQUS. *Eng Struct* 98:38–48. <https://doi.org/10.1016/j.engstruct.2015.04.016>
100. Kang THK, Wallace JW, Elwood KJ (2009) Nonlinear modeling of flat-plate systems. *J Struct Eng* 135:147–158. [https://doi.org/10.1061/\(ASCE\)0733-94452009135:2147](https://doi.org/10.1061/(ASCE)0733-94452009135:2147)
101. Setiawan A, Vollum RL, Macorini L (2019) Numerical and analytical investigation of internal slab-column connections subject to cyclic loading. *Eng Struct* 184:535–554. <https://doi.org/10.1016/j.engstruct.2019.01.089>
102. Eder MA, Vollum RL, Elghazouli AY (2011) Inelastic behaviour of tubular column-to-flat slab connections. *J Constr Steel Res* 67(7):1164–1173. <https://doi.org/10.1016/j.jcsr.2011.02.009>
103. Eder MA, Vollum RL, Elghazouli AY (2012) Performance of ductile RC flat slab to steel column connections under cyclic loading. *Eng Struct* 36:239–257. <https://doi.org/10.1016/j.engstruct.2011.12.002>
104. El-Mandouh MA E-H (2022) Cyclic behavior of high strength lightweight concrete slab-edge column connections with and without openings. *Case Stud Construct Mater*. <https://doi.org/10.1016/j.cscm.2021.e00832>
105. El-Gendy MG, El-Salakawy EF (2021) Finite-element analysis of FRP-reinforced concrete slab-column edge connections subjected to reversed-cyclic lateral loads. *J Compos Constr*. [https://doi.org/10.1061/\(asce\)cc.1943-5614.0001098](https://doi.org/10.1061/(asce)cc.1943-5614.0001098)
106. Isufi B, Cismasiu I, Marreiros R, Pinho Ramos A, Lúcio V (2020) Role of punching shear reinforcement in the seismic performance of flat slab frames. *Eng Struct*. <https://doi.org/10.1016/j.engstruct.2020.110238>
107. Panahi H, Genikomsou AS (2023) A machine-learning-based model for seismic performance assessment of interior slab-column connections. *Soil Dynamics Earthq Eng*. <https://doi.org/10.1016/j.soildyn.2023.107943>
108. Coronelli D, Muttoni A, Martinelli L, del Vecchio C, Gaetani d'Aragona M, Lignola GP, Prota A, Kagermanov A, Marinković M (2023) Flat slab response for seismic and cyclic actions prediction with numerical models. *Eng Struct*. <https://doi.org/10.1016/j.engstruct.2023.116307>
109. Najafgholipour MA, Radbakhsh SH, Erfani E (2024) Evaluation of the seismic behavior of reinforced concrete structures with flat slab-column gravity frame and shear walls through nonlinear analysis methods. *Earthq Eng Vib* 23(3):713–726. <https://doi.org/10.1007/s11803-024-2267-3>
110. Tian Y, Liu X, George S (2020) Effects of vertical ground motion on seismic performance of reinforced concrete

- flat-plate buildings. *J Struct Eng.* [https://doi.org/10.1061/\(asce\)st.1943-541x.0002840](https://doi.org/10.1061/(asce)st.1943-541x.0002840)
111. Najafgholipour MA, Sarhadi F (2022) A finite element study on the ultimate lateral drift capacity of interior reinforced concrete flat slab-column connections. *Structures* 46:913–926. <https://doi.org/10.1016/j.istruc.2022.10.128>
 112. Muttoni A (2008) Punching shear strength of reinforced concrete slabs without transverse reinforcement. *ACI Struct J* 105(4):440–450. <https://doi.org/10.14359/19858>
 113. El-Shafiey TF, Atta AM, Hassan A, Elnasharty M (2022) Effect of opening shape, size and location on the punching shear behaviour of RC flat slabs. *Structures* 44:1138–1151. <https://doi.org/10.1016/j.istruc.2022.08.076>
 114. Alnemrawi BR, Al-Rousan RZ, Ababneh AN (2024) The structural behavior of heat-damaged flat slabs with openings of different sizes and locations. *Arab J Sci Eng* 49:5403–5430. <https://doi.org/10.1007/s13369-023-08411-6>
 115. Bursać S, Bešević M, Purčar MV, Kozarić L, Đurić N (2021) Experimental analysis of punching shear strength of eccentrically loaded slab with the opening along the face of the internal column. *Eng Struct* 249:113359. <https://doi.org/10.1016/j.engstruct.2021.113359>
 116. Mostofinejad D, Jafarian N, Naderi A, Mostofinejad A, Salehi M (2020) Effects of openings on the punching shear strength of reinforced concrete slabs. *Structures* 25:760–773. <https://doi.org/10.1016/j.istruc.2020.03.061>
 117. Mercimek Ö, Akkaya ST, Türer A, Erbas W, Bıçakçıoğlu K, Anil Ö (2024) Experimental investigation of the effectiveness of strengthening methods in reinforced concrete flat slabs with multiple openings under punching load. *Structures* 63:106386. <https://doi.org/10.1016/j.istruc.2024.106386>
 118. Mercimek O, Ghoroubi R, Ozdemir A, Anil O, Baran M (2022) Punching behaviour of two-way RC slabs having different multiple opening locations and sizes strengthened with TRM. *Structures* 42:531–549. <https://doi.org/10.1016/j.istruc.2022.06.038>
 119. Santos JB, Souza RM, Melo GD, Gomes RB (2022) Punching resistance of flat slabs with openings adjacent to the columns. *ACI Struct J* 119(1):41–53. <https://doi.org/10.14359/51734216>
 120. Augustin T, Fillo L, Halvonik J (2020) Punching resistance of slab-column connections with openings. *Struct Concr* 21:278–290. <https://doi.org/10.1002/suco.201900158>
 121. Genikomsou AS, Polak AM (2017) Effect of openings on punching shear strength of reinforced concrete slabs-finite element investigation. *ACI Struct J* 114(5):1249–1261. <https://doi.org/10.14359/51689871>
 122. Vieira PHSM, Sanabria Díaz R, Marques MG, Liberati EAP, Trautwein LM (2023) Nonlinear analysis of the impact of openings on punching shear strength. *Struct Concr* 24(6):7334–7354. <https://doi.org/10.1002/suco.202300476>
 123. Marques MG, Liberati EAP, Gomes RB, Almeida LC, Trautwein LM (2020) Study of failure mode of reinforced concrete flat slabs with openings and studs. *ACI Struct J* 117(4):39–48. <https://doi.org/10.14359/51723518>
 124. Liberati EAP, Marques MG, Leonel ED, Almeida LC, Trautwein LM (2019) Failure analysis of punching in reinforced concrete flat slabs with openings adjacent to the column. *Eng Struct* 182:331–343. <https://doi.org/10.1016/j.engstruct.2018.11.073>
 125. Yousef AM, El-Metwally SE, Aska HH, El-Mandouh MA (2019) Behavior of high-strength concrete interior slab-column connections with openings under seismic loading. *Constr Build Mater* 214:619–630. <https://doi.org/10.1016/j.conbuildmat.2019.04.143>
 126. Abdulrahman BQ, Wu Z, Cunningham LS (2017) Experimental and numerical investigation into strengthening flat slabs at corner columns with externally bonded CFRP. *Constr Build Mater* 139:132–147. <https://doi.org/10.1016/j.conbuildmat.2017.02.056>
 127. Al-Tameemi A, Habelalmateen MA, Alalikhhan AA (2021) Effect of embedded pipelines within slab thickness on punching shear capacity of flat slabs haider. *KSCE J Civ Eng* 25(1):208–218. <https://doi.org/10.1007/s12205-020-5145-1>
 128. Oukaili NK, Merie HD (2021) CFRP strengthening efficiency on enhancement punching shear resistance of RC bubbled slabs with openings. *Case Stud Constr Materials.* <https://doi.org/10.1016/j.cscm.2021.e00641>
 129. Akkaya ST, Mercimek Ö, Ghoroubi R, Anil Ö, Erbaş Y, Yılmaz T (2022) Experimental, analytical, and numerical investigation of punching behaviour of two-way RC slab with multiple openings. *Structures* 43:574–593. <https://doi.org/10.1016/j.istruc.2022.06.070>
 130. El-Salakawy EF, Polak MA, Soliman MH (1999) Reinforced concrete slab-column edge connections with openings. *ACI Struct J.* <https://doi.org/10.14359/598>
 131. Balomenos GP, Genikomsou AS, Polak AM (2018) Investigation of the effect of openings of interior reinforced concrete flat slabs. *Struct Concr* 19(6):1672–1681. <https://doi.org/10.1002/suco.201700201>
 132. Sahoo S, Singh B (2021) Punching shear capacity of recycled-aggregate concrete slab-column connections. *J Build Eng* 41:102430. <https://doi.org/10.1016/j.jobbe.2021.102430>
 133. Sahoo S, Singh B (2020) Recycled aggregate concrete slab punching shear capacity. *Structures* 24:426–443. <https://doi.org/10.1016/j.istruc.2019.11.010>
 134. Francesconi L, Pani L, Stochino F (2018) Punching shear strength of reinforced recycled concrete slabs. *Constr Build Mater* 127:248–263. <https://doi.org/10.1016/j.conbuildmat.2016.09.094>
 135. Ferreira MP, Damasceno IIR, Pereira Filho MJM, Lima Neto AF, Oliveira MH, Ribeiro Júnior JA (2024) Effect of recycled concrete aggregates on the punching strength of slab-column connections without shear reinforcement. *J Build Eng* 95:110174. <https://doi.org/10.1016/j.jobbe.2024.110174>
 136. Leelatano S, Imjai T, Setkit M, Garcia R, Kim B (2022) Punching shear capacity of recycled aggregate concrete slabs. *Buildings* 12:1584. <https://doi.org/10.3390/buildings12101584>
 137. Mahmoud ZI, El Tony M, Saeed KS (2018) Punching shear behavior of recycled aggregate reinforced concrete slabs. *Alex Eng J* 57:841–849. <https://doi.org/10.1016/j.aej.2015.12.004>
 138. McGinnis MJ, Gangone MV, Nogales A, Gomez-Santana LM, Weldon B, Reihl A, Tošić N, Kurama Y (2024) Experimental and numerical investigation of the bending, shear, and punching shear behavior of recycled aggregate concrete precast/prestressed hollow core slabs. *Struct Concr.* <https://doi.org/10.1002/suco.202400008>
 139. Reis N, Brito J, Correia JR, Arruda MRT (2015) Punching behaviour of concrete slabs incorporating coarse recycled concrete aggregates. *Eng Struct* 100(1):238–248. <https://doi.org/10.1016/j.engstruct.2015.06.011>
 140. Mukhtar F, Lardhi M (2024) Punching shear of sustainable recycled aggregate FRC slabs strengthened with NSM FRP bars. *Case Stud Constr Mater* 21:e03468. <https://doi.org/10.1016/j.cscm.2024.e03468>
 141. Youssf O, Hassanli R, Elchalakani M, Mills JE, Tayeh BA, Agwa IS (2023) Punching shear behaviour and repair efficiency of reinforced eco-friendly lightweight concrete slabs. *Eng Struct* 281:115805. <https://doi.org/10.1016/j.engstruct.2023.115805>
 142. Mohammad SH, Gülsan ME, Çevik A (2022) Punching shear behaviour of geopolymer concrete two-way slabs reinforced by FRP bars under monotonic and cyclic loadings. *Adv Struct Eng* 25(3):453–472. <https://doi.org/10.1177/13694332211052349>
 143. Chen C, Zhang X, Hao H (2023) Investigation on the impact resistance of reinforced geopolymer concrete slab. *J Clean Prod* 406:137144. <https://doi.org/10.1016/j.jclepro.2023.137144>

144. Chen C, Zhang X, Hao H, Sarker PK (2025) Experimental and numerical study of steel fibre reinforced geopolymer concrete slab under impact loading. *Eng Struct*. <https://doi.org/10.1016/j.engstruct.2024.119096>
145. Azevedo E, Oliveira D, Nzambi A (2024) Geopolymer flat slabs with synthetic coarse aggregate under punching. *ACI Struct J* 121(4):161–172. <https://doi.org/10.14359/51740573>
146. Huang X-N, Li Q-H, Tong J-Z, Xu S-L, Lu Y-C (2023) Punching shear behavior and strength prediction of UHTCC-enhanced RC slab-column joints. *Eng Struct* 286:116162. <https://doi.org/10.1016/j.engstruct.2023.116162>
147. Elsayed M, Tayeh BA, Kamal D (2021) Effect of crumb rubber on the punching shear behaviour of reinforced concrete slabs with openings. *Constr Build Mater* 311:125345. <https://doi.org/10.1016/j.conbuildmat.2021.125345>
148. Ye B, Pan P, Xiao G, Han J, Zhang Y (2021) Comparative study of reinforced-engineered cementitious composites and reinforced-concrete slab–column connections under a vertical monotonic load. *Eng Struct* 244:112740. <https://doi.org/10.1016/j.engstruct.2021.112740>
149. Ye B, Pan P, Xiao G, Zhang Y, He Z (2021) Experimental investigation on failure modes of RECC slab-column connections under concentric gravity loading. *Eng Struct* 230:111559. <https://doi.org/10.1016/j.engstruct.2020.111559>
150. Hamoda A, Abadel AA, Ahmed M, Wang Z, Vrcelj Z, Liang QQ (2025) Punching shear performance of reinforced concrete slab-to-steel column connections incorporating ECC and UHPECC. *Eng Struct* 322:119145. <https://doi.org/10.1016/j.engstruct.2024.119145>
151. Su Y, Wang X, Wu C, Xu M, Ma L, Jin C (2025) Punching shear behavior of RC slab-column connections enhanced with ECC in core area. *Structures* 74:108567. <https://doi.org/10.1016/j.istruc.2025.108567>
152. Mohammed BK, Al-Numan BS (2024) Effectiveness of limestone powder as a partial replacement of cement on the punching shear behavior of normal and high-strength concrete flat slabs. *Sustainability* 16:2151. <https://doi.org/10.3390/su16052151>
153. Mahmood LJ, Rafiq SK, Muhammad MA (2022) Punching shear behavior of reinforced concrete flat slab column connection containing recycled eggshell powder. *Structures* 46:1016–1028. <https://doi.org/10.1016/j.istruc.2022.10.133>
154. Ahmed FR, Muhammad MA, Ibrahim RK (2022) Effect of alum sludge on concrete strength and two way shear capacity of flat slabs. *Structures* 40:991–1001. <https://doi.org/10.1016/j.istruc.2022.04.086>
155. Elsayed M, Tayeh BA, Mohamed M, Elymany M, Mansi AH (2021) Punching shear behaviour of RC flat slabs incorporating recycled coarse aggregates and crumb rubber. *J Build Eng* 44:103363. <https://doi.org/10.1016/j.jobe.2021.103363>
156. Elsamak G, Abdullah A, Salama MI, Hu JW, El-Mandouh MA (2023) Punching shear behavior of slabs made from different types of concrete internally reinforced with SHCC-filled steel tubes. *Materials*. <https://doi.org/10.3390/ma16010072>
157. Inácio MMG, Lapi M, Ramos AP (2020) Punching of reinforced concrete flat slabs– Rational use of high strength concrete. *Eng Struct* 206:110194. <https://doi.org/10.1016/j.engstruct.2020.110194>
158. Mostafa AM, El-Salakawy EF (2018) Behavior of GFRP-RC slab–column edge connections with high-strength concrete and shear reinforcement. *J Compos Constr* 22(2):04018001. [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0000831](https://doi.org/10.1061/(ASCE)CC.1943-5614.0000831)
159. Barakat S, Alateyat A, Junaid MT, Altoubat S, Maalej M, Awad R, Metawally A (2025) Experimental study on the punching shear of GFRP-RC slabs with normal and high strength concrete. *J Build Eng* 102:111975. <https://doi.org/10.1016/j.jobe.2025.111975>
160. Zhou K, Qi J, Wang J (2023) Post-cracking punching shear behavior of concrete flat slabs partially reinforced with full-depth UHPC: Experiment and mechanical model. *Eng Struct* 275:115313. <https://doi.org/10.1016/j.engstruct.2022.115313>
161. Qi J, Cheng Z, Zhou K, Zhu Y, Wang J, Bao Y (2021) Experimental and theoretical investigations of UHPC-NC composite slabs subjected to punching shear-flexural failure. *J Build Eng* 44:102662. <https://doi.org/10.1016/j.jobe.2021.102662>
162. Zohrevand P, Yang X, Jiao X, Mirmiran A (2015) Punching shear enhancement of flat slabs with partial use of ultrahigh-performance concrete. *J Mater Civil Eng*. [https://doi.org/10.1061/\(ASCE\)MT.1943-5533.0001219](https://doi.org/10.1061/(ASCE)MT.1943-5533.0001219)
163. Awad R, Altoubat S, Al Ateyat A, Junaid MT, Maalej M, Barakat S, Al-Sadoon Z (2024) Punching shear capacity of fiber-reinforced concrete suspended slabs: Database analysis and models assessments. *J Build Eng* 83:108433. <https://doi.org/10.1016/j.jobe.2023.108433>
164. Altoubat S, Maalej M, Nassif N, Barakat S (2023) Punching shear behavior of RC slabs incorporating macro-synthetic fibers. *J Build Eng* 67:105983. <https://doi.org/10.1016/j.jobe.2023.105983>
165. Bassurucu M, Turk K, Turgut P (2023) The effect of hybrid fiber and shear stud on the punching performance of flat-slab systems. *J Build Eng* 77:107555. <https://doi.org/10.1016/j.jobe.2023.107555>
166. Hernández Fraile D, Faccin D, Minelli F, Plizzari G, Muttoni A (2024) Fibre orientation in SFRC slabs and consequences for punching shear and flexural resistance. *Eng Struct* 302:117364. <https://doi.org/10.1016/j.engstruct.2023.117364>
167. Alves V, Oliveira L, Santos A, Trautwein L (2021) Experimental evaluation of the load carrying capacity of SFRC flat slabs varying the rectangularity index of columns. *KSCE J Civ Eng* 25(2):574–585. <https://doi.org/10.1007/s12205-021-0456-4>
168. Tan KH, Venkateshwaran A (2019) Punching shear in steel fiber-reinforced concrete slabs with or without traditional reinforcement. *ACI Struct J* 116(3):107–118. <https://doi.org/10.14359/51713291>
169. Gouveia ND, Faria DMV, Ramos AP (2019) Assessment of SFRC flat slab punching behaviour–part I: monotonic vertical loading. *Mag Concr Res* 71(11):587–598. <https://doi.org/10.1680/jmacr.17.00343>
170. Gouveia ND, Lapi M, Orlando M, Faria DMV, Ramos AP (2018) Experimental and theoretical evaluation of punching strength of steel fiber reinforced concrete slabs. *Struct Concr* 19:217–229. <https://doi.org/10.1002/suco.201700136>
171. Di Prisco M, Felicetti R (1997) Some results on punching shear in plain and fibre-reinforced micro-concrete slabs. *Mag Concr Res* 49(180):201–219. <https://doi.org/10.1680/jmacr.1997.49.180.201>
172. Mostafaei H, Vecchio FJ, Gauvreau P, Semelawy M (2011) Punching shear behavior of externally prestressed concrete slabs. *J Struct Eng* 137(1):100. [https://doi.org/10.1061/\(ASCE\)ST.1943-541X.0000283](https://doi.org/10.1061/(ASCE)ST.1943-541X.0000283)
173. Barros JAO, Sanz B, Filho M, Kabele P, Yu RC, Meschke G et al (2024) Blind competition on the numerical simulation of slabs reinforced with conventional flexural reinforcement and fibers subjected to punching loading configuration. *Struct Concr*. <https://doi.org/10.1002/suco.202400061>
174. Sahoo S, Singh B (2022) Punching shear capacity of steel-fibre recycled aggregate concrete slab. *Mag Concr Res* 74(17):865–878. <https://doi.org/10.1680/jmacr.21.00059>
175. Isufi B, Relvas JP, Marchão C, Ramos AP (2022) Behavior of flat slabs with partial use of high-performance fiber reinforced concrete under monotonic vertical loading. *Eng Struct* 264:114471. <https://doi.org/10.1016/j.engstruct.2022.114471>

176. Ramos A, Isufi B, Marreiros R, Bolesová M, Gajdosová K (2022) Rational use of HPFRC in slab column connections under reversed horizontal cyclic loading. *Eng Struct* 270:114903. <https://doi.org/10.1016/j.engstruct.2022.114903>
177. Sanabria Díaz R, Isufi B, Trautwein LM, Ramos AP (2022) Non-linear analysis of flat slab-column connections to optimize the use of HPFRC under monotonic vertical loading. *Struct Concr* 24(5):5787–5807. <https://doi.org/10.1002/suco.202200826>
178. Lampropoulos A, Tsioulou O, Mina A, Nicolaides D, Petrou MF (2024) Punching shear and flexural performance of ultra-high performance fibre reinforced concrete (UHPFRC) slabs. *Eng Struct* 281:115808. <https://doi.org/10.1016/j.engstruct.2023.115808>
179. Goldyn M, Urban T (2022) UHPFRC hidden capitals as an alternative method for increasing punching shear resistance of LWAC flat slabs. *Eng Struct* 271:114906. <https://doi.org/10.1016/j.engstruct.2022.114906>
180. Shoukry ME, Tarabia AM, Yassin AM (2020) Punching shear strength of ultra-high-performance fibre concrete slab-column connections. *Procedia Instit Civil Eng—Struct Build* 175(8):644–656. <https://doi.org/10.1680/jstbu.19.00201>
181. Elsayed M, Abdel-Hady I, Abdel-Hafez LM, Tawfic YR (2022) Strengthening of slab-column connections using ultra high-performance fiber concrete. *Case Stud Constr Mater* 17:e07110. <https://doi.org/10.1016/j.cscm.2022.e07110>
182. Kadhim MAM, Saleh AR, Cunningham LS, Semendary AA (2021) Numerical investigation of non-shear-reinforced UHPC hybrid flat slabs subject to punching shear. *Eng Struct* 241:112444. <https://doi.org/10.1016/j.engstruct.2021.112444>
183. Wu X, Yu S, Xue S, Kang TH-K, Hwang H-J (2019) Punching shear strength of UHPFRC-RC composite flat plates. *Eng Struct* 184(1):278–286. <https://doi.org/10.1016/j.engstruct.2019.01.099>
184. Sousa AMD, Lantsoght EOL, Genikomsou AS, Krahli PA, El Debs MK (2021) Behavior and punching capacity of flat slabs with the rational use of UHPFRC: NLFEA and analytical predictions. *Eng Struct* 244:112774. <https://doi.org/10.1016/j.engstruct.2021.112774>
185. El Zareef MA, Abdulrahman AG, Alnemari A (2023) Experimental investigation of punching shear behaviour of ultra-high performance self-compacting concrete slabs. *Case Stud Constr Mater* 19:e02307. <https://doi.org/10.1016/j.cscm.2023.e02307>
186. Zamri NF, Mohamed RN, Awalluddin D, Abdullah R (2022) Experimental evaluation on punching shear resistance of steel fibre reinforced self-compacting concrete flat slabs. *J Build Eng* 52:104441. <https://doi.org/10.1016/j.jobbe.2022.104441>
187. Nguyen TN, Nguyen TT, Pansuk W (2017) Experimental study of the punching shear behavior of high performance steel fiber reinforced concrete slabs considering casting directions. *Eng Struct* 131:564–573. <https://doi.org/10.1016/j.engstruct.2016.10.031>
188. Teixeira MDE, Barros JAO, Cunha VMCF, Moraes-Neto BN, Ventura-Gouveia A (2015) Numerical simulation of the punching shear behaviour of self-compacting fibre reinforced flat slabs. *Constr Build Mater* 74(15):25–36. <https://doi.org/10.1016/j.conbuildmat.2014.10.003>
189. El-kassas AI, Bashandy AA, Eied F-AM, Arab MAE-S (2022) Effect of fibers type on the behavior of fibrous high-strength self-compacted reinforced concrete flat slabs in punching with and without shear reinforcement. *Constr Build Mater* 360:129625. <https://doi.org/10.1016/j.conbuildmat.2022.129625>
190. Bastien-Masse M, Bruhwiler E (2016) Experimental investigation on punching resistance of R-UHPFRC–RC composite slabs. *Mater Struct* 49:1573–1590. <https://doi.org/10.1617/s11527-015-0596-4>
191. Lin C, Shi Z, Kanstad T, Baghban MHM, Ji G (2025) Application of steel fiber reinforced-concrete in post-tensioned flat slabs: A numerical study. *Eng Struct* 324:119347. <https://doi.org/10.1016/j.engstruct.2024.119347>
192. Kitano Y, Ito H, Suzuki S (2021) Punching shear capacity of prestressed concrete slabs made of lightweight concrete mixed with short fibers. *J Adv Concr Technol* 19:1227–1244. <https://doi.org/10.3151/jact.19.1227>
193. Hassanli R, Youssef O, Elchalakani M, El-Naqeeb NH, Ghanbari-Ghazijahani T, Yekrangnia M, Bazli M (2024) Investigation of punching shear performance in concrete slabs reinforced with GFRP and synthetic fibers: An experimental study. *Eng Struct* 311:118215. <https://doi.org/10.1016/j.engstruct.2024.118215>
194. Al Hamaydeh M, Orabi MA (2021) Punching shear behavior of synthetic fiber-reinforced self-consolidating concrete flat slabs with GFRP bars. *J Compos Constr*. [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0001131](https://doi.org/10.1061/(ASCE)CC.1943-5614.0001131)
195. Xiao RY, Chin CS (2007) Flat slabs at slab-column connection: nonlinear finite element modelling and punching shear capacity design criterion. *Adv Struct Eng* 10(5):567–579. <https://doi.org/10.1260/136943307782417717>
196. Labib WA (2020) Evaluation of hybrid fibre-reinforced concrete slabs in terms of punching shear. *Constr Build Mater* 260:119763. <https://doi.org/10.1016/j.conbuildmat.2020.119763>
197. Choi K-K, Truong GT, Kim SD, Choi I-R (2015) Punching shear behaviour of shear reinforced concrete slabs. *Proc Inst Civ Eng Struct Build* 168(SB6):402–420. <https://doi.org/10.1680/stbu.14.00062>
198. Chanthabouala K, Teng S, Chandra J, Tan K-H, Ostertag CP (2018) Punching tests of double-hooked-end fiber reinforced concrete slabs. *ACI Struct J* 115(6):1777–1789. <https://doi.org/10.14359/5170689>
199. Barros JAO, Moraes Neto BN, Melo GSSA, Frazão CMV (2015) Assessment of the effectiveness of steel fibre reinforcement for the punching resistance of flat slabs by experimental research and design approach. *Compos Part B* 78:8–25. <https://doi.org/10.1016/j.compositesb.2015.03.050>
200. Afshani Z, Rofooei FR (2025) Experimental study of lightweight and normal-weight concrete slabs under impact loads: The influence of prestressing and fiber reinforcement. *Constr Build Mater* 472:140863. <https://doi.org/10.1016/j.conbuildmat.2025.140863>
201. Carmo RNF, Costa H, Rodrigues M (2016) Experimental study of punching failure in LWAC slabs with different strengths. *Mater Struct* 49:2611–2626. <https://doi.org/10.1617/s11527-015-0671-x>
202. Ahmed FR (2023) Punching shear capacity of expanded polystyrene (EPS) lightweight concrete flat slabs without shear reinforcement. *Adv Mech Eng* 15(9):1–11. <https://doi.org/10.1177/16878132231199885>
203. Caratelli A, Imperatore S, Meda A, Rinaldi Z (2016) Punching shear behavior of lightweight fiber reinforced concrete slabs. *Compos B* 99:257–265. <https://doi.org/10.1016/j.compositesb.2016.06.045>
204. Youm K-S, Kim JJ, Moon J (2014) Punching shear failure of slab with lightweight aggregate concrete (LWAC) and low reinforcement ratio. *Constr Build Mater* 65(29):92–102. <https://doi.org/10.1016/j.conbuildmat.2014.04.097>
205. Yan JB, Wang J-Y, Liew JYL, Qian X, Zhang W (2016) Reinforced ultra-lightweight cement composite flat slabs: Experiments and analysis. *Mater Des* 95:148–158. <https://doi.org/10.1016/j.matdes.2016.01.097>
206. Urban T, Goldyn M, Krawczyk L, Sowa L (2019) Experimental investigations on punching shear of lightweight aggregate concrete flat slabs. *Eng Struct* 197:109371. <https://doi.org/10.1016/j.engstruct.2019.109371>
207. Boules P, Fam A, Genikomsou A (2023) Punching shear of flat slabs with pultruded GFRP stay-in-place structural forms. *J Compos Constr*. <https://doi.org/10.1061/JCCOF2.CCENG-4076>

208. Elbady A, Mousa S, Mohamed HM, Benmokrane B (2023) Effect of test parameters on the punching-shear strength of precast tunnel segments reinforced with GFRP bars. *Eng Struct* 297:117016. <https://doi.org/10.1016/j.engstruct.2023.117016>
209. Kim MS, Lee YH (2021) Punching shear strength of reinforced concrete flat plates with GFRP vertical grids. *Appl Sci* 11:2736. <https://doi.org/10.3390/app11062736>
210. Eladawy M, Hassan M, Benmokrane B, Fam A (2020) Effect of stirrups on lateral cyclic behavior of interior glass fiber-reinforced polymer-reinforced concrete slab column connections. *ACI Struct J*. <https://doi.org/10.14359/51724671>
211. Hassan M, Ahmed EA, Benmokrane B (2015) Punching shear behavior of two-way slabs reinforced with FRP shear reinforcement. *J Compos Constr*. [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0000493](https://doi.org/10.1061/(ASCE)CC.1943-5614.0000493)
212. Said M, Adam MA, Arafa AE, Moatasem A (2020) Improvement of punching shear strength of reinforced lightweight concrete flat slab using different strengthening techniques. *J Build Eng* 32:101749. <https://doi.org/10.1016/j.jobbe.2020.101749>
213. Zhang R, Zhao X, Jin L, Du X (2023) Effect of elevated temperature on the low-velocity impact performances of GFRP reinforced concrete slabs. *Constr Build Mater* 395:132285. <https://doi.org/10.1016/j.conbuildmat.2023.132285>
214. Jafarian N, Mostofinejad D, Naderi A (2020) Effects of FRP grids on punching shear behavior of reinforced concrete slabs. *Structures* 28:2523–2536. <https://doi.org/10.1016/j.istruc.2020.10.061>
215. Hamoda A, Hossain KMA (2019) Numerical assessment of slab-column connection additionally reinforced with steel and CFRP bars. *Arab J Sci Eng* 44:8181–8204. <https://doi.org/10.1007/s13369-019-03846-2>
216. Al-Rousan RZ, Alhassan M, Al-wadi R (2020) Nonlinear finite element analysis of full-scale concrete bridge deck slabs reinforced with FRP bars. *Structures* 27:1820–1831. <https://doi.org/10.1016/j.istruc.2020.08.024>
217. Akhundzada H, Donchev T, Petkova D (2019) Punching shear resistance of flat slabs strengthened with near surface-mounted CFRP bars. *J Compos Constr*. [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0001146](https://doi.org/10.1061/(ASCE)CC.1943-5614.0001146)
218. Huang Z, Zhao Y, Zhang J, Wu Y (2020) Punching shear behaviour of concrete slabs reinforced with CFRP grids. *Structures* 26:617–625. <https://doi.org/10.1016/j.istruc.2020.04.047>
219. Qian K, Weng Y-H, Liang S-L, Fu F (2024) Punching shear behavior and corrosion resistance of composite slab-column connections reinforced by BFRP and steel rebar. *Eng Struct* 304:117589. <https://doi.org/10.1016/j.engstruct.2024.117589>
220. Zheng Y, Zhou L, Xia L, Luo Y, Taylor SE (2018) Investigation of the behaviour of SCC bridge deck slabs reinforced with BFRP bars under concentrated loads. *Eng Struct* 171:500–515. <https://doi.org/10.1016/j.engstruct.2018.05.105>
221. Elgabbas F, Ahmed EA, Benmokrane B (2016) Experimental testing of concrete bridge-deck slabs reinforced with basalt-FRP reinforcing bars under concentrated Loads. *J Bridg Eng* 21(7):04016029. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0000892](https://doi.org/10.1061/(ASCE)BE.1943-5592.0000892)
222. El Zareef MA, Badawi M, Elbisy AM (2023) Review and validation of code expressions and developed models in last three decades estimating concrete shear strength of FRP-reinforced members without stirrups. *Case Stud Constr Mater* 19:e02352. <https://doi.org/10.1016/j.cscm.2023.e02352>
223. Liao JJ, Zeng JJ, Lin X-C, Zhuge Y, He SH (2023) Punching shear behavior of FRP grid-reinforced ultra-high performance concrete slabs. *J Compos Constr*. <https://doi.org/10.1061/JCCOF2.CCENG-4148>
224. Qian K, Liu J-G, Yu X-H, Weng Y-H (2022) Experimental and numerical investigation of punching shear capacity of corroded reinforced concrete slab-column connections. *Structures* 43:1548–1557. <https://doi.org/10.1016/j.istruc.2022.07.065>
225. Gomaa AM, Lotfy EM, Khafaga SA, Hosny S, Ahmed MA (2023) Experimental, numerical, and theoretical study of punching shear capacity of corroded reinforced concrete slab-column joints. *Eng Struct* 289:116280. <https://doi.org/10.1016/j.engstruct.2023.116280>
226. Huang X-N, Tong J-Z, Li Q-H, Xu S-L (2023) Numerical and analytical investigations on punching shear performance of UHTCC-enhanced RC slab-column joints. *J Build Eng* 79:107900. <https://doi.org/10.1016/j.jobbe.2023.107900>
227. Alkhattabi L, Ayash NM, Hassan M, Gouda A (2024) Investigation of key parameters influencing shear behavior in glass-fiber-reinforced polymer (GFRP)-reinforced concrete (RC) interior slab-column connections. *Buildings* 14(5):1251. <https://doi.org/10.3390/buildings14051251>
228. Oliveira DRC, Regan PE, Melo GSSA (2004) Punching resistance of RC slabs with rectangular columns. *Mag Concr Res* 56(3):123–138. <https://doi.org/10.1680/macrc.2004.56.3.123>
229. Madkour H, Maher M, Ali O (2022) Finite element analysis for interior slab-column connections reinforced with GFRP bars using damage plasticity model. *J Build Eng* 48:104013. <https://doi.org/10.1016/j.jobbe.2022.104013>
230. Yehia E, Khalil AH, Mostafa E-E, El-Nazzer MA (2023) Experimental and numerical investigation on punching behavior of ultra-high performance concrete flat slabs. *Ain Shams Eng J* 14(10):102208. <https://doi.org/10.1016/j.asej.2023.102208>
231. Mostafa HMF, Mahmoud AA, Mostafa TS, Khater ANM (2024) Experimental and analytical punching shear behavior of reinforced concrete flat slabs having internal GFRP gratings. *Structures* 60:105882. <https://doi.org/10.1016/j.istruc.2024.105882>
232. Ors DM, Ramadan M, Maree AMF, Zaher AH, Afifi A, Ebid AM (2024) Machine learning base models to predict the punching shear capacity of posttensioned UHPC flat slabs. *Sci Rep* 14:3969. <https://doi.org/10.1038/s41598-024-54358-5>
233. Esfahani MR, Kianoush MR, Moradi AR (2009) Punching shear strength of interior slab column connections strengthened with carbon fiber reinforced polymer sheets. *Eng Struct* 31:15351542. <https://doi.org/10.1016/j.engstruct.2009.02.021>
234. Soudki K, El-Sayedb AK, Vanzwol T (2012) Strengthening Concrete Slab-column connections using CFRP strips. *J King Saud Univ Eng Sci* 24:25–33. <https://doi.org/10.1016/j.jksues.2011.07.001>
235. Sahib MQA, Aghayari R, Moradi MJ (2024) Roudsari MT (2024) Experimental investigation on the rehabilitation of RC flat slabs using CFRP sheets to enhance punching shear capacity. *Buildings* 14:153. <https://doi.org/10.3390/buildings14010153>
236. Ghayeb HH, Atea RS, Al-Kannoon MAA, Lee FW, Wong LS, Mo KH (2024) Performance of reinforced concrete flat slab strengthened with CFRP for punching shear. *Case Stud Constr Mater* 18:e01801. <https://doi.org/10.1016/j.cscm.2022.e01801>
237. Torabian A, Mostofinejad D (2024) Strengthening of two-way slabs with fiber-reinforced polymer composites: A new system using grooving technique and fans. *Structures* 62:106228. <https://doi.org/10.1016/j.istruc.2024.106228>
238. Mansour W, Li W, Ghalla M, Badawi M, El-Zareef MA (2024) Improving the punching capacity of two-way RC flat slabs via external strengthening using various configurations of aluminum sheets. *Constr Build Mater* 420:135611. <https://doi.org/10.1016/j.conbuildmat.2024.135611>
239. Ghalla M, Mansour W, Li W, Wang P, Badawi M, El-Zareef MA (2024) Enhancing the punching performance of two-way RC flat slabs using different configurations of embedded aluminum sections: Experimental program and numerical analysis. *Constr Build Mater* 434:136737. <https://doi.org/10.1016/j.conbuildmat.2024.136737>

240. Mercimek O, Ghoroubi R, Erbaş Y, Anil O (2022) Comparison of strengthening methods to improve punching behavior of two-way RC flat slabs. *Structures* 46:1495–1516. <https://doi.org/10.1016/j.istruc.2022.11.018>
241. Santos GS, Melo GSSA, Barros JAO (2019) Punching CFRP-based strengthening solutions for reinforced concrete flat slabs. *Compos Struct* 224:111077. <https://doi.org/10.1016/j.composstruct.2019.111077>
242. Saleh H, Kalfat R, Abdouka K, Al-Mahaidi R (2019) Punching shear strengthening of RC slabs using L-CFRP laminates. *Eng Struct* 194:274–289. <https://doi.org/10.1016/j.engstruct.2019.05.050>
243. Erdogan H, Binici B (2010) Ozcebe G (2010) Punching shear strengthening of flat-slabs with CFRP dowels. *Mag Concr Res* 62(7):465–478. <https://doi.org/10.1680/mac.2010.62.7.465>
244. Gherdaoui M, Guenfoud M, Madi R (2018) Punching behavior of strengthened and repaired RC slabs with CFRP. *Constr Build Mater* 170:272–278. <https://doi.org/10.1016/j.conbuildmat.2018.03.093>
245. Saadoun AM, Mashrei MA, Oumari KAA (2022) Punching shear strength of recycled aggregate-steel fibrous concrete slabs with and without strengthening. *Adv Struct Eng* 25(10):2175–2190. <https://doi.org/10.1177/13694332221090288>
246. Farghaly AS, Ueda T (2010) Prediction of punching shear strength of two-way slabs strengthened externally with FRP sheets. *J Compos Constr*. [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0000177](https://doi.org/10.1061/(ASCE)CC.1943-5614.0000177)
247. Radnić L, Matešan D, Grgić N, Baloević G (2015) Impact testing of RC slabs strengthened with CFRP strips. *Compos Struct* 121:90–103. <https://doi.org/10.1016/j.compstruct.2014.10.033>
248. Barros JAO, Rezazadeh M, Laranjeira JPS, Hosseini MRM, Mastali M (2017) Ramezansefat H (2017) Simultaneous flexural and punching strengthening of RC slabs according to a new hybrid technique using U-shape CFRP laminates. *Compos Struct* 159(1):600–614. <https://doi.org/10.1016/j.compstruct.2016.10.009>
249. Silva MAL, Dedigamuwa KV, Gamage JCPH (2021) Performance of severely damaged reinforced concrete flat slab-column connections strengthened with Carbon Fiber Reinforced Polymer. *Compos Struct* 255:112963. <https://doi.org/10.1016/j.composstruct.2020.112963>
250. Koppitz R, Kenel A, Keller T (2014) Punching shear strengthening of flat slabs using prestressed carbon fiber-reinforced polymer straps. *Eng Struct* 76:283–294. <https://doi.org/10.1016/j.engstruct.2014.07.017>
251. Meisami MH, Mostofinejad D, Nakamura H (2014) Punching shear strengthening of two-way flat slabs with CFRP grids. *J Compos Constr*. [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0000443](https://doi.org/10.1061/(ASCE)CC.1943-5614.0000443)
252. Abdullah A, Bailey CG (2018) Punching behaviour of column-slab connection strengthened with non-prestressed or prestressed FRP plates. *Eng Struct* 160(1):229–242. <https://doi.org/10.1016/j.engstruct.2018.01.030>
253. Abdullah A, Bailey CG, Wu ZJ (2013) Tests investigating the punching shear of a column-slab connection strengthened with non-prestressed or prestressed FRP plates. *Constr Build Mater* 48:1134–1144. <https://doi.org/10.1016/j.conbuildmat.2013.07.012>
254. Durucan C, Anil O (2015) Effect of opening size and location on the punching shear behavior of interior slab-column connections strengthened with CFRP strips. *Eng Struct* 105:22–36. <https://doi.org/10.1016/j.engstruct.2015.09.033>
255. Abbas H, Siddiqui N, Almusallam T, Abadel A, Al-Salloum Y. (2022) Prediction of ballistic limit of strengthened reinforced concrete slabs using quasi-static punching test. *Buildings*, 12: 1815, <https://doi.org/10.3390/buildings12111815>
256. Le-Nguyen K, Nguyen X-H, Nguyen HC, Cao M-Q, Larbi AS, Djama ZI (2023) Experimental and numerical investigations of punching shear behavior of FRCM-strengthened two-way RC slabs. *J Compos Constr*. [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0001282](https://doi.org/10.1061/(ASCE)CC.1943-5614.0001282)
257. Hamoda A, Sennah K, Abadel AA, Emara M (2024) Strengthening of interior circular column-slab region incorporating combinations of bonded UHPECC layer, CFRP sheets, and threaded steel bolts. *Structures* 69:107545. <https://doi.org/10.1016/j.istruc.2024.107545>
258. Ghalla M, Badawi M, Hu JW, Elsamak G, Mlybari EA, Emara M (2025) Ultimate performance of two-way reinforced concrete flat slabs enhanced by SHCC drop panels mitigating punching failure. *J Build Eng* 99:111574. <https://doi.org/10.1016/j.jobbe.2024.111574>
259. Elbakry HMF, Allam SM (2015) Punching strengthening of two-way slabs using external steel plates. *Alex Eng J* 54:1207–1218. <https://doi.org/10.1016/j.aej.2015.09.005>
260. Taresh HR, Yatim MYM, Azmi MR (2021) Punching shear behaviour of interior slab-column connections strengthened by steel angle plates. *Eng Struct* 238:112246. <https://doi.org/10.1016/j.engstruct.2021.112246>
261. Ammash HK, Kadhim SS, Dhahir MK (2022) Repairing half-loaded flat slabs against punching shear using steel stiffeners. *Case Stud Constr Mater* 16:e01032. <https://doi.org/10.1016/j.cscm.2022.e01032>
262. Agbossou A, Michel L, Lagache M, Hamelin P (2008) Strengthening slabs using externally-bonded strip composites: analysis of concrete covers on the strengthening. *Compos B Eng* 39(7–8):1125–1135. <https://doi.org/10.1016/j.compositesb.2008.04.002>
263. Siqueira JPA, Melo GS (2022) Punching shear in flat slabs with open stirrups as shear reinforcement. *Struct Concr* 24(2):2827–2843. <https://doi.org/10.1002/suco.202200050>
264. Baig ZI, Alsayed SH, Abbas H (2016) Punching of slab-column connections strengthened using external steel shear bolts. *Mag Concr Res* 68(2):55–68. <https://doi.org/10.1680/mac.14.00434>
265. Saleh H, Kalfat R, Abdouka K, Al-Mahaidi R (2018) Experimental and numerical study into the punching shear strengthening of RC flat slabs using post-installed steel bolts. *Constr Build Mater* 188:28–39. <https://doi.org/10.1016/j.conbuildmat.2018.08.064>
266. Navarro M, Ivorra S, Varona FB (2020) Parametric finite element analysis of punching shear behaviour of RC slabs reinforced with bolts. *Comput Struct* 228:106147. <https://doi.org/10.1016/j.compstruct.2019.106147>
267. Jiang M, Shi Q (2024) Force mechanism and parametric finite element analysis of punching shear behavior of flat slabs with welded shear reinforcements. *KSCE J Civ Eng* 28:3477–3488. <https://doi.org/10.1007/s12205-024-1371-2>
268. Liu Y, Zhang W, Zhou M, Hou W, Zhang Y (2023) Punching shear performance of star-shaped steel plates reinforced RC slab-column connections. *Case Stud Constr Mater* 19:e02570. <https://doi.org/10.1016/j.cscm.2023.e02570>
269. Shatarat N, Salman D (2022) Investigation of punching shear behavior of flat slabs with different types and arrangements of shear reinforcement. *Case Stud Constr Mater* 16:e01028. <https://doi.org/10.1016/j.cscm.2022.e01028>
270. Al-Gasham TS, Mhalhal JM, Jabir HA (2019) Improving punching behavior of interior voided slab-column connections using steel sheets. *Eng Struct* 199:109614. <https://doi.org/10.1016/j.engstruct.2019.109614>
271. Habeeb M, Al-Azzawi AA, Al-Zwainy FMS (2021) Punching shear behavior of LWA bubble deck slab with different types of shear reinforcement. *J King Saud Univ Eng Sci* 33:15–22. <https://doi.org/10.1016/j.jksues.2020.01.001>

272. Sarvari S, Esfahani MR (2020) An experimental study on post-punching behavior of flat slabs. *Structures* 27:894–902. <https://doi.org/10.1016/j.istruc.2020.06.036>
273. Yankelevsky DZ, Karinski YS, Feldgun VR (2020) Dynamic punching shear failure of a RC flat slab-column connection under a collapsing slab impact. *Int J Impact Eng* 135:103401. <https://doi.org/10.1016/j.ijimpeng.2019.103401>
274. Najmi AQ, Allouzi R, Al-Ateyat A (2021) Analysis of reinforced concrete plates with swimmer bars as punching shear reinforcement. *Proceed Institut Civil Eng—Struct Build* 174(11):920. <https://doi.org/10.1680/jstbu.18.00135>
275. Papadopoulos V, Murcia-Delso J, Benson PS (2018) Development of headed bars in slab-column joints of reinforced concrete slab bridges. *ACI Struct J* 115(5):1393–1406. <https://doi.org/10.14359/51702247>
276. Polo GE, Bayrak O, Hrynyk TD (2020) Shear-resisting performance of reinforced concrete flat plates with different headed stud layouts. *ACI Struct J*. <https://doi.org/10.14359/51728087>
277. Schmidt P, Kueres D, Hegger J (2020) Punching shear behavior of reinforced concrete flat slabs with a varying amount of shear reinforcement. *Struct Concr* 21:235–246. <https://doi.org/10.1002/suco.201900017>
278. Nicácio WG, Barros JAO, Melo GSSA (2020) Punching behavior of BubbleDeck type reinforced concrete slabs. *Struct Concr* 21:262–277. <https://doi.org/10.1002/suco.201900176>
279. Said M, Mahmoud AA, Salah A (2020) Performance of reinforced concrete slabs under punching loads. *Mater Struct*. <https://doi.org/10.1617/s11527-020-01509-5>
280. Setiawan A, Vollum RL, Bassam LM, Izzuddin A (2021) Numerical modelling of punching shear failure of reinforced concrete flat slabs with shear reinforcement. *Mag Concr Res* 73(23):1205–1224. <https://doi.org/10.1680/jmacr.19.00562>
281. Ewees MH, Gabr ASA, Farrag MR (2024) Effect of tension and compression flexural reinforcement on punching shear strength of reinforced concrete flat slab. *Alex Eng J* 99:282–302. <https://doi.org/10.1016/j.aej.2024.05.019>
282. Bolešová M, Gajdošová K (2023) Experimental evaluation of flat slabs strengthened with post-installed shear bolts with pre-loading. *Eng Struct* 295:116659. <https://doi.org/10.1016/j.engstruct.2023.116659>
283. Balomenos GP, Genikomsou AS, Polak MA, Pandey MD (2015) Efficient method for probabilistic finite element analysis with application to reinforced concrete slabs. *Eng Struct* 103:85–101. <https://doi.org/10.1016/j.engstruct.2015.08.038>
284. Genikomsou AS, Polak MA (2016) Finite-element analysis of reinforced concrete slabs with punching shear reinforcement. *J Struct Eng*. [https://doi.org/10.1061/\(ASCE\)ST.1943-541X.0001603](https://doi.org/10.1061/(ASCE)ST.1943-541X.0001603)
285. Wörle P (2014) Enhanced shear punching capacity by the use of post installed concrete screws. *Eng Struct* 60:41–51. <https://doi.org/10.1016/j.engstruct.2013.12.015>
286. Spiegl M, Walkner R, Feix J (2021) Concrete screws as a post-installed punching reinforcement. *Struct Concr* 22:2289–2308. <https://doi.org/10.1002/suco.202100073>
287. Walkner R, Spiegl M, Feix J (2021) Concrete screws as a post-installed punching shear reinforcement. *Struct Concr* 22:709–725. <https://doi.org/10.1002/suco.202000427>
288. Feiri T, Schulze-Ardey JP, Ricker M, Hegger J (2023) Using system reliability concepts to derive partial safety factors for punching shear with shear reinforcement: an explorative analysis. *Appl Sci* 13:1360. <https://doi.org/10.3390/app13031360>
289. Lewiński PM, Więch PP (2020) Finite element model and test results for punching shear failure of RC slabs. *Arch Civil Mech Eng*. <https://doi.org/10.1007/s43452-020-00037-x>
290. Oliveira VHD, Lima HJN, Melo GS (2022) Punching shear resistance of flat slabs with different types of stirrup anchorages such as shear reinforcement. *Eng Struct* 253:113671. <https://doi.org/10.1016/j.engstruct.2021.113671>
291. Lima H, Palhares R, Sales de Melo G, Oliveira M (2021) Experimental analysis of punching shear in flat slabs with variation in the anchorage of shear reinforcement. *Struct Concr* 22:1165–1182. <https://doi.org/10.1002/suco.202000158>
292. Valivonis J, Skuturna T, Daugevicius M, Šneideris A (2017) Punching shear strength of reinforced concrete slabs with plastic void formers. *Constr Build Mater* 145:518–527. <https://doi.org/10.1016/j.conbuildmat.2017.04.057>
293. Oukaili N, Merie H, Allawi A, Wardeh G (2024) Reduced volume approach to evaluate biaxial Bubbled slabs' resistance to punching shear. *Buildings* 14:676. <https://doi.org/10.3390/buildings14030676>
294. Sagadevan R, Rao BN (2019) Experimental and analytical investigation of punching shear capacity of biaxial voided slabs. *Structures* 20:340–352. <https://doi.org/10.1016/j.istruc.2019.03.013>
295. Schnellenbach-Held M, Pfeffer K (2002) Punching behavior of biaxial hollow slabs. *Cement Concr Compos* 24(6):551–556. [https://doi.org/10.1016/S0958-9465\(01\)00071-3](https://doi.org/10.1016/S0958-9465(01)00071-3)
296. Chung J-H, Bae B-I, Choi H-K, Jung H-S, Choi C-K (2018) Evaluation of punching shear strength of voided slabs considering the effect of the ratio b_0/d . *Eng Struct* 164:70–81. <https://doi.org/10.1016/j.engstruct.2018.02.085>
297. Valivonis J, Šneideris A, Šalna R, Popov V, Daugevicius M, Jonaitis B (2017) Punching strength of biaxial voided slabs. *ACI Struct J* 114(6):1373–1383. <https://doi.org/10.14359/51700912>
298. Han C-X, He H-L, Luo Q-J, Zhao Y-L, Zhang J-C (2024) Experimental and theoretical study on static performance of prefabricated steel-bar truss frame-precast concrete composite hollow-core slabs. *Structures* 61:105971. <https://doi.org/10.1016/j.istruc.2024.105971>
299. Souza SSM, Shehata IAEM, Shehata LCD (2016) Shear resistance of reinforced concrete waffle flat slabs around the solid panel. *Mater Struct* 49:1367–1380. <https://doi.org/10.1617/s11527-015-0582-x>
300. Al-Bayati AF, Lau TL, Clark LA (2015) Concentric punching shear of waffle slab. *ACI Struct J* 112(5):533–542. <https://doi.org/10.14359/51687906>
301. Al-Bayati AF, Lau TL, Clark LA (2023) Edge punching shear of waffle slabs subjected to moment parallel to the slab's free edge. *Mag Concr Res* 75(2):55–65. <https://doi.org/10.1680/jmacr.21.00215>
302. Grabski M, Ambroziak A (2023) Experimental and analytical analysis of punching shear in flat slabs supported on column topped with concrete head. *Constr Build Mater* 364:129961. <https://doi.org/10.1016/j.conbuildmat.2022.129961>
303. Kang S-M, Na S-J, Hwang H-J (2022) Punching shear strength improved by upward panel in reinforced concrete transfer slabs. *J Build Eng* 46:103753. <https://doi.org/10.1016/j.jobbe.2021.103753>
304. Qian K, Li B (2015) Load-resisting mechanism to mitigate progressive collapse of flat slab structures. *Mag Concr Res* 67(7):349–363. <https://doi.org/10.1680/jmacr.14.00293>
305. Yankelevsky DZ, Karinski YS, Brodsky A, Feldgun VR (2021) Dynamic punching shear of impacting RC flat slabs with drop panels. *Eng Fail Anal* 129:105682. <https://doi.org/10.1016/j.engfailanal.2021.105682>
306. Grabski M, Ambroziak A (2022) Influence of the shear cap size and stiffness on the distribution of shear forces in flat slabs. *Materials* 15:188. <https://doi.org/10.3390/ma15010188>
307. Chen J, Zhang R, Wang Z, Wu S (2023) Punching shear behaviour of steel-concrete composite slab-column connections with spatial steel drop panel. *Eng Struct* 28:115875. <https://doi.org/10.1016/j.engstruct.2023.115875>

308. Sousa AMD, Lantsoght EOL, El Debs MK (2023) Shear and punching capacity predictions for one-way slabs under concentrated loads considering the transition between failure mechanisms. *Buildings* 13:434. <https://doi.org/10.3390/buildings13020434>
309. Sousa AMD, Lantsoght EOL, El Debs MK (2023) Failure mechanism of one-way slabs under concentrated loads after local reinforcement yielding. *Eng Struct* 291:116396. <https://doi.org/10.1016/j.engstruct.2023.116396>
310. Sousa AMD, Lantsoght EOL, El Debs MK (2023) Transition between shear and punching in reinforced concrete slabs: review and predictions with ACI Code expressions. *ACI Struct J* 120(2):115–128. <https://doi.org/10.14359/51738350>
311. Lantsoght EOL, Veen CVD, Walraven JC (2013) Shear in one-way slabs under concentrated load close to support. *ACI Struct J* 110(2):275–284. <https://doi.org/10.14359/51684407>
312. Díaz RS, Trautwein LM, de Almeida LC (2021) Numerical investigation of the punching shear capacity of unbonded post tensioned concrete flat slabs. *Struct Concr* 22:1205–1222. <https://doi.org/10.1002/suco.202000448>
313. Ramos AP, Lúcio VJG, Faria DMV (2014) The effect of the vertical component of prestress forces on the punching strength of flat slabs. *Eng Struct* 76:90–98. <https://doi.org/10.1016/j.engstruct.2014.06.039>
314. Yang T, Liu Z, Lian J (2021) Progressive collapse of RC flat slab substructures with unbonded post tensioning strands after the loss of an exterior column. *Eng Struct* 234:111989. <https://doi.org/10.1016/j.engstruct.2021.111989>
315. Amir S, van der Veen C, Walraven JC, de Boer A (2019) Numerical investigation of the punching shear capacity of transversely prestressed concrete deck slabs. *Struct Concr* 20:1109–1122. <https://doi.org/10.1002/suco.201800285>
316. Jiao Z, Li Y, Guan H, Diao M, Sun H, Zhao Z, Gilbert BP (2024) Experimental and numerical investigation of punching and post-punching shear capacities of post-tensioned flat plate slab-column joints influenced by prestressing levels. *Eng Struct* 313:118259. <https://doi.org/10.1016/j.engstruct.2024.118259>
317. Thai D-K, Kim S-E (2017) Numerical simulation of pre-stressed concrete slab subjected to moderate velocity impact loading. *Eng Fail Anal* 79:820–835. <https://doi.org/10.1016/j.engfailanal.2017.05.020>
318. Wang Z, Guo Q, Zhang Z (2023) Dynamic response of bidirectional bonded prestressed concrete slabs subject to low velocity impact: Impact energy and prestress degree. *Eng Struct* 289:116355. <https://doi.org/10.1016/j.engstruct.2023.116355>
319. Abandah MR, Issa MA (2022) Influence of reinforcement parameters on punching shear capacity of laterally restrained FRP-reinforced concrete bridge deck slabs. *Structures* 41:434–446. <https://doi.org/10.1016/j.istruc.2022.04.074>
320. Zheng Y, Robinson D, Taylor S, Cleland D (2012) Non-linear finite-element analysis of punching capacities of steel–concrete bridge deck slabs. *Struct Build* 165(SB5):255–269. <https://doi.org/10.1680/stbu.10.00003>
321. Cook W, Barr PJ, Brackus TR (2014) Flexural and punching shear capacity comparison of full-scale ABC steel bridge girders. *J Perform Constr Facil*. [https://doi.org/10.1061/\(ASCE\)CF.1943-5509.0000545](https://doi.org/10.1061/(ASCE)CF.1943-5509.0000545)
322. Hon A, Taplin G, Al-Mahaidi RS (2005) Strength of reinforced concrete bridge decks under compressive membrane action. *ACI Struct J* 102(3):393–401. <https://doi.org/10.14359/14410>
323. Belletti B, Walraven JC, Trapani F (2015) Evaluation of compressive membrane action effects on punching shear resistance of reinforced concrete slabs. *Eng Struct* 95:25–39. <https://doi.org/10.1016/j.engstruct.2015.03.043>
324. Setiawan A, Cantone R, Ruiz MF, Muttoni A (2024) Verification of shear failures of cantilever bridge deck slabs subjected to concentrated loads. *Eng Struct*, 303, 117491. <https://orcid.org/0000-0003-2791-6118>
325. Ebeido T, Kennedy JB (1996) Punching strength of deck slabs in skew composite bridges. *J Bridge Eng*. [https://doi.org/10.1061/\(ASCE\)1084-0702\(1996\)1:2\(59\)](https://doi.org/10.1061/(ASCE)1084-0702(1996)1:2(59))
326. Belletti B, Pimentel M, Scolari M, Walraven JC (2015) Safety assessment of punching shear failure according to the level of approximation approach. *Struct Concr* 16:366–380. <https://doi.org/10.1002/suco.201500015>
327. Makita T, Ichinomiya T, Watanabe Y (2024) Experimental investigation of fatigue behavior of UHPFRC – RC composite deck slabs subjected to moving wheel loading. *Eng Struct* 303:117460. <https://doi.org/10.1016/j.engstruct.2024.117460>
328. Elgabbas F, Ahmed EA, Benmokrane B (2016) Experimental testing of concrete bridge-deck slabs reinforced with basalt-FRP reinforcing bars under concentrated loads. *J Bridge Eng* 21(7):04016129. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0000892](https://doi.org/10.1061/(ASCE)BE.1943-5592.0000892)
329. Lantsoght EOL, Koekkoek R, Van der Veen C, Sliedrecht H (2019) Fatigue assessment of prestressed concrete slab-between-girder bridges. *Appl Sci* 9:2312. <https://doi.org/10.3390/app9112312>
330. Lantsoght E, Van der Veen C, Koekkoek R, Sliedrecht H (2019) Punching capacity of prestressed concrete bridge decks under fatigue. *ACI Struct J* 116(4):209–218. <https://doi.org/10.14359/51715563>
331. Yan P, Wang YC (2016) Hybrid steel tubular column/flat slab construction — Development of a shearhead system to improve punching shear resistance. *J Constr Steel Res* 119:154–168. <https://doi.org/10.1016/j.jcsr.2015.12.024>
332. Luu BT, Ngo-Huu C, Nguyen-Minh L (2022) Punching shear capacity of unbonded post-tensioned slab—Concrete filled steel tube column joints with innovative connections: Experiment and a new prediction model. *J Build Eng* 59:104974. <https://doi.org/10.1016/j.jobe.2022.104974>
333. Yan P, Wang YC (2014) Behaviour of steel tube reinforced concrete flat slab shearhead systems. *Struct Build* 167(SB11):667–677. <https://doi.org/10.1680/stbu.12.00091>
334. Eder MA, Vollum RL, Elghazouli AY A-F (2010) Modelling and experimental assessment of punching shear in flat slabs with shearheads. *Eng Struct* 32:3911–3924. <https://doi.org/10.1016/j.engstruct.2010.09.004>
335. Kim J-W, Lee C-H, Kang TH-K (2014) Shearhead reinforcement for concrete slab to concrete filled tube column connections. *ACI Struct J* 111(3):629–638. <https://doi.org/10.14359/51686623>
336. Bompá DV, Elghazouli AY (2016) Structural performance of RC flat slabs connected to steel columns with shear heads. *Eng Struct* 117:161–183. <https://doi.org/10.1016/j.engstruct.2016.03.022>
337. Yu JL, Wang YC (2018) Punching shear behavior and design of an innovative connection from steel tubular column to flat concrete slab. *J Struct Eng*. [https://doi.org/10.1061/\(ASCE\)ST.1943-541X.0002111](https://doi.org/10.1061/(ASCE)ST.1943-541X.0002111)
338. Wang Z, Yan J, Lin Y, Fan F, Gebbeken N (2024) Impact response of SCS sandwich slabs with ultra-high-performance concrete: Failure mechanism and influence of shear connector configuration. *Int J Impact Eng* 186:104889. <https://doi.org/10.1016/j.ijimpeng.2024.104889>
339. Wang Z, Yan J, Lin Y, Fan F (2021) Influence of shear connectors on the ultimate capacity of steel-UHPC-steel slabs subjected to concentrated loads. *Eng Struct* 231:111763. <https://doi.org/10.1016/j.engstruct.2020.111763>
340. Wang Z, Yan J, Lin Y, Fan F (2022) Punching shear performance of steel-fiber reinforced concrete-steel sandwich plates under different patch loading heads. *J Build Eng* 48:103955. <https://doi.org/10.1016/j.jobe.2021.103955>

341. Takahashi R, Sato Y, Konno K, Ueda T (2005) 3D Nonlinear punching shear simulation of steel-concrete composite slab. *J Adv Concr Technol* 3(2):297–307. <https://doi.org/10.3151/jact.3.297>
342. Farghaly A, Furuuchi H, Ueda T (2005) Punching shear failure mechanism of open sandwich slab and its parameters' effects. *J Adv Concr Technol* 3(2):283–296. <https://doi.org/10.3151/jact.3.283>
343. Mohammad SH, Gülsan ME, Çevik A (2022) Behaviour of geopolymer concrete two-way slabs reinforced by FRP bars after exposure to elevated temperatures. *Arab J Sci Eng* 47:12399–12421. <https://doi.org/10.1007/s13369-021-06411-y>
344. Arna'ot FH, Abbass AA, Abualtemen AA, Abid SR, Özakça M (2017) Residual strength of high strength concentric column-SFRC flat plate exposed to high temperatures. *Constr Build Mater* 154:204–218. <https://doi.org/10.1016/j.conbuildmat.2017.07.141>
345. Bing L, Zhou-lin Y, Zong-fu Z, Zheng-shang L, Ai-ping T, Yang Z (2024) Punching shear performance of reinforced flat slabs considering in-plane restraints under and after a fire. *Constr Build Mater* 457:139311. <https://doi.org/10.1016/j.conbuildmat.2024.139311>
346. Zhang C, Ma W, Liu X, Tian Y, Orton SL (2019) Effects of high temperature on residual punching strength of slab-column connections after cooling and enhanced post-punching load resistance. *Eng Struct* 199:109580. <https://doi.org/10.1016/j.engstruct.2019.109580>
347. Liu S, Xu X, Zhou B, Yang K (2023) Punching shear failure analysis of reinforced concrete slabs under close-in explosion. *Materials* 16:6339. <https://doi.org/10.3390/ma16186339>
348. Sagaseta J, Olmati P, Micallef K, Cormie D (2017) Punching shear failure in blast-loaded RC slabs and panels. *Eng Struct* 147:177–194. <https://doi.org/10.1016/j.engstruct.2017.04.051>
349. Song S, Xu X, Ren W, Liu S, Jiang J (2024) Determination and application of the RHT constitutive model parameters for ultra-high-performance concrete. *Structures* 69:107488. <https://doi.org/10.1016/j.istruc.2024.107488>
350. Yılmaz T, Kırac N, Anil O, Erdem RT, Sezer C (2018) Low-velocity impact behaviour of two way RC slab strengthening with CFRP strips. *Constr Build Mater* 186:1046–1063. <https://doi.org/10.1016/j.conbuildmat.2018.08.027>
351. Goswami A, Adhikary SD, Li B (2019) Predicting the punching shear failure of concrete slabs under low velocity impact loading. *Eng Struct* 184:37–51. <https://doi.org/10.1016/j.engstruct.2019.01.081>
352. Şengel S, Erol H, Yılmaz T, Anil O (2022) Investigation of the effects of impactor geometry on impact behavior of reinforced concrete slabs. *Eng Struct* 263:114429. <https://doi.org/10.1016/j.engstruct.2022.114429>
353. Batarlar B, Saatci S (2022) Numerical investigation on the behavior of reinforced concrete slabs strengthened with carbon fiber textile reinforcement under impact loads. *Structures* 41:1164–1177. <https://doi.org/10.1016/j.istruc.2022.05.057>
354. Yankelevsky DZ, Karinski YS, Feldgun VR (2023) Damage and failure of a column-supported RC flat slab subjected to impulsive loading. *Appl Sci*. <https://doi.org/10.3390/app13031933>
355. Jawdhari A, Orton S, Al-Zuheriy ASJ, Tian Y (2019) Response of slab-column connections under dynamic collapse load rates. *ACI Struct J* 116(5):55–66. <https://doi.org/10.14359/51715638>
356. Xiao Y, Li B, Fujikake K (2017) Behavior of reinforced concrete slabs under low-velocity impact. *ACI Struct J* 114(3):643–658. <https://doi.org/10.14359/51689565>
357. Saito H, Imamura A, Takeuchi M, Okamoto S, Kasai Y, Tsubota H, Yoshimura M (1995) Loading capacities and failure modes of various reinforced-concrete slabs subjected to high speed loading. *Nucl Eng Des* 156(1–2):277–286. [https://doi.org/10.1016/0029-5493\(94\)00953-V](https://doi.org/10.1016/0029-5493(94)00953-V)
358. Heckotter C, Sievers J (2013) Simulation of impact tests with hard, soft and liquid filled missiles on reinforced concrete structures. *J Appl Mech* 80(3):031805. <https://doi.org/10.1115/1.4023391>
359. Thai D-K, Nguyen D-L, Kim SE (2019) Numerical investigation on local damage of proposed RC panels under impact loading. *Nucl Eng Des* 341:377–389. <https://doi.org/10.1016/j.nucengdes.2018.11.025>
360. Saarenheimo A, Tuomala M, Calonius K (2015) Shear punching studies on an impact loaded reinforced concrete slab. *Nucl Eng Des* 295:730–746. <https://doi.org/10.1016/j.nucengdes.2015.04.018>
361. Orbovic N, Tarallo F, Rambach JM, Sagals G, Blahoiianu A (2015) IRIS2012 OECD/NEA/CSNI benchmark: Numerical simulations of structural impact. *Nucl Eng Des* 295:700–715. <https://doi.org/10.1016/j.nucengdes.2015.06.009>
362. Orbovic N, Sagals G, Blahoiianu A (2015) Influence of transverse reinforcement on perforation resistance of reinforced concrete slabs under hard missile impact. *Nucl Eng Des* 295:716–729. <https://doi.org/10.1016/j.nucengdes.2015.06.007>
363. Mizushima Y, Iino N (2022) Behavior of thin reinforced concrete slabs and effect of reinforcement bars subjected low-velocity impact. *Structures* 38:832–847. <https://doi.org/10.1016/j.istruc.2022.02.036>
364. Othman H, Marzouk H (2016) An experimental investigation on the effect of steel reinforcement on impact response of reinforced concrete plates. *Int J Impact Eng* 88:12–21. <https://doi.org/10.1016/j.ijimpeng.2015.08.015>
365. Gao S, Yang J, Kang S, Li F, Shi X (2024) Experimental and numerical studies on deformation performance of square concrete-filled steel tubular columns under repeated lateral impacts. *Eng Struct* 308:117909. <https://doi.org/10.1016/j.engstruct.2024.117909>
366. Liu K, You Z-Y, Elchalakani M, Kang S-B (2025) Dynamic behaviour of sandwich panel-reinforced concrete composite slabs under low-velocity impact. *Eng Struct* 333:120131. <https://doi.org/10.1016/j.engstruct.2025.120131>
367. Yang Z, Guo X, Huang W, Li Y, Guan H, Diao M, Wang J (2023) Progressive collapse mode induced by slab-column joint failure in RC flat plate substructures: A new perspective. *Eng Struct* 292:116506. <https://doi.org/10.1016/j.engstruct.2023.116506>
368. Pérez-Díaz JA, Ríos JD, Sánchez-González E, Ponce-Torres A (2025) Analysis of the progressive collapse of a parking garage concrete structure due to punching shear. *Eng Fail Anal* 167:108902. <https://doi.org/10.1016/j.engfailanal.2024.108902>
369. Diao M, Li Y, Guan H, Lu X, Xue H, Hao Z (2021) Post-punching mechanisms slab-column joint under upward and downward punching actions. *Mag Concr Res* 73(6):302–314. <https://doi.org/10.1680/jmacr.19.00217>
370. Xue H, Guan H, Gilbert BP, Lu X, Li Y (2021) Simulation of punching and post-punching shear behaviours of RC slab-column connections. *Mag Concr Res* 73(22):1135–1150. <https://doi.org/10.1680/jmacr.19.00465>
371. Yankelevsky DZ, Karinski YS, Feldgun VR, Tsemakh D (2024) Column core response during progressive collapse of RC flat slabs. *Eng Fail Anal* 158:108028. <https://doi.org/10.1016/j.engfailanal.2024.108028>
372. Yang Z, Xu L, Li Y, Guan H, Diao M, Gilbert BP, Sun H (2022) Dynamic response and collapse resistance of RC flat plate structures subjected to instantaneous removal of an interior column. *Eng Struct* 264:114469. <https://doi.org/10.1016/j.engstruct.2022.114469>
373. Cuong TQ, Linh NN, Hung HM, Hieu NT, Dat PD (2018) Experiments on the collapse response of flat slab structures subjected

- to column loss. *Mag Concr Res* 71(5):228–243. <https://doi.org/10.1680/jmacr.17.00394>
374. Ma F, Gilbert BP, Guan H, Lub X, Li Y (2020) Experimental study on the progressive collapse behaviour of RC flat plate substructures subjected to edge-column and edge-interior-column removal scenarios. *Eng Struct* 209:110299. <https://doi.org/10.1016/j.engstruct.2020.110299>
 375. Yang Z, Guo X, Li Y, Guan H, Diao M, Gilbert PB, Sun H (2025) Real collapse responses of an RC flat plate structure under extreme overloading condition: Reversing load redistribution. *Eng Struct* 322:119120. <https://doi.org/10.1016/j.engstruct.2024.119120>
 376. Garzón-Roca J, Sagaseta J, Micallef K (2023) Sudden column removal in tall buildings with flat slabs supported on core and perimeter columns. *Structures*. <https://doi.org/10.1016/j.istruc.2023.104919>
 377. Weng Y-H, Qian K, Fu F, Fang Q (2020) Numerical investigation on load redistribution capacity of flat slab substructures to resist progressive collapse. *J Build Eng* 29:101109. <https://doi.org/10.1016/j.jobe.2019.101109>
 378. Russell JM, Owen JS, Hajirasouliha I (2018) Nonlinear behaviour of reinforced concrete flat slabs after a column loss event. *Adv Struct Eng* 21(14):2169–2183. <https://doi.org/10.1177/1369433218768968>
 379. Qian K, Weng Y-H, Li B (2018) Impact of two columns missing on dynamic response of RC flat slab structures. *Eng Struct* 177:598–615. <https://doi.org/10.1016/j.engstruct.2018.10.011>
 380. Martinelli P, Colombo M, Ravasini S, Belletti B (2022) Application of an analytical method for the design for robustness of RC flat slab buildings. *Eng Struct* 258:114117. <https://doi.org/10.1016/j.engstruct.2022.114117>
 381. Ulaeto NW, Sagaseta J, Chryssanthopoulos M (2021) Horizontal collapse propagation of concrete flat slabs supported on columns. *J Struct Eng*. [https://doi.org/10.1061/\(ASCE\)ST.1943-541X.0003245](https://doi.org/10.1061/(ASCE)ST.1943-541X.0003245)
 382. Peng Z, Orton SL, Liu J, Tian Y (2017) Effects of in-plane restraint on progression of collapse in flat-plate structures. *J Performance Constr Facilities*. [https://doi.org/10.1061/\(asce\)cf.1943-5509.0000984](https://doi.org/10.1061/(asce)cf.1943-5509.0000984)
 383. Liu J, Tian Y, Orton SL, Said AM (2015) Resistance of flat-plate buildings against progressive collapse. I: modeling of slab-column connections. *J Struct Eng* 141(12):04015053. [https://doi.org/10.1061/\(ASCE\)ST.1943-541X.0001294](https://doi.org/10.1061/(ASCE)ST.1943-541X.0001294)
 384. Olmati P, Sagaseta J, Cormie D, Jones AEK (2017) Simplified reliability analysis of punching in reinforced concrete flat slab buildings under accidental actions. *Eng Struct* 130:83–98. <https://doi.org/10.1016/j.engstruct.2016.09.061>
 385. Liu J, Tian Y, Orton SL (2015) Resistance of flat-plate buildings against progressive collapse. II: system response. *J Struct Eng* 141(12):04015054. [https://doi.org/10.1061/\(ASCE\)ST.1943-541X.0001295](https://doi.org/10.1061/(ASCE)ST.1943-541X.0001295)
 386. Keyvani L, Sasani M, Mirzaei Y (2013) Compressive membrane action in progressive collapse resistance of RC flat plates. *Eng Struct* 59:554–564. <https://doi.org/10.1016/j.engstruct.2013.10.040>
 387. Isufi B, Ramos AP (2021) A review of tests on slab-column connections with advanced concrete materials. *Structures* 32:849–860. <https://doi.org/10.1016/j.istruc.2021.03.036>
 388. Ramos A, Marchão C, Pacheco JN, Enfedaque A, Coronelli D, Faria D et al (2024) A review of punching behavior of slab column connections with recycled coarse aggregate concrete. *Struct Concr*. <https://doi.org/10.1002/suco.202400435>
 389. El Zareef MA, Ghalla M, Hu JW, Elbisy AM (2024) Machine learning approaches for estimating concrete shear strength in FRP-reinforced members without shear reinforcement. *Steel Composite Struct* 53(3):327–341. <https://doi.org/10.12989/scs.2024.53.3.327>
 390. Xu W, Shi X (2023) Machine Learning-based predictive models for punching shear strength of FRP-reinforced concrete slabs: a comparative study. *Buildings* 14:2492. <https://doi.org/10.3390/buildings14082492>
 391. Abood EA, Abdallah MH, Alsaadi M, Imran H, Bernardo LFA, De Domenico D, Henedy SN (2024) Machine Learning based prediction models for punching shear strength of fiber-reinforced polymer reinforced concrete slabs using a gradient-boosted regression tree. *Materials* 17:3964. <https://doi.org/10.3390/ma17163964>
 392. Babiker A, Abbas YM, Khan MI, Ismail FI (2025) From robust deep-learning regression to refined design formulas for punching shear strength of internal GFRP-reinforced flat slab-column connections. *Eng Struct* 326:119534. <https://doi.org/10.1016/j.engstruct.2024.119534>
 393. Albostami AS, Mohamad SA, Alzabeebee S, Al-Hamd RKS, Al-Bander B (2025) Optimized punching shear design in steel fiber-reinforced slabs: Machine learning vs evolutionary prediction models. *Eng Struct* 322:119150. <https://doi.org/10.1016/j.engstruct.2024.119150>

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